R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM

A Thesis
Presented to the Faculty of
Department of Computing and Informatics of the
City College of Calamba
Calamba, City

In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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2025

ABSTRACT

The study entitled "R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM" was designed and developed to enhance the efficiency of inventory monitoring and sales recording within R.P. Habana Auto Repair Shop. Currently, the business relies heavily on manual processes such as handwritten records and physical logs. These traditional methods often result in inaccurate inventory counts, difficulty in tracking sales transactions, delays in data retrieval, and risks of data loss or damage. These issues negatively impact the business' ability to maintain stock levels, monitor product movement, and evaluate financial performance.

The researchers were able to gather information about the current inventory and sales process, as well as the needs and expectations of the shop owner and staff, through structured interviews, on-site observations, and ISO 25010-based assessments. To streamline the process, the researchers created a web-based Inventory and Sales Management System to automate stock monitoring, sales recording, and report generation while guaranteeing data quality, secure record storage, and restricted user access in order to address the concerns and expectations they found. Using ISO 25010-based criteria, a post-evaluation was conducted with the clients and staff across all branches to determine the system's effectiveness.

The overall weighted mean of 4.75, interpreted as Excellent, signifies that the developed system aligns with the ISO 25010 software quality standards, demonstrating its effectiveness, reliability, and efficiency in supporting the operations of R.P. Habana Auto Repair Shop.

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

In today's era, businesses are already using digital systems as the modern solution to streamline their operations, making them capable of performing fast-paced tasks that help their employees complete their responsibilities. One specific scenario involves Inventory and sales management, especially for businesses that operate multiple branches. It is common for small and medium-sized businesses to struggle with tracking their inventory and sales when they already have multiple branches. These challenges lead to various problems, such as inaccurate stocks due to human error, lost sales opportunities due to overstocking and understocking of products.

The burden of handling sales and inventory is a significant challenge to employees, who are required to update records manually, coordinate data between different branches, and examine repeatedly to determine whether or not goods are available. The human effort takes up precious time and increases stress levels as well as error possibilities. With growing businesses, the complexity of handling sales and inventory across different locations makes it difficult for employees to track day-to-day operations. Therefore, productivity can be affected, and employees become overwhelmed, leading to burnout as well as a loss of morale.

Project Context

The R.P. Habana Auto Repair Shop is an automotive service provider that specializes in vehicle repair and maintenance, as well as markets vehicle components. The shop was established on October 08, 2007, and currently has seven (7) branches, with its main branch located at Brgy. Bucal, Bypass Road, Calamba City, Laguna. Each of those business branches offers the same range of products and services. According to an interview with the owner, Mrs. Riza P. Habana, the main branch typically serves five to ten (5-10) customers per day, and it has more than twenty (20) employees. The shop guarantees smooth service operation everywhere.

The current process of managing inventory and sales at R.P. Habana is time-consuming and labor-intensive. Each branch typically relies on two to three staff members for inventory recording. One to two staff members are assigned to manually count the quantity of each product, which varies depending on the product category. R.P. Habana carries a large number of products, as observed by the researchers from the main inventory sheet—managed solely by Mrs. Habana. After counting, the final staff member records the data in a handwritten inventory notebook using tally marks. This tallying method is used to easily track and deduct inventory for each sales transaction. The counting and recording process continues until all products are accounted for. To monitor daily sales, staff deduct sold items from the tallied inventory. At the end of the day, they cross-reference their sales using the original receipts (ORs) and the tally in their inventory notebook. This information is then manually totaled to determine the overall sales for the branch.

R.P. Habana's manual systems tend to create inefficiencies and errors, which bring about inventory discrepancies and inaccurate sales and inventory reports several times a month. They are the product of manual counting, tally records, and notebooks, which subject them to miscounts, omissions, and wrong deductions upon sales. Reliable records become difficult to keep, with small discrepancies becoming astronomical inventory and financial mismatches. In addition, the time-consuming process makes routine audits across branches cumbersome, with scanning handwritten logs and receipts being ineffective. This intermittent monitoring slows down decision-making, lowers accountability, and slows down operational efficiency.

By analyzing the gaps found in the existing information from the owner, Mrs. Habana, the researchers intend to develop a centralized web-based inventory and sales management system for R.P. Habana Auto Repair Shop that enables the business to monitor real-time inventory and sales across multiple branches, manage branch operations and stock levels, and integrate Point-Of-Sale System (P.O.S) that automatically updates the inventory in real-time. After this research study, the researchers must be able to improve how R.P. Habana Auto Repair Shop handles its inventory and sales across all its branches.

Objectives of the Study

The objective of the study is to develop a centralized web-based inventory and sales management system for R.P Habana Auto Repair Shop that facilitates real-time tracking and reporting of sales and inventory data. The system being proposed aims to address the limitations of the current manual process by providing accurate sales and inventory data. Specifically, the study seeks:

1. To develop a centralized web-based system that accurately manages sales and inventory reports real-time, including product tracking, stock level monitoring, sales transactions, and comprehensive reporting.
2. To create a system that tracks the number of service jobs completed by R.P. Habana Auto Repair Shop, with reporting capabilities on a monthly, quarterly, and yearly basis.
3. To implement an automated notification system that provides real-time updates on stock levels, keeping users informed about low-stock and out-of-stock products to ensure timely action and improve inventory management processes.
4. To evaluate the feasibility of the proposed system through a cost-benefit analysis.
5. To evaluate the effectiveness of R.P Habana Inventory and Sales Management System by using ISO-25010 2023 in terms of:

    5.1 Interaction Capability;

    5.2 Performance Efficiency;

    5.3 Functional Suitability;

    5.4 Reliability; and

    5.5 Security.

Research Framework

This study aims to developed a centralized inventory and sales management for R.P Habana. The researchers provide a conceptual framework to address the inefficiencies of the current manual process. These frameworks help to understand and guide the development of the proposed system for improving the overall efficiency.

Conceptual Framework

Inaccurate sales and mismatch inventory is the problem faced by businesses who rely on manual processing of managing inventory and sales. Leading to operational inefficiencies. The researchers considered this to develop a system that automatically monitor the inventory stocks and sales transaction to minimize the said issue.

The researchers introduce the conceptual framework of the proposed system which is the Input-Process-Output Model. That provides a comprehensive perspective on the researcher's propose system.

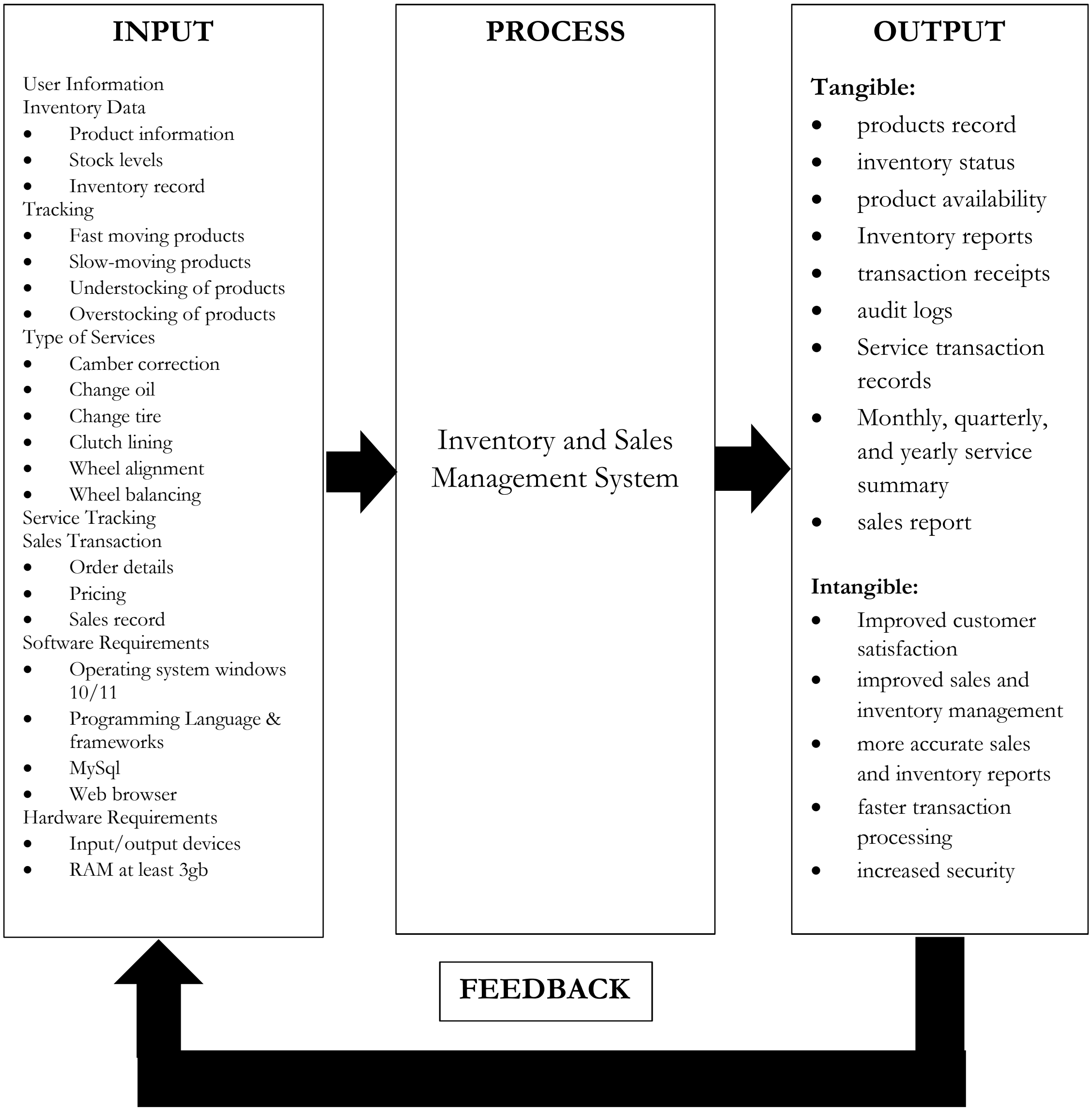

Figure 1- 1. Structure of the Study

Figure 1-1 Illustrates the structure of the study based on the Input-Process-Output (IPO) model of the proposed inventory and sales management system. Highlighting how the system works. The system begins when user gathers and manages inventory data, tracking, service-related details and sales transaction. Additionally, system requirements such as software and hardware components are necessary to support the system functionality.

In the process stage, once the data is collected the inventory and sales management system handles, analyze and updates this information, automating the manual process by recording the inventory records and sales transaction, monitoring the stocks movement, tracking sales, and generating receipts. Through this process, the system reduces human error, improves the processing speed, and provides more reliable data handling.

Following the process of inputs the system produces both tangible and intangible outputs, improving both inventory and sales management. Tangible outputs include product records, updated inventory status, product availability, inventory and sales reports, transaction receipts, audit logs, and sales reports. On the other hand, intangible outputs refer to the operational improvements such as enhanced inventory and sales management, more accurate reporting, faster transaction processing, and heightened security of business data.

Finally, feedback from outputs such as data analytics, improves decision making and indicates when to restock a product. Audit logs also help to identify operational inefficiencies, and reports that monitor the business's performances. This ensures that the system continuously supports accurate inventory and sales management processes.

Significance of the Study

This study aims to improve the managing of inventory and sales for R.P Habana Auto Repair Shop. That rely on the traditional manual processing of business operation that leads to negative impact to business performance. The proposed system that introduces was to manage sales and inventory in real-time reducing the human errors. Implementing this system helps R.P Habana Auto Repair Shop to manage and effectively monitor the status of their inventory. The findings of this study benefited the following;

Customer – With the elimination of delays brought by manual processing, transactions were processed quickly, allowing customers to have a faster service and a better experience.

Employees – Employee were able to work efficiently eliminating the time consuming of manual process and experienced the speeds up of transaction. This allows them to focus on their important task and make the work environment more organized.

R.P Habana Auto Repair Shop – This study improved the management of inventory and sales of the shop. Productivity became more efficient, the stocks and sales reports were easily managed and track. Additionally, it simplified the daily operations for both the owner and its employees, allowing them to focus on business growth rather than handling the manual errors.

Researchers – The proposed study improved the researchers' knowledge and skills in developing systems, which proved useful in their future careers.

Future Researchers - The results of the study served as a resource for future researchers as they encounter similar problems.

Scope and limitation

The researchers' scope is to provide an inventory and sales management system to R.P. Habana Auto Repair Shop to manage the monitoring of inventory stocks and sales transactions more accurately and efficiently. The system objectives are to ensure real-time tracking of fast-moving and slow-moving items, insufficient stock, overstocking, and understocking, sales transactions, data analytics, and reports. Additionally, it is compared to the shop's manual process of inventory and sales to tailor to the specific needs of the shop. User Role and Access Control is implemented to utilize a role-based access control (RBAC) feature that allocates certain dashboards and permissions to a particular user role, thus providing secure and efficient management of inventory and sales. Next, real-time data analytics shows general information such as total sales and inventory of out-of-stock products and low-in-stock products.

There are three user levels in the system, Admin, Staff, and Stockman, wherein each of them has its own functionality and permission to access certain modules in the system. User Dashboard grants Admin access to the central dashboard, which allows the Admin to monitor real-time inventory of all branches, manage user accounts and permissions, monitor activity logs, Service Tracking, Archive files, sales History of all branches, and inspect real-time business data such as fast-moving and slow-moving items and Service tracking to make informed decisions.

While Staff are provided access to only their respective branch dashboard, via which they can only access and run the Point of Sale (POS) system for transaction processing, as well as Sales history to view past transactions. This access control discipline provides Admins with complete control of operations, while Staff focus on effective sales transactions in their respective branch.

Last, Stockman also has access to only their respective branch dashboard, via which they can manage their respective branch inventory, as well as add new products, and make stock transfer requests. Furthermore, Centralized Inventory and Sales Management admin can monitor the inventory and sales of R.P. Habana Auto Repair Shop from any branch as long as they are logged in to system. Lastly, the Point-of-Sale System enables staff to carry out functions like sales processing, which entails choosing products for every transaction, computing totals, and printing receipts. It also facilitates real-time inventory management by automatically updating and deducting stock levels immediately after every sale transaction, ensuring accurate and up-to-date inventory records. On the other hand, the study is limited to the development and implementation of an Inventory and Sales Management System for R.P. Habana Auto Repair Shop.

While the system effectively fulfills its functionality and improves the business operation of R.P. Habana, there are several limitations in the system. One of the shortcomings is the absence of a Purchase Order (PO) Module, even with the presence of suppliers. Without this functionality, the system is unable to manage purchase orders and facilitate automated restocking of the inventory. Another limitation is the lack of communication or a built-in chat module. This affects the internal communication among users, particularly between Admin, Staff, and Stockman. Lastly, the system does not support any cashless payment such as GCash, Maya, or credit/debit card transactions. It restricts the range of available payment options for customers and may affect their satisfaction.

Definition of Terms

This section defines the technical terms used in the study. To help readers to understand its concept. These are provided to ensure clear and accurate comprehension. The following terminologies are defined both operationally and conceptually:

Conceptual Terms

The following phrases are conceptually defined to provide a better understanding of the study being proposed:

Inventory. A list of stocks on hand.

Inventory Management. A process of stocking and tracking goods to ensure there is enough materials.

Monitoring. It refers to an activity of observing and tracking activities.

Sales. The process of selling products and providing services to customers, where money is exchanged.

Sales Management. Process of analyzing sales transactions.

Sales report. The documented summary of all sales transaction over a certain period.

Staff. A person or employee that carried out the work of an establishment.

Stockman. A person who organized and assist in livestock.

Web-based. Application or services that is hosted on web servers and accessed by users through internet-connected devices.

Operational Terms

Centralized Management System. Allows the head admin to monitor inventory and sales across all branches in real-time.

Manual Process. A process of R.P Habana Auto Repair Shop currently used to record and track their inventory and sales without any used of machine or system.

MySQL. A database used to organize, store and managing inventory and sales records for ensuring secure data handling.

PHP. A primary language used for developing R.P Habana Inventory and Sales Management System

Point of Sale System. Used to process sales, track payments and printing receipts.

R.P Habana Auto Repair Shop. The business shop who offers an auto service that featured in the system.

R.P Habana Inventory and Sales Management System. A system responsible for managing the inventory stocks and sales transaction of R.P Habana Auto Repair Shop.

Researchers. The individuals conducting the study and developing the system.

Staff. The employee that handles sales history, process sales transaction and access POS and sales report in the proposed system.

Chapter 2

LITERATURE REVIEW

This chapter represents the ideas, articles, completed projects, and conclusions that contribute to the proposed system. It provides understanding from the previous studies, highlighting the technological advancement related to the topic. Those that are included in this chapter help the researchers to support their study's objectives.

Current Inventory Management Practices in Philippine Auto Repair Shops

The Philippine auto repair industry continues to grapple with significant inventory management challenges that directly impact operational efficiency and profitability. Recent comprehensive studies that the auto repair industry is at a critical juncture, where traditional manual processes are increasingly inadequate and digital transformation encounters multiple barriers. Briones' [1] 2023 nationwide survey of 420 auto repair establishments provides sobering statistics about current practices, showing that 87% of shops still rely entirely on paper-based inventory systems. This reliance on manual methods results in an average of ₱12,500 in monthly losses per shop due to stock discrepancies, misplaced parts, and recording errors. The problem becomes particularly acute for high-usage items like engine oils, filters, and brake components, where tracking inaccuracies account for nearly 35% of total inventory losses across the sector.

Delving deeper into these operational challenges, the Philippine Auto Repair Industry Report [2] conducted a six-month observational study that yielded critical insights about workflow inefficiencies. Their researchers documented that mechanics in shops using manual systems spend an average of 2.3 hours daily – approximately 29% of their productive time – either searching for parts or verifying inventory records. This lost time translates directly into reduced service capacity and longer customer wait times. The study found that shops with digital inventory solutions could accommodate 42% more customer vehicles daily compared to their manual-system counterparts, demonstrating the substantial opportunity cost of outdated practices.

The human dimension of these challenges emerges clearly from TESDA's [3] comprehensive skills assessment of the auto repair workforce. Their findings reveal that only 28% of practicing auto mechanics have received any formal training in inventory management systems, creating a significant adoption barrier for digital solutions. Even more concerning, the assessment showed that 63% of shop owners and managers lack basic computer literacy skills needed to operate digital inventory systems effectively. This skills gap helps explain why many shops continue with manual methods despite their demonstrated inefficiencies, highlighting the need for systems with exceptionally intuitive interfaces and minimal training requirements.

Inventory Management Strategies

According to recent studies, the Service strategy and effective inventory management heavily improve the service process [4] of a Service Company. Effective stock control minimizes wastage of resources, while strategic service planning ensures maximum service excellence. Both of these aspects in combination result in a remarkable improvement in an organization's service delivery processes. Interestingly, the CSI strategy has also been recognized as a great method for improving the quality of IT services and enhancing customer satisfaction. However, most businesses need help to recognize and implement CSI-associated activities. Therefore, business owners and managers need to develop and implement inventory management practices based on scientific and data-driven methodologies. These initiatives can improve competitiveness, optimize service efficiency, and optimize overall organizational performance.

In line with the significance of scientific inventory practices, another study centered on using classification techniques to enhance inventory management, specifically as a method for the detection and prevention of obsolescence. Inventory management is essential in achieving optimal service levels at lower costs in terms of excess or obsolete inventory. One research study proposed a system where items susceptible to obsolescence in an auto spare parts store are determined through inventory classification methods. The system proposed a concurrent ABC-XYZ and FSN analysis [5] for spare parts classification based on criticality and volatility of demand. ABC classification revealed that A-class items, which comprised only 10.39% of the inventory, had the highest value of inventory, which reflected their high dollar value. The XYZ analysis revealed that 52.7% of the products belonged to category Z, which reflected very volatile demand and the need for close monitoring. Furthermore, the FSN analysis classified as non-moving products N-class items, which reflected the urgent need for action to avoid further overstocking. When these techniques were used together, the system managed to discriminate between non-critical and non-moving products with volatile demand, thereby facilitating better inventory control.

Research on food processing companies in Ghana examined the most common inventory management strategies and their effects on operational efficiency [6]. Using a quantitative research design and descriptive approach based on the Theory of Constraints, the research collected primary data on 104 food processing companies using structured questionnaires. It was discovered that Economic Order Quantity (EOQ) and Strategic Supplier Partnership (SSP) were the most frequently utilized strategies in inventory management. These strategies were discovered to have significant effects in ensuring effective inventory control, hence allowing companies to reduce waste, improve the utilization of resources, and improve supply chain reliability. The research underscored the practical significance of their use, and it is recommended that they are essential in fostering firm performance, competitiveness, and contribution toward overall economic development objectives. This research contributes to the emerging literature on inventory management by focusing on an industry setting, hence providing important information for policymakers and business managers to guide their operations in the same setting.

In line with these findings, yet another research study focusing on Oman-based logistics firms highlighted the strategic importance of inventory management as a way of maximizing organizational performance [7]. The study examined different practices such as demand forecasting, procurement optimization, and inventory control practices, and how they influence organizational performance metrics such as order fulfillment, inventory turnover, on-time delivery, and customer satisfaction. The conclusion was that an effective inventory strategy can have a vast impact on the utilization of resources, prevention of stockouts, and overall supply chain responsiveness.

Advancements in Real-Time Inventory Management and Forecasting Technologies

Recent studies have identified a growing need for advanced inventory management systems that can respond to inefficiencies in stock tracking and control, particularly in fast-changing and multi-branch organizational setups. Companies aiming to maintain their competitive advantage in the context of growing changes and dynamics in the current business environment have witnessed a rise in demand for efficient stock inventory management. Recent advancements in technologies have enabled the possibility of developing real-time inventory systems that employ state-of-the-art technologies to address traditional inventory issues. A study revealed a Real-Time Stock Inventory Management System that integrates cloud computing, data analytics, and Internet of Things (IoT) technologies to provide holistic and flexible inventory control solutions [8]. The system enables real-time tracking, monitoring, and management of stock inventory, thereby considerably improving operational accuracy and efficiency. Utilization of IoT devices complements the cloud-based setup by enabling real-time and accurate tracking of inventory amounts, locations, and statuses, as well as improving scalability, accessibility, and flexibility of the system.

The increasing importance of RFID (Radio Frequency Identification) technology in optimizing logistics functions and supply chain management. RFID has been successfully implemented in inventory management systems, which enable stocks to be tracked online in real-time and replace manual traditional practices with computerized procedures to improve efficiency, accuracy, and monitoring. A case study reported in [9] explored the design, implementation, and effectiveness of an RFID-based real-time stock monitoring system, demonstrating the shift from manual tracking to technology-based, automated systems. The study was conducted to integrate the RFID system into existing warehouse infrastructures to evaluate its efficiency and effectiveness. The findings revealed that the RFID system made an evident contribution to inventory control by providing accurate real-time information. The study also demonstrated the potential for future integration of real-time RFID monitoring and inventory optimization models to further enhance purchasing decisions and policy accuracy. The findings illustrate the pivotal importance of real-time technologies in modern inventory management models.

In addition, barcode remains one of the most widely used tools in retail and inventory system. It enhances accuracy in recording sales and inventory management. While requiring significantly less cost and Infrastructure unlike RFID. Barcode is affordable and easy to implement. As it suitable to small and medium enterprises (SMEs). [] barcode speed up the process. By reducing the manual encoding, providing a better visibility in stock movement within and across the branches. This makes barcode technology a practical and sustainable solution for SMEs that seek to modernize inventory processes without the high costs of IoT infrastructures.

Furthermore, forecasting and safety stock level determination are key elements of efficient supply chain planning. Demand forecasting entails studying historical records together with different internal and external factors to forecast future demand for products or services. Correct forecasts assist in avoiding stockouts and overproduction, thus facilitating more efficient production, inventory, and logistics planning. Safety stock is held by companies to act as a buffer against demand volatility or supply interruptions—another critical element for guaranteeing service continuity without unwarranted carrying costs. As noted in [10], conventional methods of forecasting are mainly aimed at reducing errors in forecasts without considering the impact on inventory costs. The research suggests the application of Key Performance Indicators (KPIs) that include inventory cost implications in the forecasting process. In addition, it suggests an innovative approach to determining optimal safety stock levels through the incorporation of supply reliability and seasonality factors obtained from historical demand patterns. Based on real-world data analysis, the suggested approach provides forecast accuracy and inventory efficiency improvements, effectively reducing the risk of stockouts and excessive inventory levels.

Unique Challenges of Multi-Branch Inventory Coordination

The complexities of inventory management multiply exponentially in multi-branch operations like R.P. Habana, creating challenges that single-location shops seldom encounter. The ASEAN Automotive Journal's [11] comparative analysis of regional auto repair chains provides valuable perspective on these multi-branch difficulties. Their study of 42 Philippine multi-branch operations revealed that these businesses experience 35% more inventory discrepancies than their Malaysian or Thai counterparts, suggesting systemic issues with current management approaches.

Adding regional depth to this issue, the ASEAN Automotive Research Group [12] conducted a comparative study of inventory practices across five Southeast Asian countries. Philippine shops, the study found, had the lowest adoption rate of synchronized inventory tracking tools (just 9%) and the highest reported incidence of stock duplication and emergency part transfers. These inefficiencies inflate operating costs by an estimated 17% annually and disrupt workflow continuity.

Through detailed case studies, the research identified three specific pain points that plague Philippine multi-branch auto repair businesses. First, inconsistent record-keeping practices between branches occur in 78% of cases, leading to confusion and stockouts when parts are needed unexpectedly. Second, uncoordinated purchasing decisions result in 22% overstocking of slow-moving items, tying up capital in inventory that may take months or years to use. Third, inefficient inter-branch transfers account for 18% of total operational costs, as shops struggle to move parts between locations in a timely manner.

Rodriguez's [13] in-depth supply chain analysis of Manila-based auto repair chains provides concrete examples of these challenges in practice. One particularly illuminating case study followed a three-branch operation that lost ₱350,000 annually due to duplicate orders of specialty tools that could easily have been shared between locations. Another revealed that 42% of emergency part deliveries between branches – costing ₱1,200 per delivery on average – could have been avoided with real-time inventory visibility. These findings directly informed our system's centralized inventory dashboard design, which provides branch managers with instant visibility of all locations' stock levels while maintaining separate accountability controls to prevent unauthorized transfers.

Sales Tracking and Documentation Deficiencies

The literature reveals equally significant challenges in sales tracking and documentation practices across Philippine auto repair shops. The Automotive Service Association of the Philippines' [14] comprehensive audit of 215 shops provides the most authoritative data on current practices and their shortcomings. Their research found that 71% of shops still use triplicate carbon-copy receipts, a method prone to numerous documentation failures. Most alarmingly, these manual systems result in 28% reconciliation discrepancies between services actually performed and revenue properly recorded.

Further emphasizing the problem, Rodriguez [15] conducted a nationwide audit of manual POS systems across 120 auto repair shops and identified a 34% mismatch rate between service logs and actual payments received. Her findings suggest that hand-written receipts are not only prone to loss and damage but also vulnerable to internal fraud, particularly in multi-shift operations where oversight is minimal. These deficiencies reduce revenue reliability and hinder shops' ability to make accurate financial forecasts.

The audit identified specific service categories particularly vulnerable to underreporting. Lubrication services and minor repairs show the highest rates of documentation failure, with 23% of these transactions never making it into formal records. The study attributes this to several factors: the high volume of small-ticket services, the casual nature of many quick-service transactions, and the lack of systematic tracking for complimentary services that often accompany paid work.

Reyes' [16] groundbreaking payment processing study introduced another critical dimension to this problem, documenting that shop's using manual invoicing experience average collection periods of 17.5 days – nearly double the 9.2-day average for shops with digital systems. The research traced this disparity to three key factors: lost or incomplete paper invoices (occurring in 31% of transactions), calculation errors in manual billing (found in 19% of invoices reviewed), and delayed invoice preparation (averaging 2.3 days after service completion). These documentation failures create cash flow challenges that 68% of shop owners in the study identified as their most pressing operational concern, often forcing them to delay payments to suppliers or reduce inventory levels below optimal quantities.

Sales Management System

Bong Jing Yee and Syahida Hassan [17], show that the SMS is effective as it boosts business productivity, streamlines sales processes and offers an essential sales performance. The automation of sales management leads to accurate sales. Lastly, the use of a sales management system ensures accuracy and reliability as it reduces employee error and mishandling. This system automatically updates the stock levels across all the branches whenever a product is sold. In accordance with Acosta et al. [18], explored the efficiency of Sales and Inventory Systems in business operations. The issue identified was manual errors in sales tracking. A mixed-methods approach, including statistical analysis and case studies, evaluated automation's impact. Findings showed reduced errors, prevented stock discrepancies, and real-time sales data access. Businesses integrating automation improved operational accuracy and minimized financial risks.

According to A Salsabila istiqlal [19] sales management software makes business more efficient as it makes the sales easier, and handles tasks automatically. Without a system it is said that many of the business owners struggle to track their sales, leading to loss of revenue. It is also stated that sales management provides real-time analytics that helps companies, businesses and owners refine their sales by identifying their most sold products. Ensuring a seamless workflow reduces manual errors. Additionally, adopting a software management system is a tactical approach when aiming for long term business success.

Point of Sale (POS)

Point of Sale System (POS) is one of the most important systems in a business operation according to Tran [20]. Tran highlights the seven (7) reasons why the Point-of-Sale system is a must in a business, and three (3) of those reasons also supports inventory management, sales tracking, and employee management. First, it eliminates human error since it eliminates the manual process of inputting the products and prices, hence, there are consistent prices across the system. It also stores sales records that can be reviewed if needed. Second, it improves the speed of the purchasing process by having a centralized database of the products, pricing, and customers. Lastly, it also provides a satisfactory record-keeping system for the inventory of stocks, preventing loss of sales.

According to Hidayati et al [21] implementing the point-of-sale system enhances the business operations. The system effectively handles both inventory and sales transactions.

They also emphasized that POS helps to make data-driven decisions. This effectively reduces the manual processing errors that may occur. However, Hidayati et al also stated that even though POS comes with disadvantages. There could be challenges also such as staff training. When implementing a POS, the employee or staff must learn how to operate the system for smooth operation. Businesses must also ensure that the system is working properly with the existing operations. This concludes that having a Point-of-Sale system is a wise strategy for growing the business and having the edge on other competitors.

Synthesis

R.P. Habana continues to rely on manual, paper-based processes for inventory and sales management, a situation common among Philippine SMEs and small auto repair businesses. Studies show that many shops still depend on spreadsheets, handwritten logs, or printed forms for inventory tracking, leading to inconsistencies, misplaced parts, and recording errors [20][21]. While previous studies primarily focused on small SMEs or single-branch operations, they rarely address multi-branch coordination issues that are central to R.P. Habana's seven-branch setup. This highlights a gap in research for centralized, real-time inventory tracking across multiple locations [5][12][13].

These operational challenges have significant implications on service capacity and profitability. Research on Philippine SMEs demonstrates that automated or web-based inventory management systems improve accuracy, reduce human errors, and streamline reporting, thereby enhancing efficiency [20], [21]. However, many prior studies either focus on high-cost IoT/RFID solutions [8], [9] or are limited to single-branch POS integration [17],

[18], [19], making them less practical for small and medium auto repair shops. In contrast, our study proposes a barcode-enabled centralized system, which is cost-effective and tailored to the shop's multi-branch structure, directly addressing issues such as overstocking, duplicate orders, and inefficient inter-branch transfers.

The integration of POS and barcode systems in prior literature has shown improvements in transaction accuracy and inventory updates [17], [18], [19]. Compared to these studies, our approach emphasizes practicality and scalability for SMEs: it provides real-time visibility of stock across all branches, supports role-based access, and does not require expensive IoT infrastructure.

In summary, while prior studies highlight the inefficiencies of manual processes and the benefits of automated inventory systems, few focus on multi-branch coordination, affordability, and SME practicality. The proposed centralized and barcode-enabled system fills this gap by providing real-time synchronization, accurate inventory tracking, and scalable solutions designed for R.P. Habana's operational needs.

Chapter 3

RESEARCH METHODOLOGY

This chapter outlines the methods and approaches used in the study, including the research design, techniques, and methodology. It illustrates the methods and procedures used by the researchers in the development of the study.

Research Design

In this study, the researchers utilized the qualitative and descriptive-quantitative research design. It aimed to understand and examine the process of sales and inventory of R.P Habana. The design focused on identifying the existing problems, delays, and inconsistencies experienced in their manual and Excel-based workflow.

The study involved both quantitative and qualitative data. Qualitative information gathered from interviews highlighted user experiences and levels of satisfaction, quantitative data collected through surveys were analyzed to identify trends and frequent patterns. This provided deeper insights into user experiences, challenges, and expectations regarding the current system.

By combining both data types, the researchers were able to describe the existing process accurately, validate the user needs, and ensure that the findings reflected the objectives. This approach helped ensure that the proposed system was aligned with actual operational needs of the shop.

Research Locale

This study focuses on R.P Habana Auto Repair Shop an automotive service provider that has seven (7) branches here in Laguna, with over twenty (20) staffs, its main branch is located at Bypass Road. However, the shop is currently using the manual process in inventory and sales processes. And to address these challenges, the system being proposed was designed to improve the inventory and sales management of all its branches.

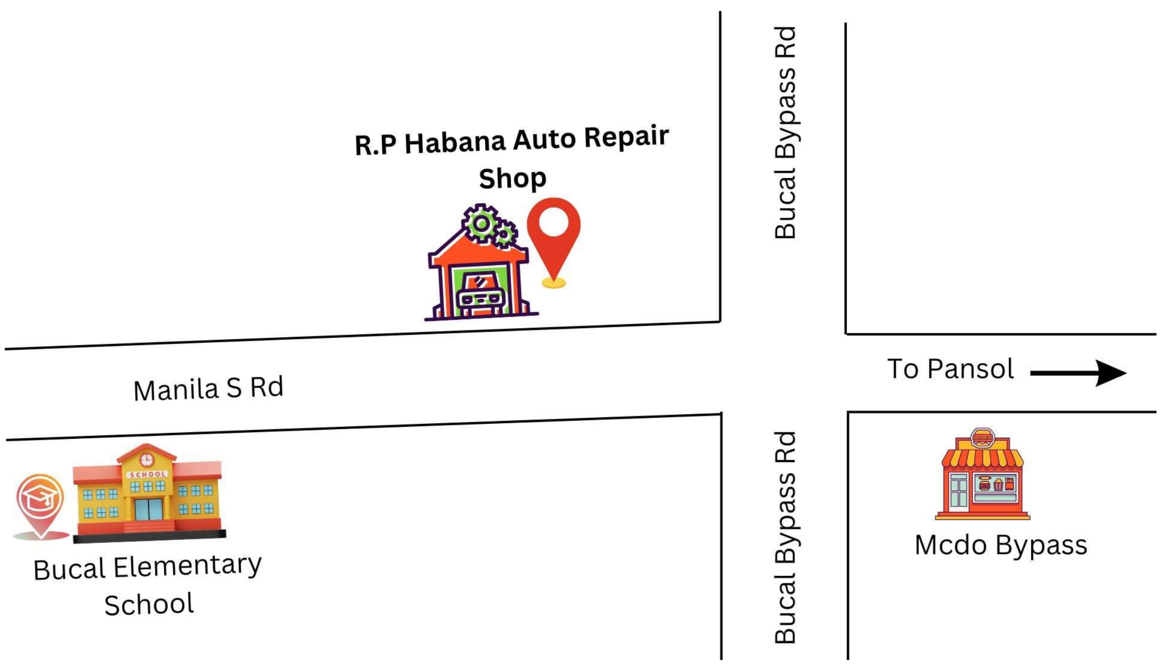

Figure 3-1. Habana Auto Repair Shop

Population of the Study

The subject of this study were the owner, and staff employee with all of its seven (7) branches. The target population consisted of individuals directly involved in managing the inventory and sales of the shop. Specifically, the Head Admin/owner was in charge of all branches, while the staff were responsible for inventory and sales of its respective location, while also in charge in processing of transaction, vehicle repairs and maintenance. A total of 22 respondents were included to ensure that key personnel who interacted with the system provided accurate feedback for assessing the system's effectiveness in improving inventory and sales management

Table 3- 1. Respondents of the Study

TYPE OF RESPONDENTS	NUMBER OF RESPONDENTS
Owner	2
Staff	20
Total	22

Table 3-1 provides information on the distribution of participants in the study. It consists a total of twenty-two (22) respondents. Specifically, the owners and its twenty (20) staff members. This highlights the users who are involved in the inventory and sales operation of the shop.

Sampling Design

Total enumeration is a type of purposive sampling technique that is commonly used if researchers intentionally choose the entire population that has a particular set of characteristics. [22]. This sampling technique considers all the individuals offering a reliable finding and it guarantees that no perspective was overlooked. The presence of all these individuals provides the most valuable insights making this approach convenient. Lastly, it is suitable for gaining a preliminary understanding of the target population. This approach ensures that the information obtained is relevant and in accordance with the system's objectives by focusing on key individuals involved in managing the inventory and sales of the shop.

Data Gathering Tools

The researchers conclude that conducting interviews, surveys, and internet research are the most suitable tools to gather the needed data. These tools can help in gathering data effectively and provide strong support for the study.

Interview. A qualitative approach that relies on asking questions in order to collect data [23]. This approach helps the researchers to collect intensive data. During the interview, the researchers used various techniques to assist in gathering information. The researchers took notes and recorded the interview session with the interviewee's permission. They also provided a guide questionnaire for the interviewee to have brief insights about the flow of the interview when held, this is to maintain the communication and preparation of the participants in an interview. The interviewer stated the questions and followed up questions with good and respectful deliberation to gain a complete and brief understanding of the response of the interviewee.

Observation. In this method, the researchers examine and analyze the existing process of R.P Habana Auto Repair Shop. By observing how the current manual inventory and sales transaction are being carried out. Through this method the researchers are able to gain information on how employee use inventory records, sales processes and how they keep an eye to the stock the levels. In order to compare the current workflow to the system being proposed

Survey. The researchers provided a survey questionnaire to the owner and staff of R.P Habana Auto Repair Shop. This method helped the researchers gain insights from the client about integrating the system into the current manual inventory and sales process.

Document review. Document review was used as a data collection method, which involved examining current R.P. Habana Auto Repair Shop documents such as sales and inventory procedures. The researchers reviewed the inventory records, sales transaction and other related documents to assess the efficiency of the previous workflow. This method provided qualitative information on stock shortages and inconsistencies. By examining these records, the researchers determined how the system could fill the existing gaps.

Internet research. The internet serves as a communication to the researchers as it serves as the guide and resource of information. As stated by Consultores [24], it provides ease to the researchers as it is easy to navigate and offers convenience in conducting research. The researchers conduct a review of related literature and studies from both local and foreign sources to gather information and create a solid base for their research study. They also follow the rules of referencing those articles and blogs they find relevant in their research to practice ethical referencing and avoiding the act of plagiarism.

Data Gathering Procedure

To ensure the successful development of a centralized inventory and sales system for R.P. Habana Auto Repair Shop. The researchers used several data collection to gather data. On February 13, 2025, the researchers conducted a face-to-face initial interview with the owner of R.P Habana Auto Repair Shop. This opportunity serves as a basis for understanding the current process of managing inventory and sales of the shop, it served as the foundation for understanding the existing manual processes in inventory and sales. A guided questionnaire was used during the interview, and with the clients' consent the interview was recorded and notes were taken. Following this on February 28, 2025 the researcher provides a formal client acceptance letter to the owner of R.P Habana Auto Repair shop to officially start to administered interviews and surveys, after that the researcher as when there are available and proceed to hand them an appointment letter for march 3 to start another interview, on this follow up interview. The researchers also conducted a direct observation examining how the client and its staff handle the inventory and process sales transaction. by comparing these practices to the proposed system, the researchers identify the areas that could benefit from the automation of the existing manual process.

A document review was also conducted, manual tracking sheets, inventory logs, and sales reports have been analyzed by the researchers to identify inconsistencies. These method helps the researchers to validate the problem raised during the interviews and observation. Internet research was also used to review related literature and studies to ensure the system adheres to the best practices in inventory and sales management. All sources that have been used by the researchers were properly cited to avoid plagiarism. On May 06, 2025 another follow-up interview was done, to gather additional information. This aims to collect a more detailed data regarding on the shops fast moving and slow-moving products. And what items they do regularly restocked.

Lastly, the survey questionnaire was administered to gather quantitative data. A 5-point Likert scale was used to evaluate system performance. Through these instruments, the researchers were able to gather comprehensive data and feedback essential to the analysis and development of the proposed system.

Data Analysis Plan

In the data analysis plan, the researchers examined and decoded the data gathered from the interview and survey. Both quantitative and qualitative methods were applied to gain a better understanding of the data results.

Table 3- 2. Data Analysis Plan of the Study.

OBJECTIVES	PROCEDURE	SOURCE OF DATA	STATISTICAL ANALYSIS
1. To develop a centralized web-based system that accurately manages sales and inventory reports real-time, including product tracking, stock level monitoring, sales transactions, and comprehensive reporting.	Survey, Interview and Observation	Interview, Observation and Questionnaire	Qualitative and Quantitative
2. To create a system that tracks the number of service jobs completed by R.P. Habana Auto Repair Shop, with reporting capabilities on a monthly, quarterly, and yearly basis.	Interview and Survey	Interview and Questionnaire	Qualitative and Quantitative
3. To implement an automated notification system that provides real-time updates on stock levels, keeping users informed about low-stock and out-of-stock products to ensure timely action and improve inventory management processes.	Interview, Observation, and Survey	Interview, Observation, and Questionnaire	Qualitative and Quantitative
4. To evaluate the feasibility of the proposed system through a cost-benefit analysis.	Interview	Riza Habana and Richard Habana (Client)	Qualitative
5. To evaluate the effectiveness of R.P Habana Inventory and Sales Management System by using ISO-25010:2023 in terms of: 5.1 Interaction Capability; 5.2 Performance Efficiency; 5.3 Functional Suitability; 5.4 Reliability; and 5.5 Security	Survey	Survey Questionnaire	Quantitative

Table 3-2 Presents the objectives of the study that help the researchers have a deeper understanding of the study using interviews, observations, document review, and internet research. This all helps to make data gathering more efficient and effective.

Statistical Treatment of Data

This section provides a systematic approach to analyze the gathered data that efficiently evaluate the survey and interview to effectively analyze the collected data. The Likert-scale was selected as a main instrument in the analysis of the survey questionnaire as it provides an accurate representation of the respondents' opinions about the system's overall effectiveness.

Weighted Mean

This formula was used to calculate the average of the survey responses. it was applied to interpret the quantitative data because it provided a precise estimation of respondent opinions. A 5-point Likert Scale was used, offering a neutral midpoint and an "excellent to poor" descriptive range appropriate for assessing system effectiveness, especially for feasibility and ISO evaluation.

To compute the weighted mean, each Likert scale value (**x**) was multiplied by the number of times it was selected (**f**). The products were added, and the sum was divided by the total number of responses (**N**).

Formula for the Weighted Mean

$$Weighted\ Mean = \frac{\sum(f \cdot x)}{N}$$

Where:

f = frequency of responses

x = Likert scale value

N = total number of responses

Five Point Likert Scale.

Table 3- 3. Five Point Likert Scale

RATING	WEIGTHED MEAN	DESCRIPTIVE VALUE
5	4.21-5.00	EXCELLENT
4	3.41-4.20	VERY SATISFACTORY
3	2.61-3.40	NEUTRAL
2	1.81-2.60	FAIR
1	1.00-1.80	POOR

This table 3-3 Presents the calculation of the weighted mean for determining the average responses gathered from the system evaluation.

Software Development Methodology

System Overview

The R.P Habana Inventory and Sales Management System is developed to address the current challenges shop faced to its multiple branches. The system provides a centralized platform for monitoring the inventory levels and sales transaction of the business.

The system provides a clear monitoring of all transaction and movements, as it offers tangible output such as product records, inventory status updates, sale transaction, transaction receipts, and audit logs. Through the intangible benefits the system offers a more accurate data reporting, faster transaction processing, and inventory and sales improvement.

The system is built using a PHP and MySQL as its database for structured and efficient data storage. It offers key features such as Role Based Access Control (RBAC) to limits permission based on user roles. While the owner/head admins have access to a centralized dashboard where it can monitor all the transaction and inventory of all branches, the Staff employee have access only in their specific branch.

Software Development Process

Figure 3-2. Agile Methodology

The Agile methodology is a framework that breakdown projects down into several dynamic phases called sprints. It is an iterative process where developers evaluate and determine what could be improved so it can modify their approach for the next sprint. [25] The researchers used this approach to ensure that the system being proposed for R.P Habana Auto Repair Shop to gather feedback from the owner, allowing them to improve the system functionality, efficiency and accuracy. Several studies also use this approach due to its flexibility and to gathered user feedback throughout the process. Using this approach, the researchers can deliver quality outcomes to ensure client satisfaction.

Phase 1. Requirements Planning

Requirements planning was an initial step in developing the system. At this stage, the researchers conducted interviews to determine the challenges in the previous workflow of managing inventory and sales. Through these interviews, the researchers gathered the necessary objectives and requirements to address the shop's needs. They also evaluated the existing workflow and identified the users who would interact with the system, ensuring that the design reflected a comprehensive understanding of what an effective and accurate system required.

Phase 1.1 Hardware and Software Requirements

This section outlined the essential hardware and software requirements for developing the system for R.P Habana Auto Repair Shop. These requirements ensured that the system operated efficiently and supported the performance and reliability needed for an inventory and sales management system.

Table 3- 4. Software Requirements for R.P Habana Inventory and Sales Management System.

System Requirement	Specifications
Operating system	Windows 10/11
Web browser	Google Chrome, Microsoft Edge and any web browser

As for table 3-4 the following are the specific software requirements that need for system to function seamlessly. This include using a window 10 and above together with its compatible web browsers.

Table 3- 5. Hardware Requirements for R.P Habana Inventory and Sales Management System.

System Requirement	Specifications
Standard input/output devices	Generic
Internet	Min. bandwidth of 25mbps
RAM	At least 3gb or higher

As for table 3-5 that provides the minimum hardware requirements for system performance. Any standard I/O devices can be used. And a RAM capacity must be 3gb or higher. The user must also have a proper internet connection to access the system since it's a web-based and also monitor the real-time inventory and sales transaction.

Phase 1.1.1 Gantt Chart

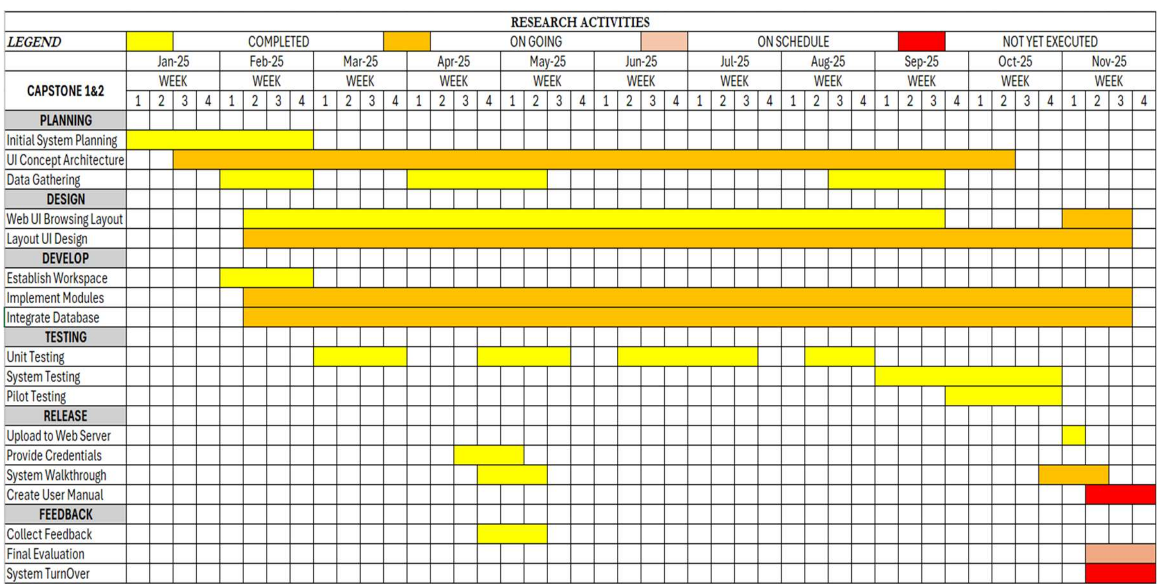

Figure 3-3. Gantt Chart

This figure shows the Gantt chart for developing the R.P Habana Inventory and Sales Management System. Which includes the planning, designing, development, testing, release, and feedback. This outlines the project timeline to guarantee that the system is developed effectively and meets the necessary functionality, usability, and deployment timeline. This strategic planning enhances the project's feasibility by providing a visual roadmap for effective project management.

Phase 2. Design

At this phase, the developers began creating wireframes to visualize the system layout and interface. These wireframes were used to allow the client to review and approve the system structure of the database was also planned to ensure efficient inventory and sales tracking. Additionally, the developers selected the appropriate framework that would support the system/s overall performance.

Phase 2.1 System Design

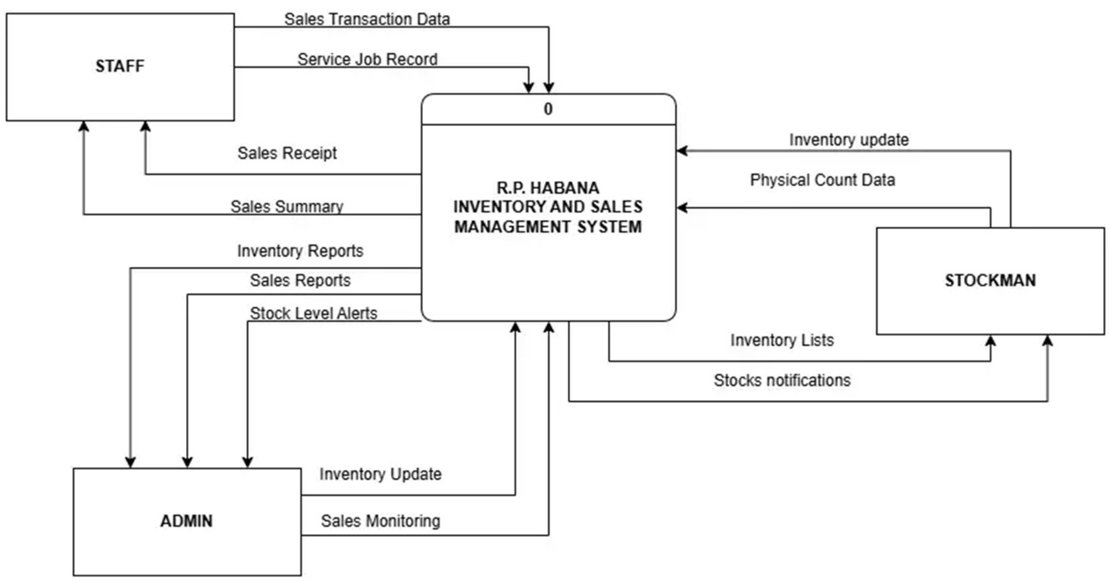

Figure 3-4. Context Diagram

The above context diagram provides a complete picture of the R.P. The Habana Inventory and Sales Management System demonstrate the connections between three important user groups—Staff, Stockman, and Admin—and the centralized system. Stockman responsibilities include entering in, adding inventory, and staff for processing sales transactions. In response, the system provides critical information such as updated total sales and inventory, a list of updated Inventory, and access to the staff and stockman dashboard. These activities are critical for ensuring real-time accuracy in stock levels and sales records, allowing employees to efficiently handle day-to-day retail or warehouse operations.

In contrast, the Admin receives Inventory Reports, Sales Reports, and Stock Level Alerts from the system to support business monitoring and decision-making. At the same time, the Admin sends Inventory Updates and oversees sales performance across all branches. These interactions show that the system acts as a centralized hub that coordinates the activities of Staff, Stockman, and Admin. Through real-time data exchange, the system improves operational efficiency, strengthens record accuracy, and ensures organized communication throughout the entire sales and inventory workflow.

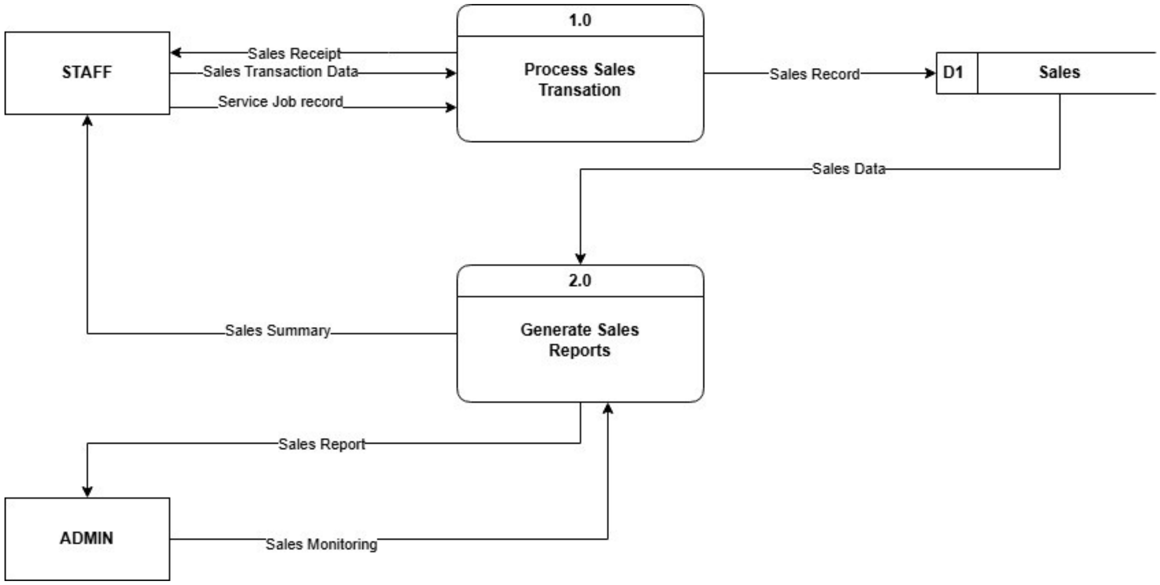

Figure 3-5. Data Flow Diagram Sales: Level 1

The Level 1 Sales Data Flow Diagram illustrates how sales information flows between the Staff, Admin, and the Sales data store (D1). The process begins with the Staff through Process 1.0 (Process Sales Transaction), where they input sales transaction data and generate sales receipts and service job records. These outputs are then stored as Sales Records in the Sales data store (D1).

From Process 1.0, the system forwards the Sales Data to Process 2.0 (Generate Sales Reports), which compiles the sales information and produces a detailed Sales Report. This report is provided to the Admin for monitoring daily sales activities and overseeing business performance. The Admin receives Sales Summaries from Staff and uses the generated reports to perform sales monitoring.

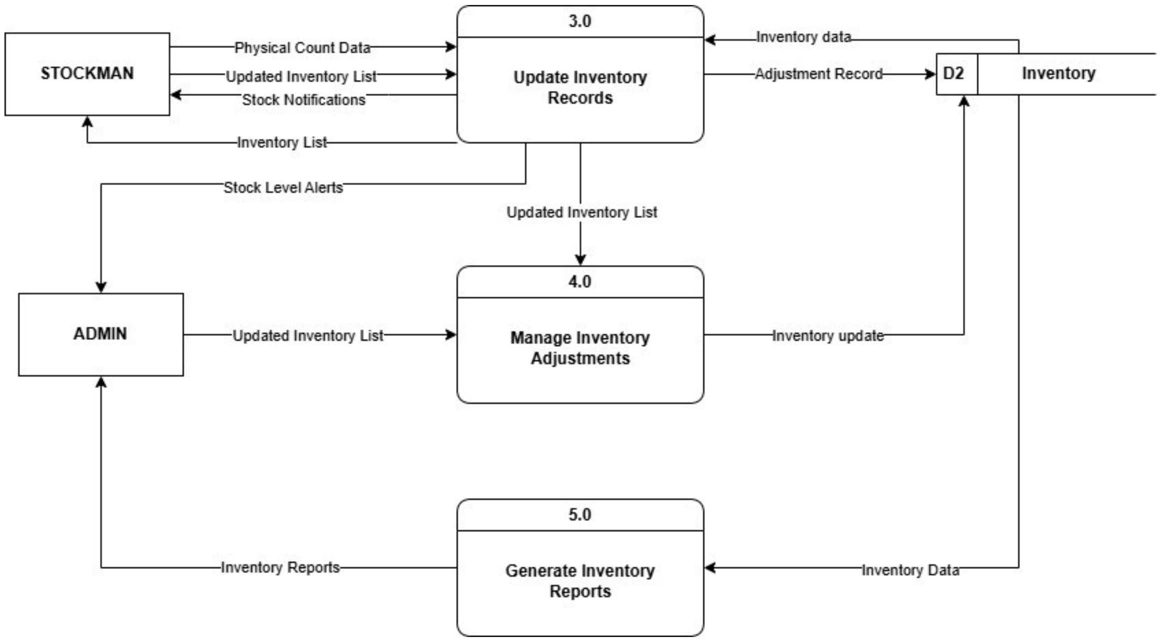

Figure 3-6. Data Flow Diagram Inventory: Level 2

The Level 2 Inventory Data Flow Diagram presents a detailed view of how inventory records are updated, adjusted, and reported. The Stockman plays a central role through Process 3.0 (Update Inventory Records), where they provide physical count data and stock notifications. These inputs generate updated inventory lists and adjustment records that are stored in the Inventory data store (D2).

Process 4.0 (Manage Inventory Adjustments) receives updated inventory lists and handles any necessary stock corrections or modifications. These adjustments ensure that the Inventory data store (D2) maintains accurate and real-time inventory information.

Process 5.0 (Generate Inventory Reports) uses data from the Inventory data store to create inventory reports for the Admin. The Admin also receives stock level alerts, enabling them to monitor inventory conditions and oversee stock replenishment decisions.

Together, the processes show a coordinated flow of inventory management tasks: The Stockman updates records, adjustments are validated, and the Admin receives comprehensive reports and alerts that support informed decision-making and efficient operational control.

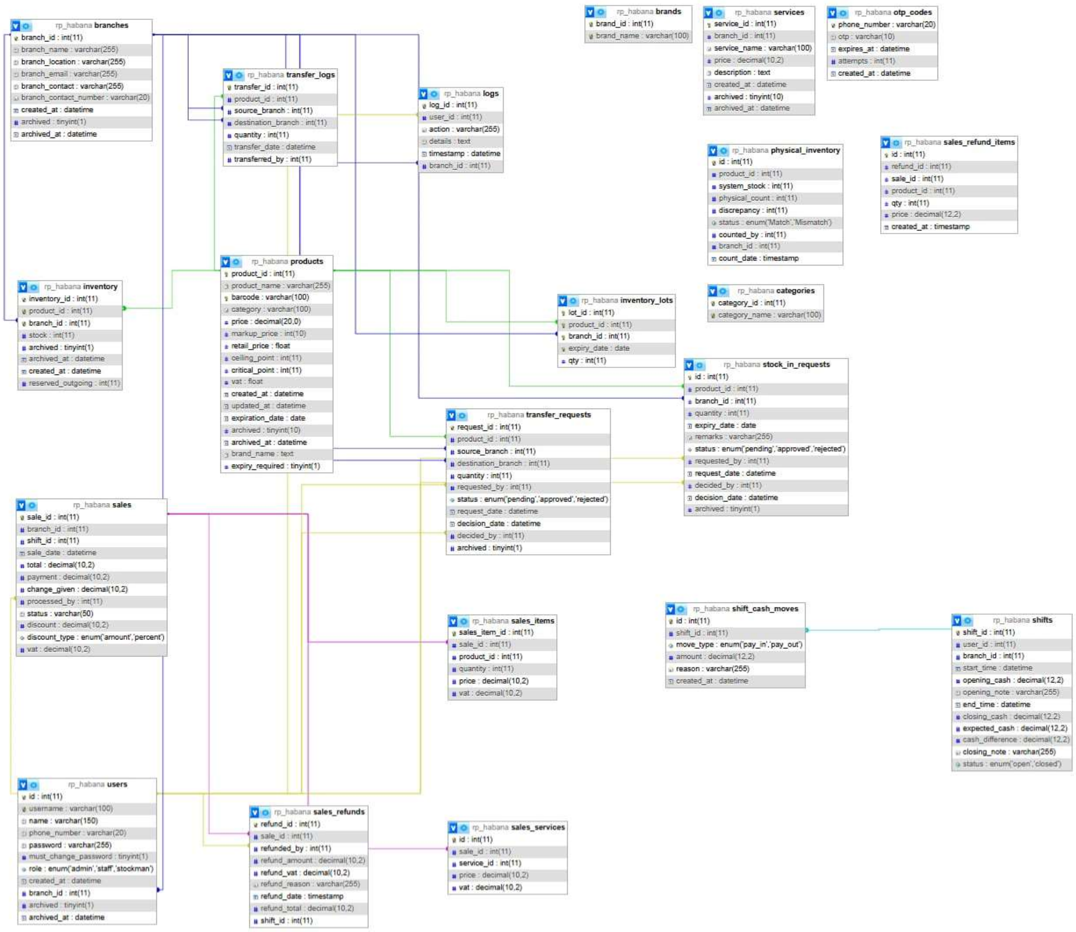

Figure 3-7. Entity Relationship Diagram

This figure presents the Entity Relationship Diagram (ERD) of the system. showing how data entities such as products, inventory records, sales transactions, stock movements, branch information, and user accounts are interconnected. The ERD ensures that the system supports accurate tracking of stock levels, branch coordination, point-of-sale processing, and audit logging across multiple locations.

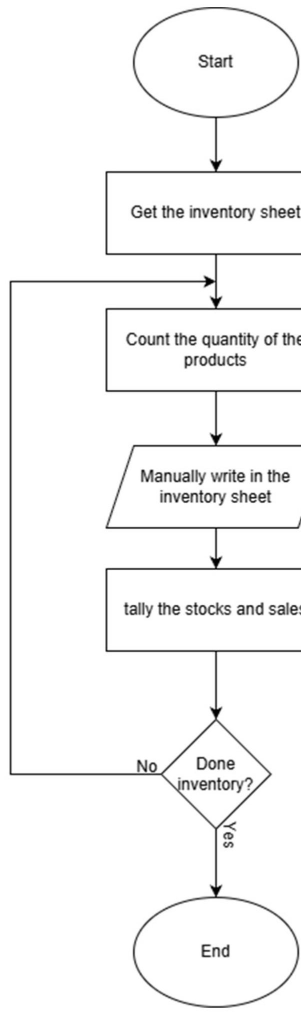

Figure 3-8. Current Existing Flowchart

This figure presents the process of manual inventory checking, starting with retrieving the inventory sheet before counting the actual quantity of products. After counting, the user records the numbers manually in the sheet and then tallies the updated stock with the recorded sales for accuracy. This process repeats until all items have been checked, and once the inventory is fully updated, the procedure ends.

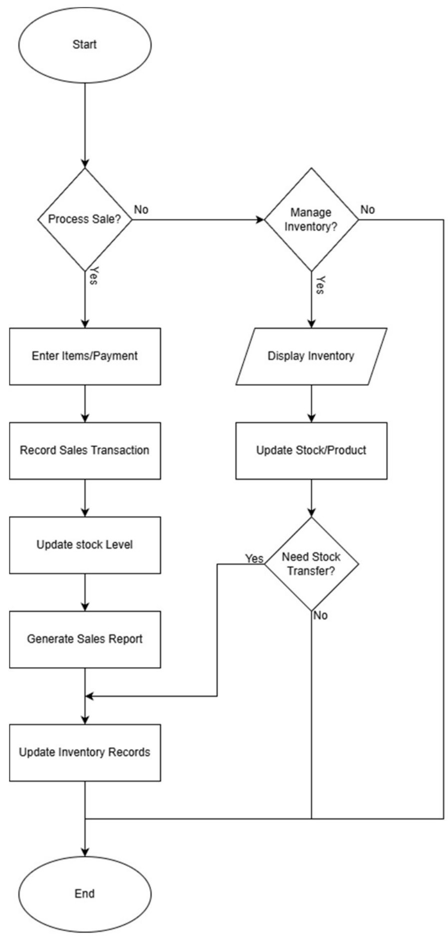

Figure 3-9. Entire Inventory and Sales Management System Flowchart

This figure illustrates the overall process flow for both sales transactions and inventory management. It begins with selecting whether the task is for sales or inventory. For sales, the system processes the sale by entering items and payments, recording the transaction, updating stock levels, generating a sales report, and updating the inventory records. For inventory tasks, the user can display current stock and add or update inventory levels, with the option to perform a stock transfer if needed. The process ends once all required updates and records have been completed.

Phase 2.2 User Interface and Function

In this phase, it introduces the proposed system and its main functions. The system is a web-based platform designed to be easy to use and help users manage data and tasks more efficiently. The researchers created this system to make work faster and more organized. The interface and features shown below may change as the system continues to improve during development.

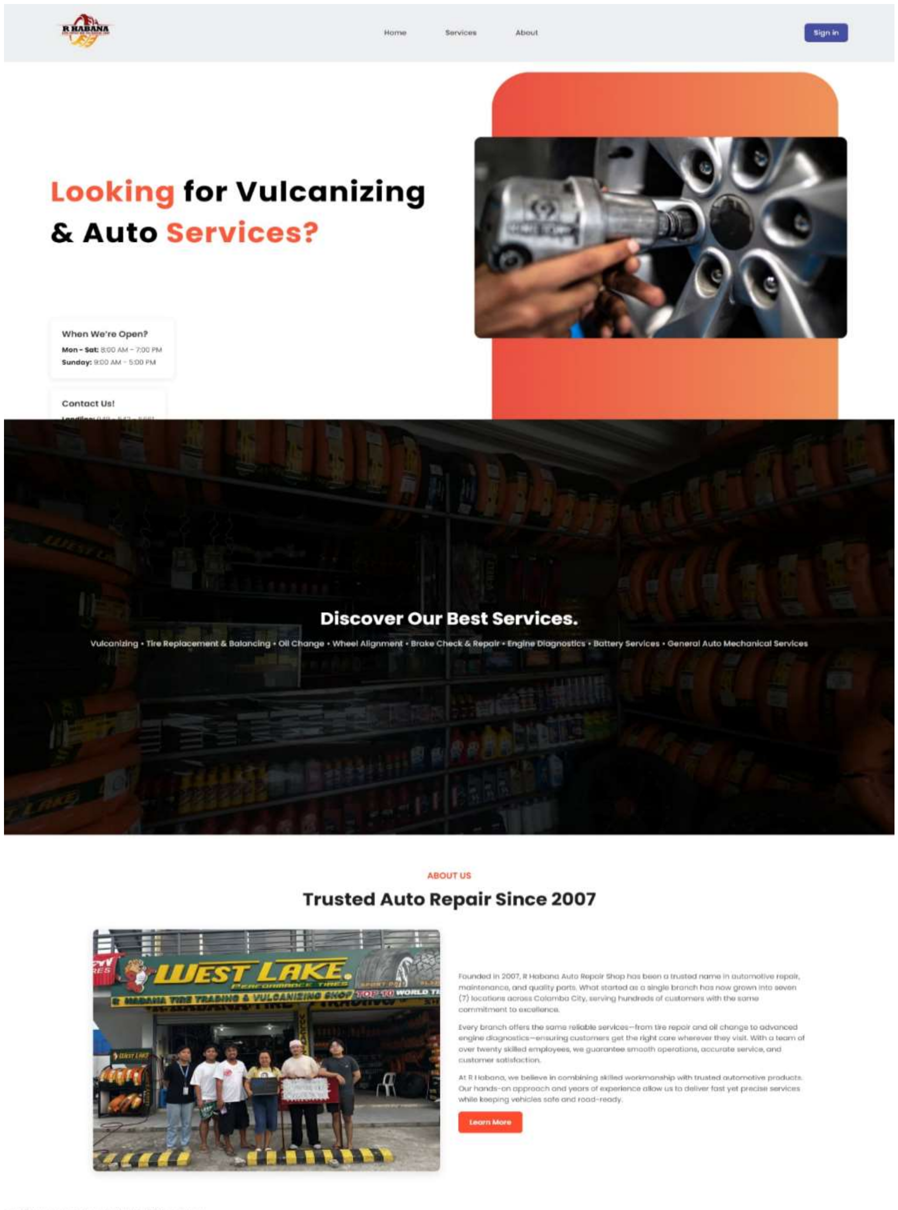

Figure 3-10. Landing Page

Figure 3-9. This figure features business hours, contact details, and a clear overview of services to help customers easily connect and learn about the shop.

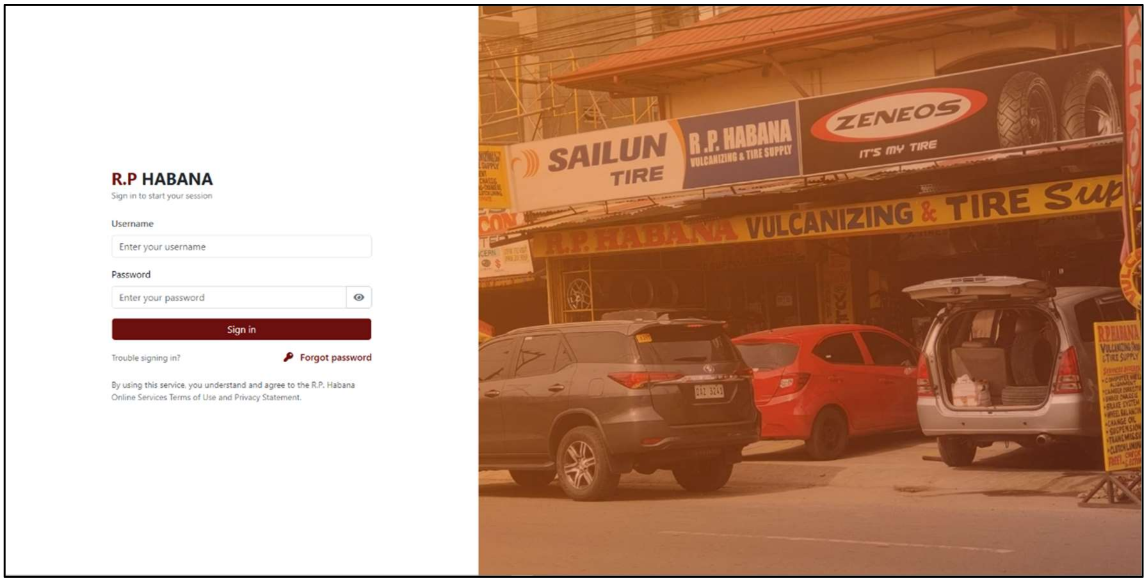

Figure 3-11. LogIn/SignIn

Figure 3-10. displays the sign-in page of the R.P. Habana Inventory and Sales Management System. The interface features a modern split-screen layout, with the left side displaying the actual storefront of R.P. Habana Tire Supply to reflect its business identity, while the right side presents the login form. To access the system, users must input their login and password. The general layout emphasizes clarity, making it simple for administrators and employees to navigate.

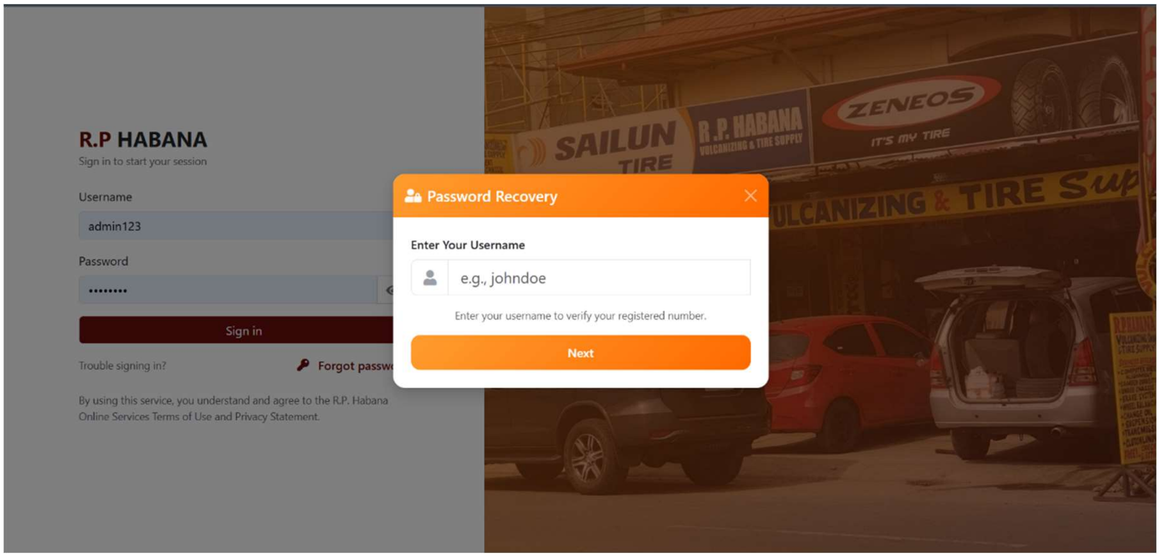

Figure 3-12. Password Recovery

Figure 3-11. shows the Password Recovery interface of the R.P. Habana Inventory and Sales Management System. This feature assists users who have forgotten their login credentials by verifying their registered account information. The form prompts users to enter their username, which the system uses to validate the corresponding registered contact number before proceeding to the next step of the recovery process. The interface is designed with clarity and accessibility in mind, providing a user-friendly experience through its organized layout and distinct color scheme. This module enhances account security and ensures that only verified users can regain access to the system.

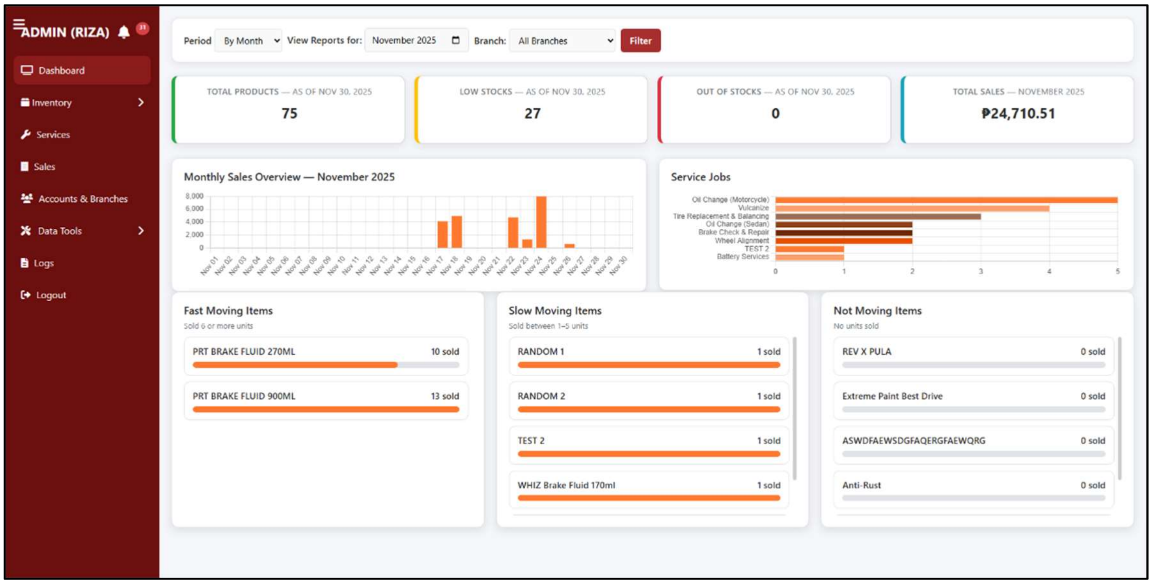

Figure 3-13. Admin Dashboard

Figure 3-12. shows the Admin Dashboard of the R.P. Habana Inventory and Sales Management System. This provides an overview of essential business metrics such as total products, low stocks, out-of-stock items, and total sales within a selected reporting period and branch. It also presents data visualizations and tables for monthly sales overview, service jobs, and item movement categories including fast-moving, slow-moving, and not-moving items. The sidebar navigation allows administrators to access various modules such as Inventory, Sales, Services, Accounts & Branches, Data Tools, and Logs.

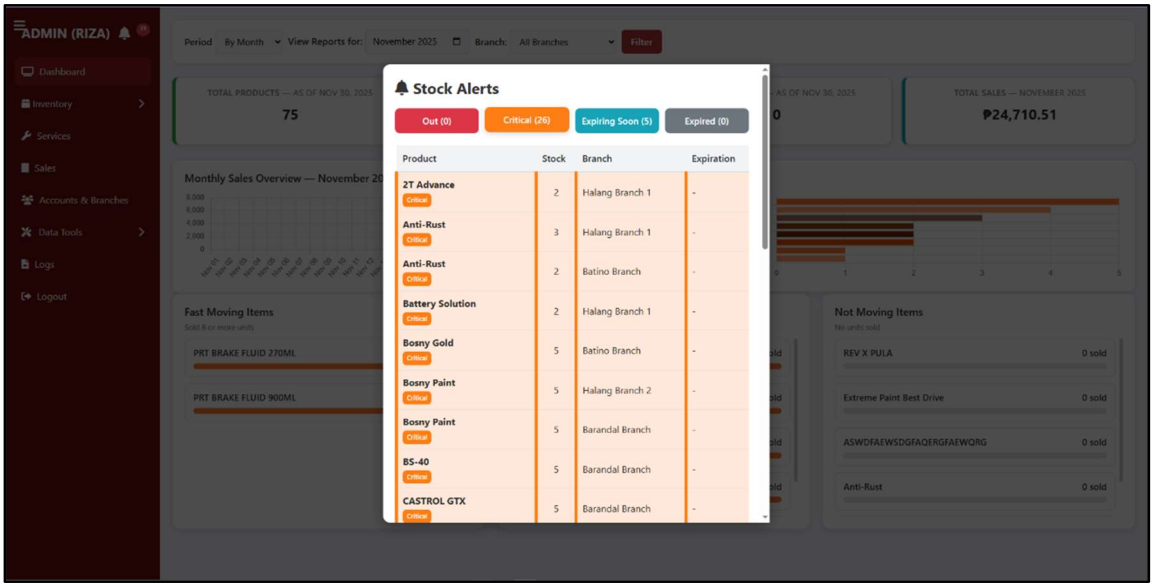

Figure 3-14. Stock Alert

Figure 3-13. displays the Stock Alerts modal of the R.P. Habana Inventory and Sales Management System. This feature provides real-time notifications to administrators regarding items that are out of stock, at critical levels, or nearing expiration. Important information including the product name, branch location, current stock count, and expiration date (if applicable) are displayed in the modal. Users may readily detect inventory difficulties thanks to the color-coded alert categories, which are red for Out of Stock, orange for Critical, and blue for Expiring Soon. By facilitating timely restocking and reducing product shortages among branches, this feature improves operational efficiency.

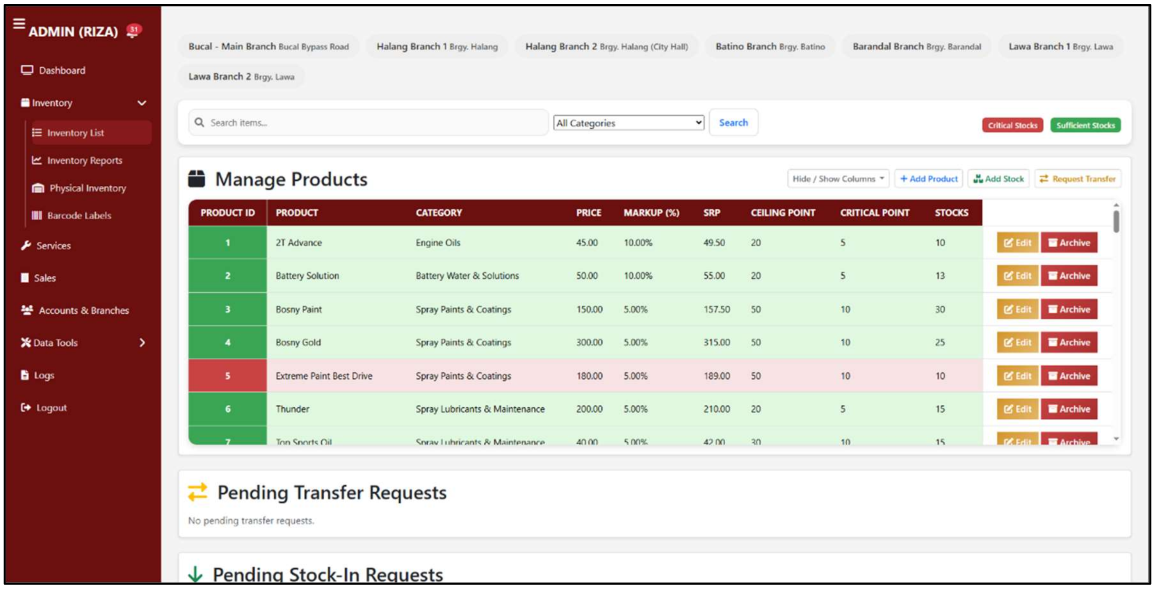

Figure 3-15. Admin Inventory List

Figure 3-14. display the Inventory Management Page of the R.P. Habana Inventory and Sales Management System. This interface allows the administrator to monitor and manage product records across multiple branches. Each product entry displays key details such as product ID, name, category, price, markup percentage, retail price, ceiling point, critical point, and available stocks. Color-coded rows indicate stock conditions—red for critical stocks and green for sufficient stocks—providing quick visual identification. The system also features options to add products, add stock, and request transfer between branches. Additionally, pending transfer and stock-in requests are displayed below for efficient tracking and coordination of inventory operations.

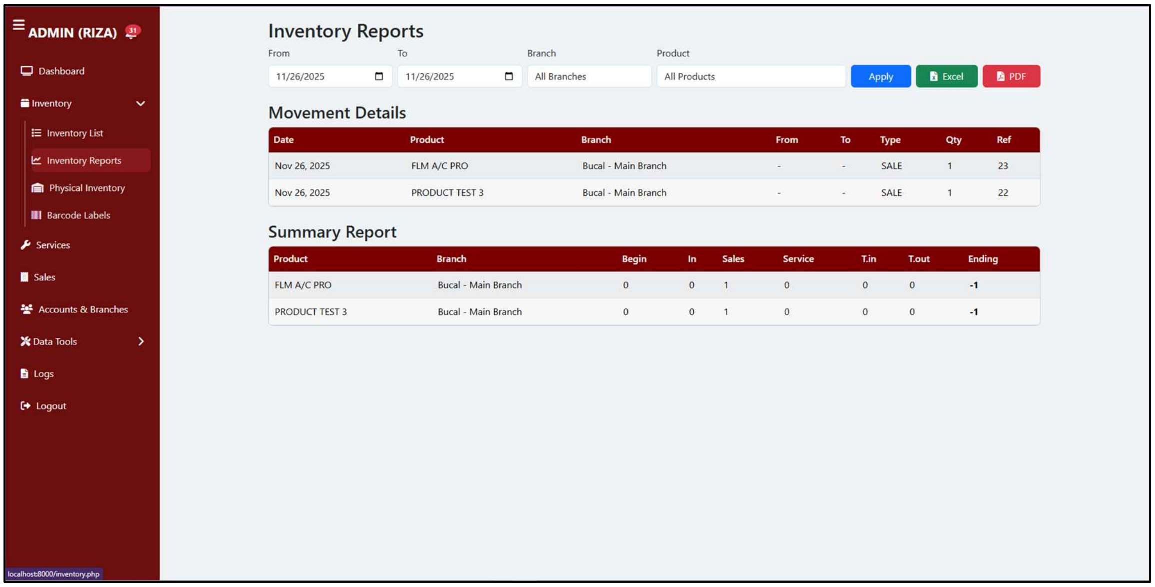

Figure 3-16. Admin Inventory Reports

Figure 3-15. Inventory Reports interface of the r.p habana Inventory and Sales Management System, showing the module's filtering capabilities for generating product movement and summary reports based on specified dates, branches, and product categories. The interface displays transaction details such as sales activity, quantities, and reference numbers, along with a consolidated summary of beginning balances, incoming and outgoing transactions, and computed ending inventory for each product. This module supports accurate monitoring and analysis of inventory performance across branches.

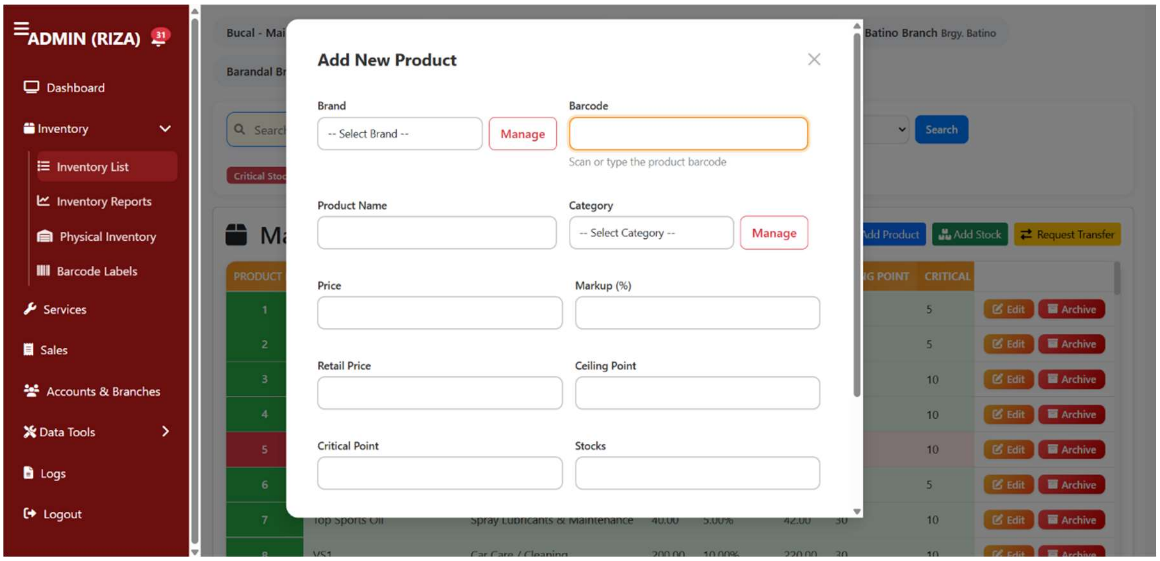

Figure 3-17. Admin (Add Product)

Figure 3-16. Shows the Add New Product form of the R.P. Habana Inventory and Sales Management System. This interface enables administrators to register new items into the inventory database by entering essential product details such as brand, barcode, product name, category, price, markup percentage, retail price, ceiling point, critical point, stock quantity, VAT, expiration date, and assigned branch. The form also includes options to add a new brand or category directly within the same window, promoting flexibility and faster data entry. A Save button is located at the bottom-right corner to confirm and store the new product record in the system database.

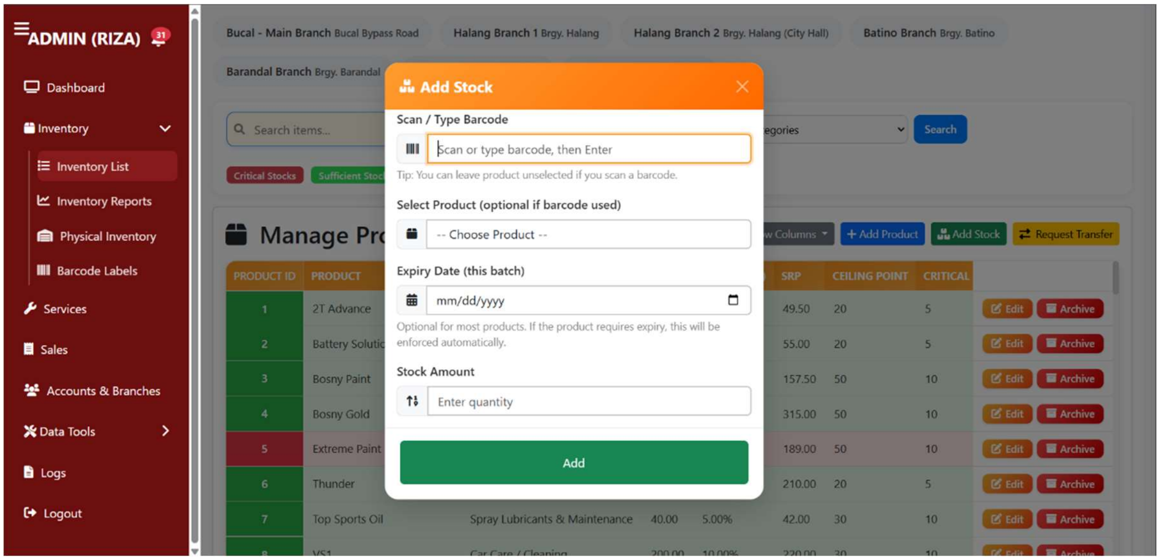

Figure 3-18. Admin (Add Stock)

Figure 3-17. illustrates the Add Stock form of the R.P. Habana Inventory and Sales Management System. This interface allows administrators to record new stock entries for existing products. Users can scan or manually enter a product barcode, or select an item from the dropdown list if no barcode is provided. The form also includes input fields for expiry date (if applicable) and stock quantity, ensuring accurate batch tracking and inventory updates. Once the necessary information is filled out, the administrator can click the Add button to update the stock database. This feature supports efficient inventory management by maintaining real-time accuracy of product quantities across branches.

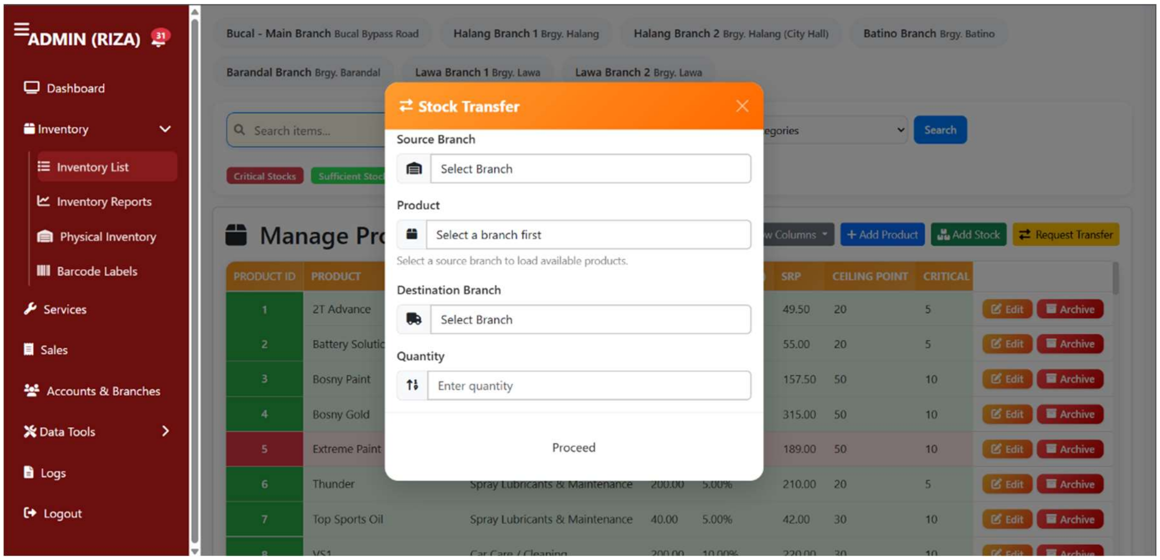

Figure 3-19. Admin (Stock Transfer)

Figure 3-18. shows the Stock Transfer form of the R.P. Habana Inventory and Sales Management System. This feature allows administrators to efficiently manage item movement between branches. The form includes dropdown menus to select the Source Branch, Product, and Destination Branch, along with a field to specify the Quantity of items to be transferred. Once completed, users can click the Proceed button to initiate the transfer request. This functionality promotes balanced inventory distribution across branches, reduces product shortages, and ensures that each branch maintains adequate stock levels to meet customer demand.

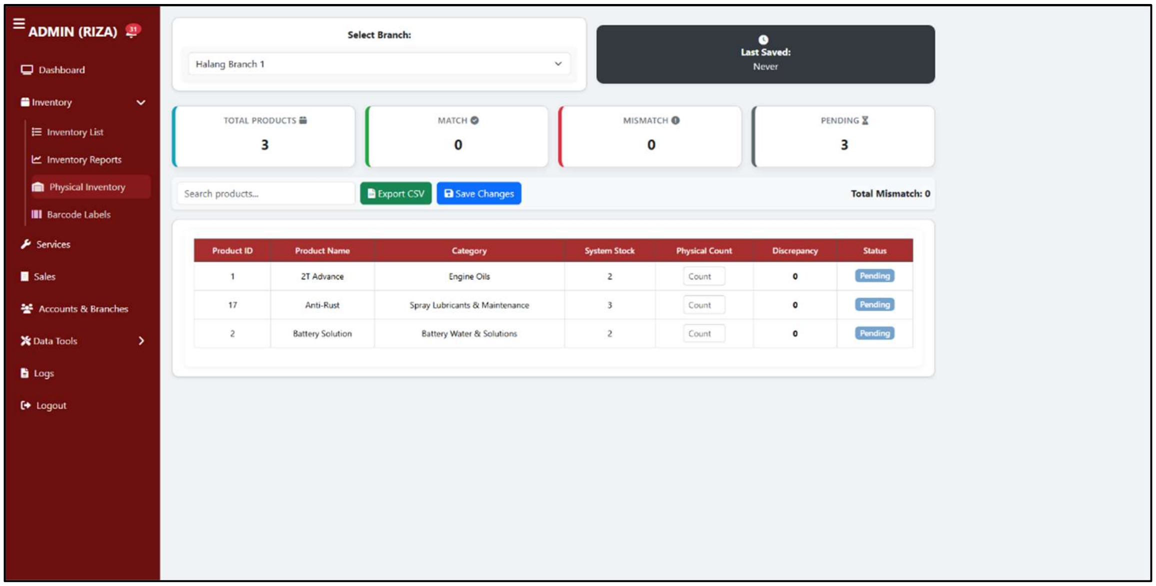

Figure 3-20. Admin (Physical Inventory)

Figure 3-19. This figure shows the System Logs interface, which records user activity within the system. Each log entry includes the timestamp, username, action taken, branch, section, and a brief description. Activities such as logging in and adding inventory items are tracked and documented in this section. This feature provides administrators with a clear and organized overview of user interactions, helping maintain system transparency and security. A button is available for downloading the logs for further review or record-keeping purposes.

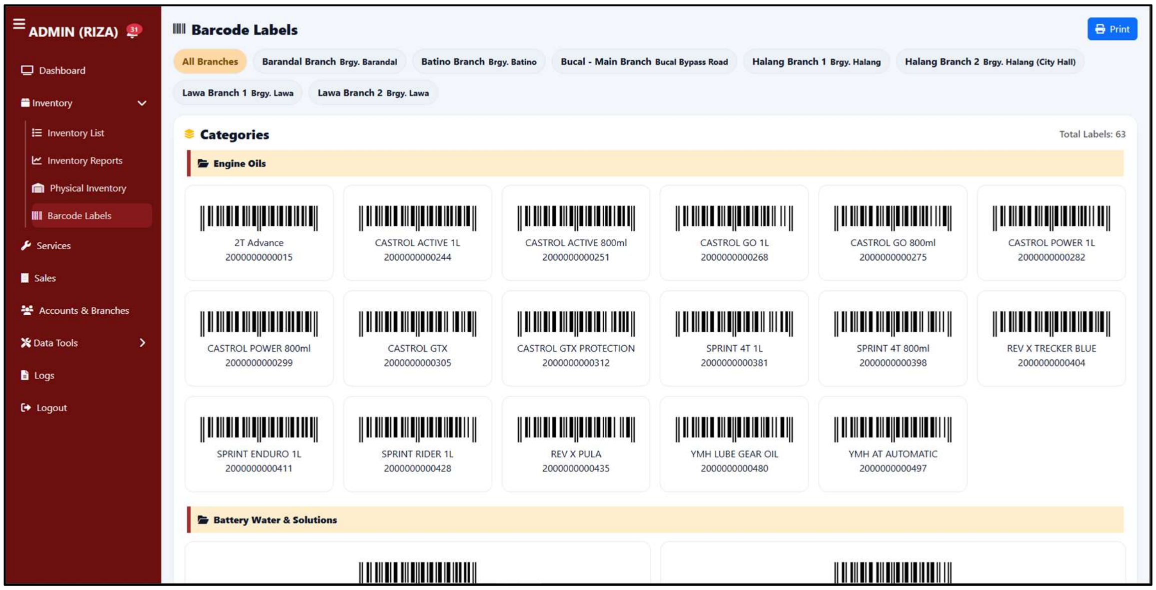

Figure 3-21. Admin (Barcode Labels)

Figure 3-20. illustrates the Barcode Labels page of the R.P. Habana Inventory and Sales Management System. This module automatically generates and displays barcodes for all registered products, categorized by type such as Solid, Liquid, and Electronics. Each product is shown with its corresponding barcode, product name, and identification number, allowing for efficient scanning and tracking during transactions and stock monitoring. Users can filter labels by branch to view location-specific inventories and use the Print button to generate physical barcode copies for labeling and operational use. This feature enhances accuracy and speed in product identification, promoting a more organized and automated inventory process.

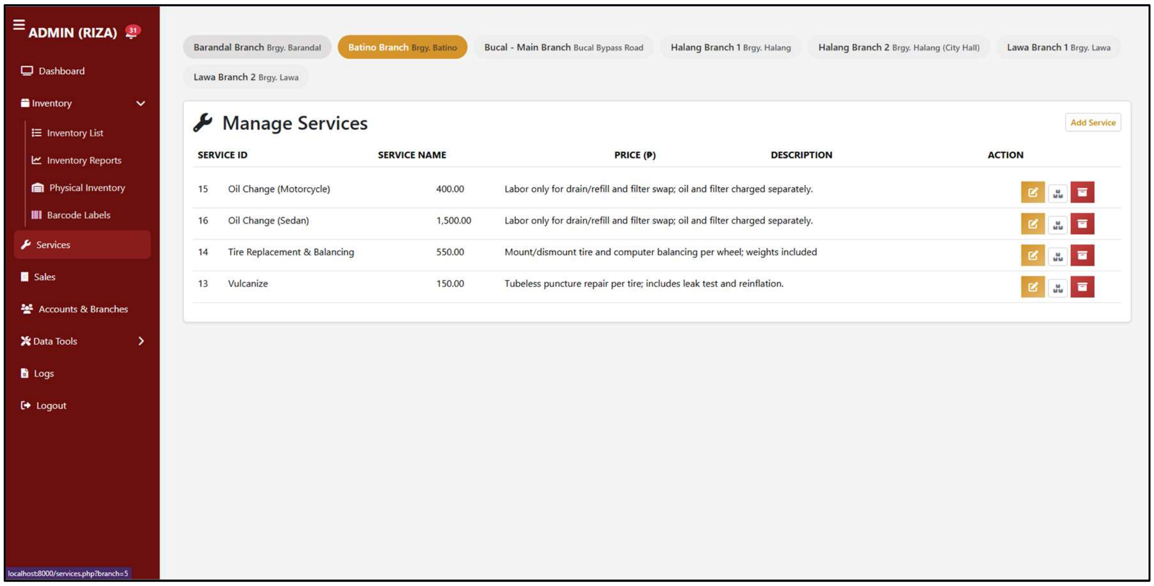

Figure 3-22. Services

Figure 3-21. presents the Manage Services page of the R.P. Habana Inventory and Sales Management System. This module enables administrators to manage and organize the various automotive services offered by the business. Each service record displays its Service ID, Service Name, Price, and Description for easy reference. The interface also includes action buttons that allow users to edit or delete existing services, as well as an Add Service button for registering new ones. This feature helps streamline the service management process, ensuring that all branches maintain consistent and up-to-date service listings.

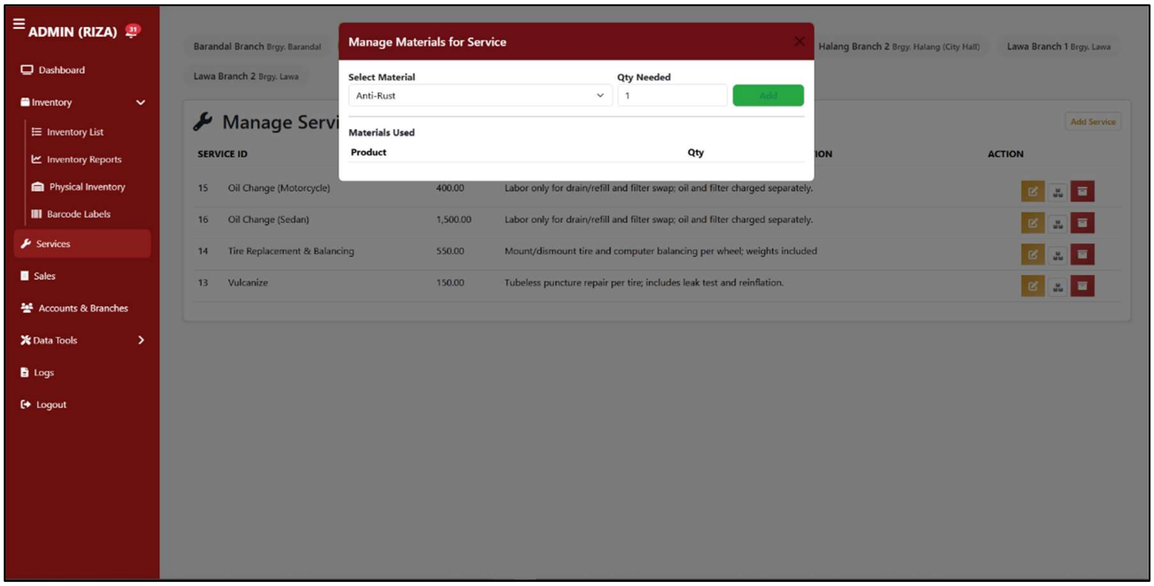

Figure 3-63. Admin Manage Material for Services

Figure 3-22. Manage Materials for Service interface within the r.p habana Inventory and Sales Management System, showing the modal window used to assign required materials for a selected service operation. The interface allows the administrator to choose a specific material, specify the quantity needed, and add it to the list of materials used for the service. This feature supports accurate tracking of material consumption, ensures proper inventory deduction during service transactions, and enhances the system's capability to manage both product-based and service-based operations across branches.

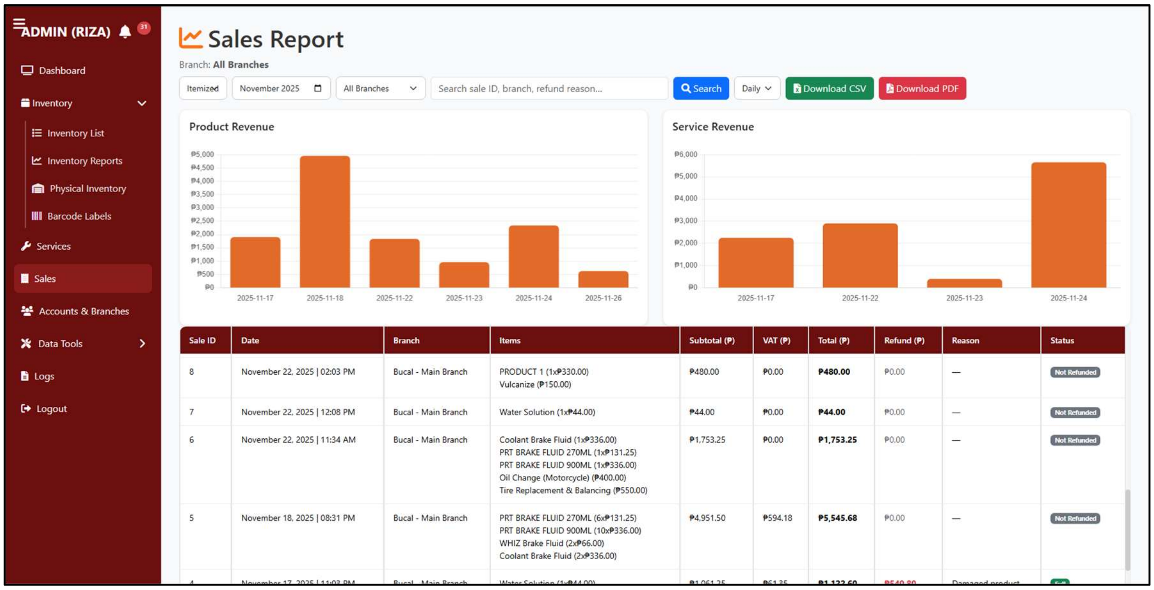

Figure 3-24. Admin (Sales)

Figure 3-23. illustrates the Sales Report page of the R.P. Habana Inventory and Sales Management System. This module provides a detailed record of all sales transactions within a specified period and branch. Each entry includes information such as Sale ID, Date, Period, Branch, Product, Category, Quantity, Price, Total Amount, Refunded Amount, and Refund Reason. The interface allows administrators to filter sales data by month and branch, facilitating efficient tracking of revenue performance and sales trends. This feature enhances business analysis by providing organized, real-time sales data for decision-making and financial reporting.

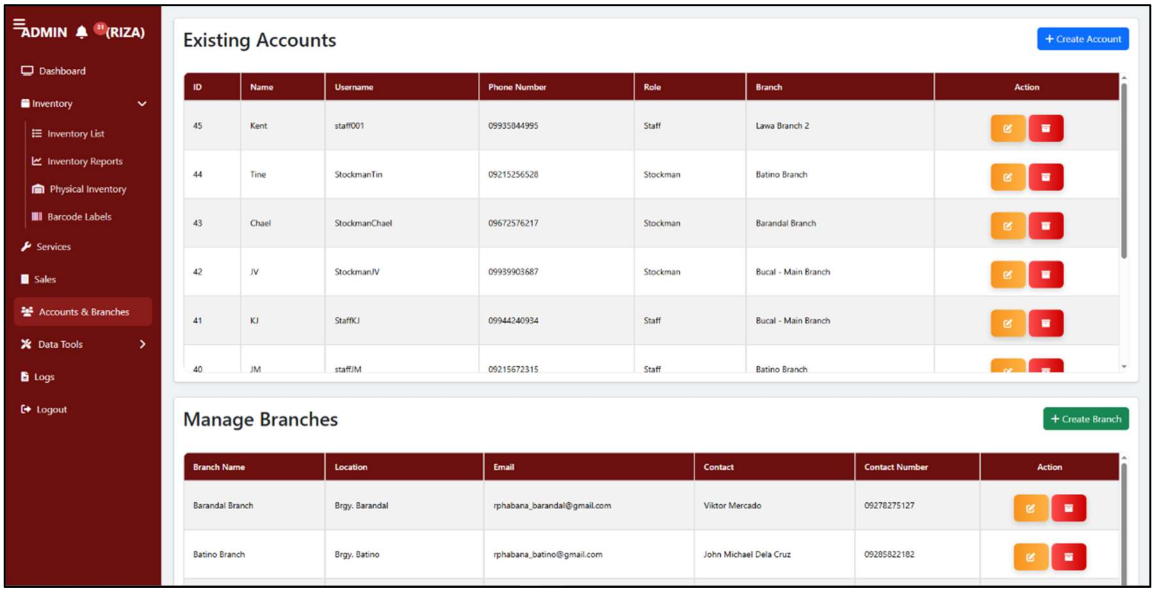

Figure 3-25. Admin (Account and Branches)

Figure 3-24. shows the Accounts and Branch Management page of the R.P. Habana Inventory and Sales Management System. This module allows administrators to oversee both user accounts and branch information within the system. The Existing Accounts section displays a list of registered users with details such as ID, Name, Username, Phone Number, Role, and Assigned Branch. Action buttons enable administrators to edit or delete user records, while a Create Account button facilitates the addition of new system users.

Below, the Manage Branches section presents each branch's Name, Location, Email, and Contact Number, with options to create, update, or remove branch records. This feature supports centralized control over multi-branch operations, ensuring that account and branch data remain synchronized across the system.

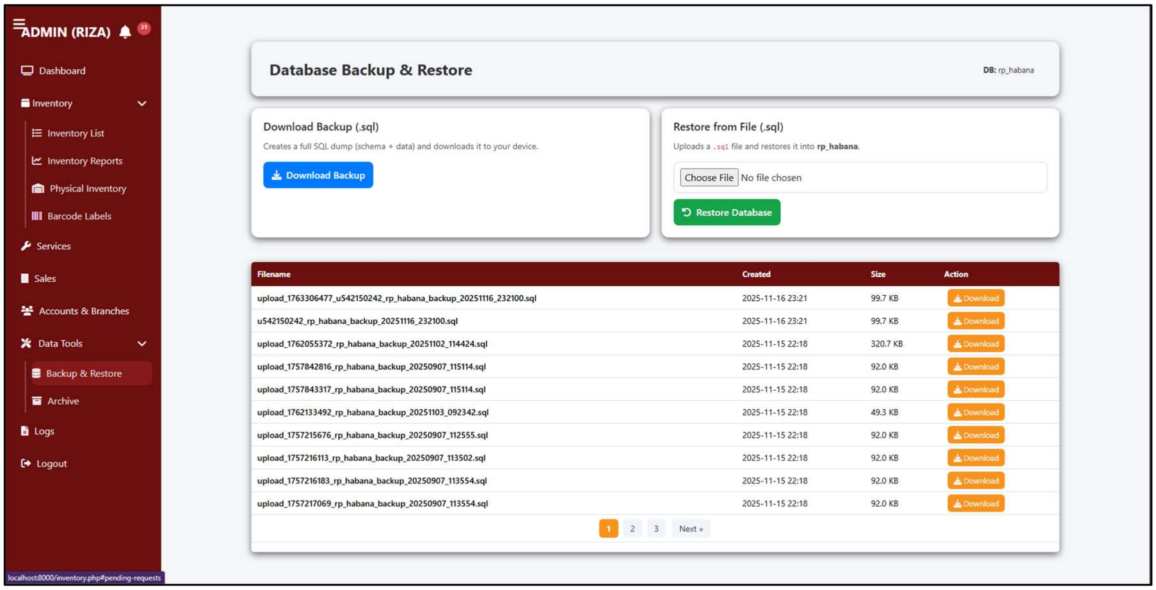

Figure 3-26. Backup and Restore (Admin Data Tools)

Figure 3-25. shows the Database Backup and Restore module of the R.P. Habana Inventory and Sales Management System. This feature enables administrators to securely manage and protect system data. The Download Backup (.sql) section allows users to create and download a complete SQL database dump, including both schema and data, ensuring data preservation and recovery readiness. Meanwhile, the Restore from File (.sql) section lets administrators upload and restore a previously saved database file in case of system failure or data loss. Below, the interface lists all existing backup files with details such as filename, creation date, file size, and a download option. This functionality strengthens data integrity and supports disaster recovery by maintaining a reliable database backup history. displays the confirmation prompt for deleting an account in the Admin Accounts module. Before proceeding, the system alerts administrators with a warning message stating that the deletion is irreversible and may impact related records. This safeguard prevents accidental removals and ensures deliberate decision-making. The interface includes a prominent "Delete Account" button, reinforcing the importance of user confirmation before finalizing the action.

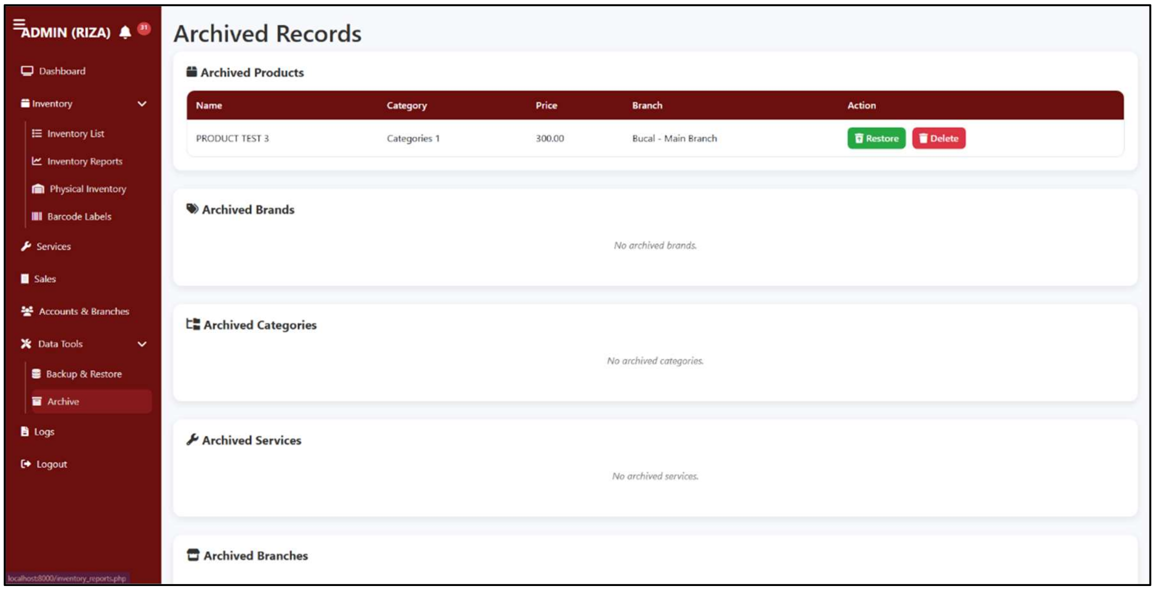

Figure 3-27. Archive (Admin Data Tools)

Figure 3-26. presents the Archived Records page of the R.P. Habana Inventory and Sales Management System. This module serves as a centralized repository for deactivated or removed data, allowing administrators to manage archived products, services, branches, and user accounts. Each record includes essential details such as name, category, price, branch, or role, depending on the data type. The interface provides two main action options—Restore, which reinstates the record into the active database, and delete, which permanently removes it from the system. This feature ensures efficient data lifecycle management by maintaining organized archival control and preventing accidental data loss.

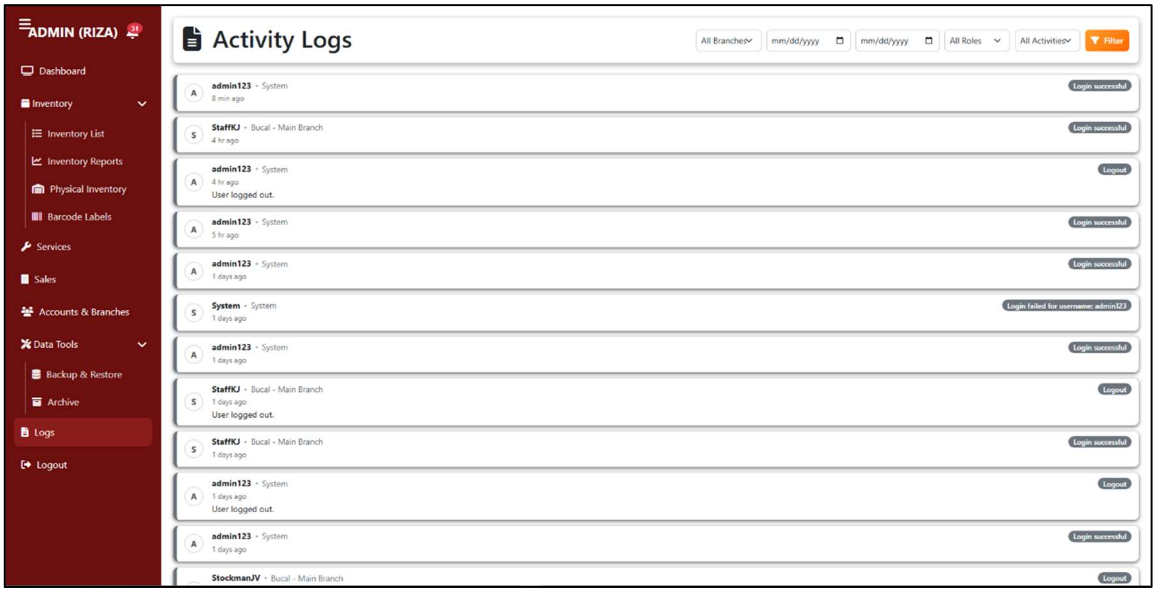

Figure 3-28. Activity Logs (Admin)

Figure 3-27. displays the Activity Logs page of the R.P. Habana Inventory and Sales Management System. This module records all user actions within the system, including logins, logouts, and modifications such as editing or updating records. Each log entry specifies the user, role, branch, activity type, and timestamp, ensuring full traceability of system events. The top section includes filter options that allow administrators to sort records by branch, date range, user role, and activity type for efficient monitoring. This feature enhances system accountability and transparency by providing a detailed audit trail of all user activities across branches.

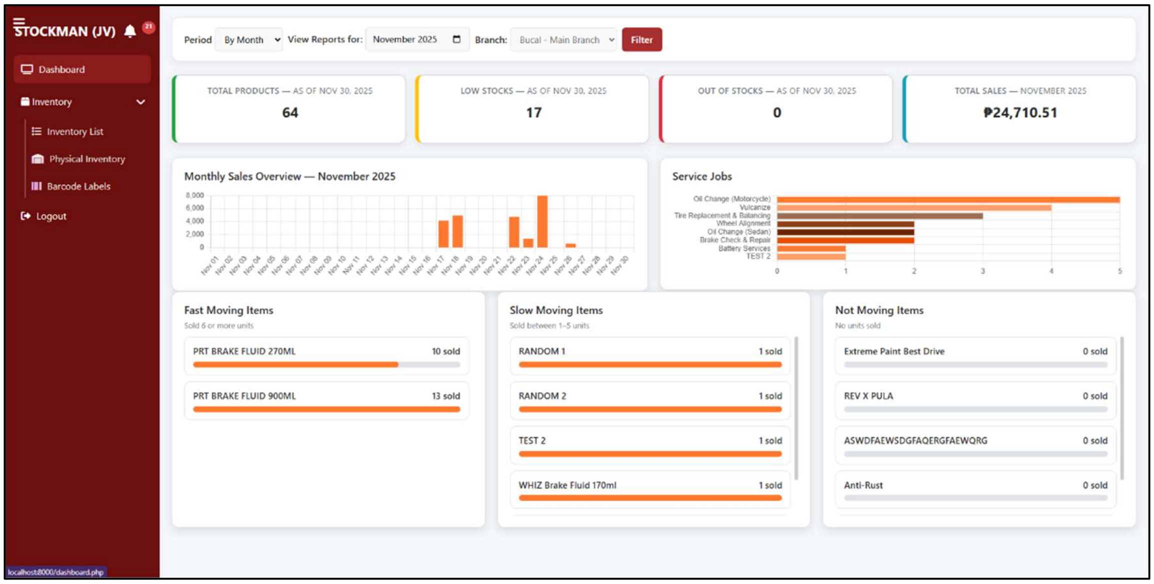

Figure 3-29. Stockman Dashboard

Figure 3-28. This figure shows the Dashboard interface, which provides an overview of product status, sales, and item movement. It displays total products, low stocks, out-of-stocks, and total sales within a selected period. Stockmen can only view data for their assigned branch, helping them monitor inventory performance and identify fast, slow, or non-moving items efficiently.

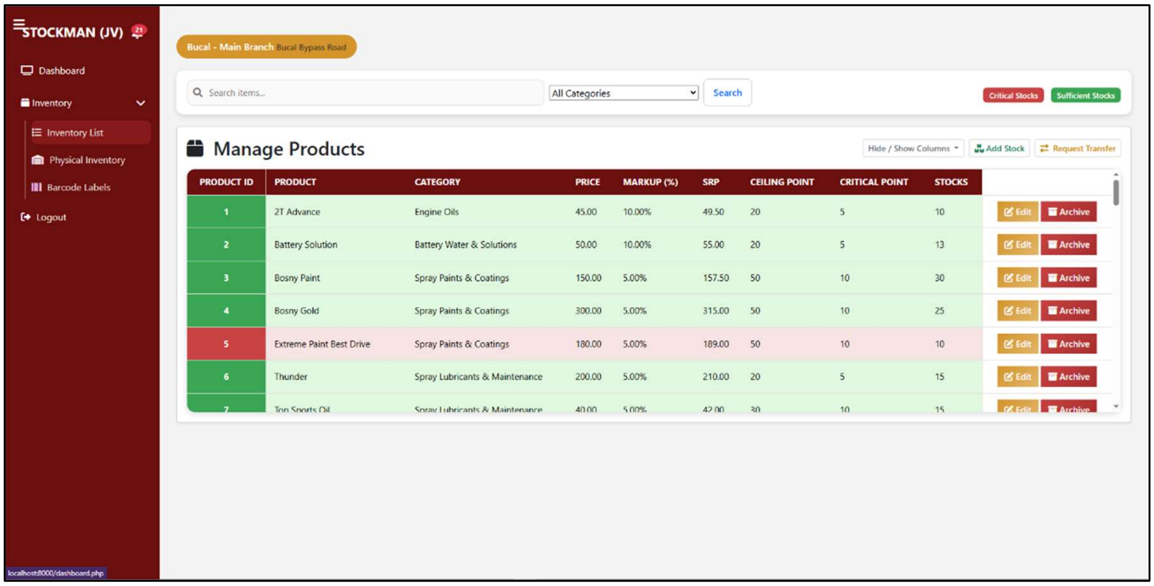

Figure 3-30. Stockman Inventory List

Figure 3-29. This figure shows the stockman inventory list interface, where stockmen can manage product details for their assigned branch. It displays product information such as category, price, and stock levels, allowing them to add, update, or request transfers to maintain accurate inventory records.

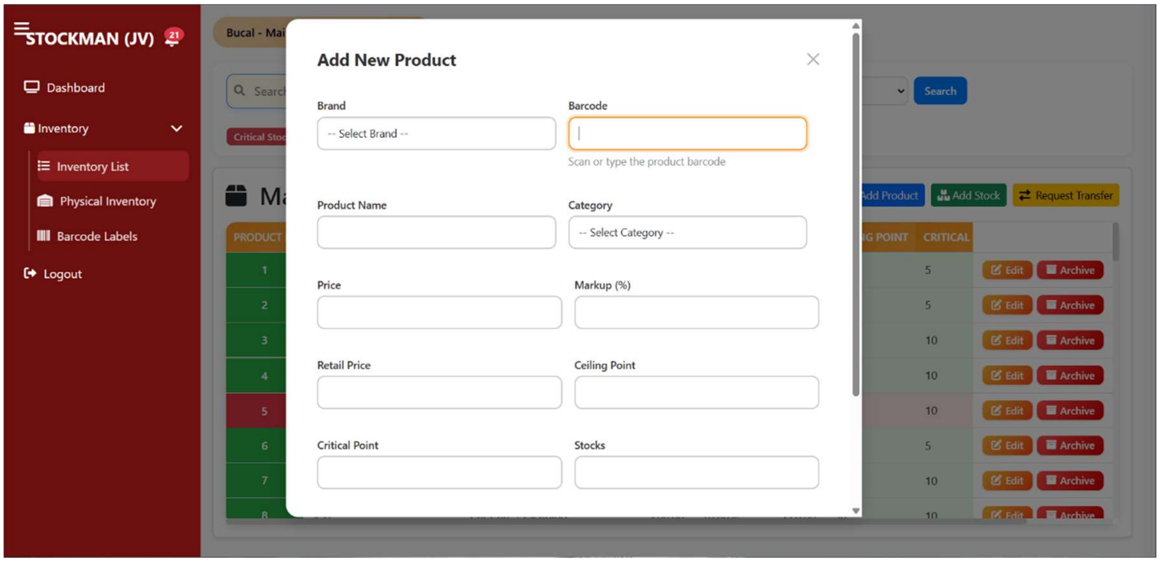

Figure 3-31. Add Product (Stockman)

Figure 3-30**.** presents the Add New Product form as viewed by a Stockman account in the R.P. Habana Inventory and Sales Management System. This interface allows stock personnel to register new products directly into the system. The form contains input fields for brand, barcode, product name, category, price, markup percentage, retail price, ceiling point, critical point, stock quantity, VAT, expiration date, and branch. However, the system restricts product addition to the branch where the stockman is assigned, ensuring controlled and branch-specific inventory management. Users may also add a new brand or category within the same modal for efficiency. Once completed, clicking the Save button records the new product in the corresponding branch's inventory database.

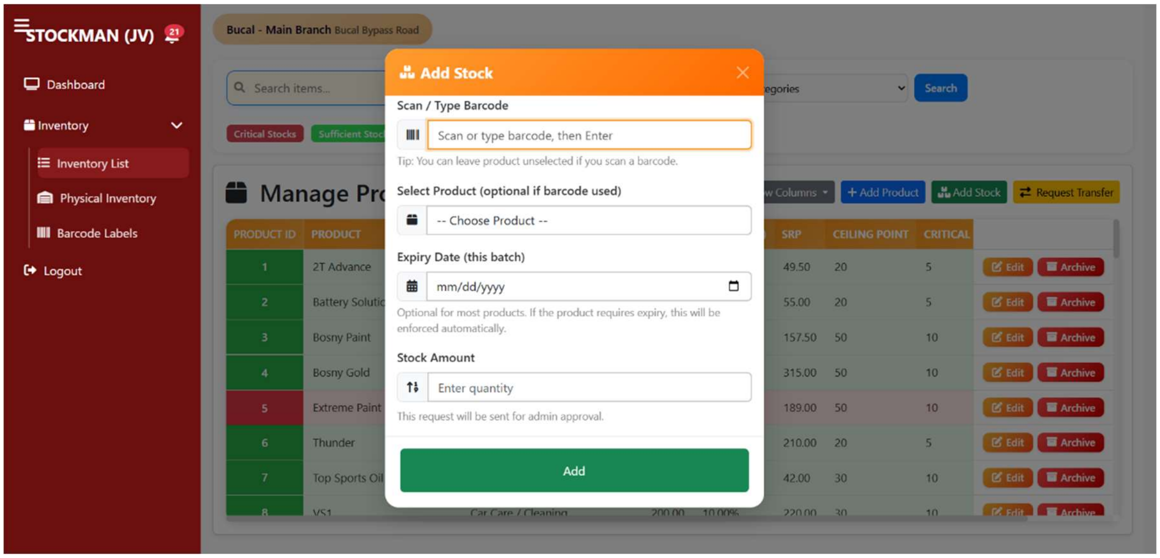

Figure 3-32. Add Stock (Stockman)

Figure 3-31. presents the Add Stock form as viewed by a Stockman account in the R.P. Habana Inventory and Sales Management System. This interface enables the stockman to record additional stock quantities for existing products. The form includes input fields for barcode scanning or typing, product selection, expiry date (if applicable), and stock amount. Once submitted, the stock-in request is automatically sent to the administrator for approval before it is reflected in the system inventory. Moreover, the stock addition is restricted to the branch where the stockman is assigned, ensuring that inventory adjustments remain accurate and branch-specific. This feature promotes accountability and prevents unauthorized stock updates across multiple branches.

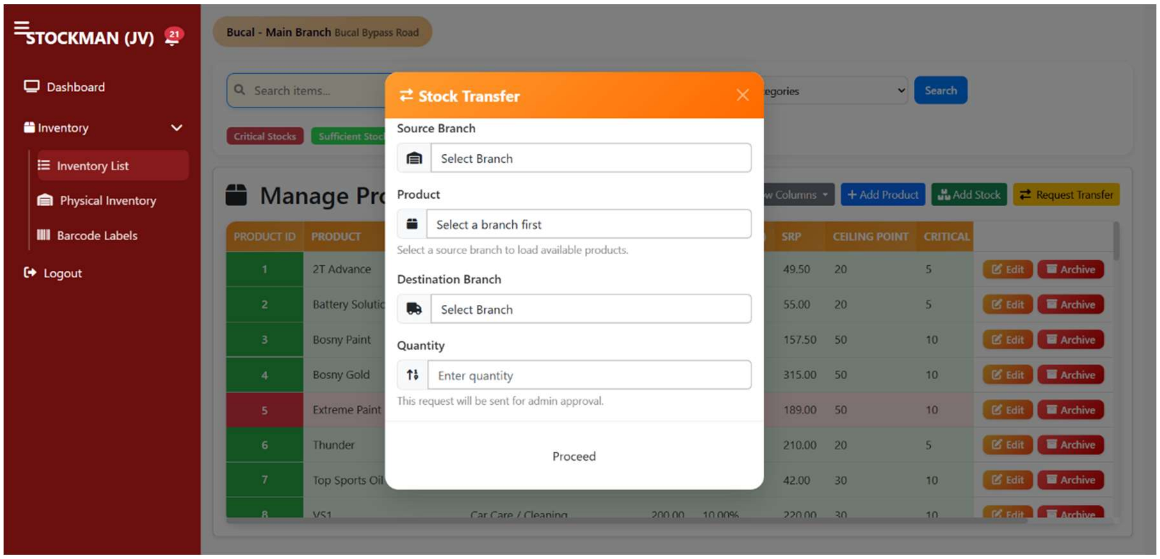

Figure 3-33. Stock Request Transfer (Stockman)

Figure 3-32**.** displays the Stock Transfer form as viewed by a Stockman account in the R.P. Habana Inventory and Sales Management System. This interface allows the stockman to initiate transfer requests of products between branches. The form contains fields for selecting the source branch, product, destination branch, and quantity to be transferred. Once submitted, the request is forwarded to the administrator for approval before the transfer is executed. Furthermore, the stockman can only create transfer requests from the branch to which they are officially assigned, ensuring that all inventory movements remain controlled, authorized, and accurately recorded in the system.

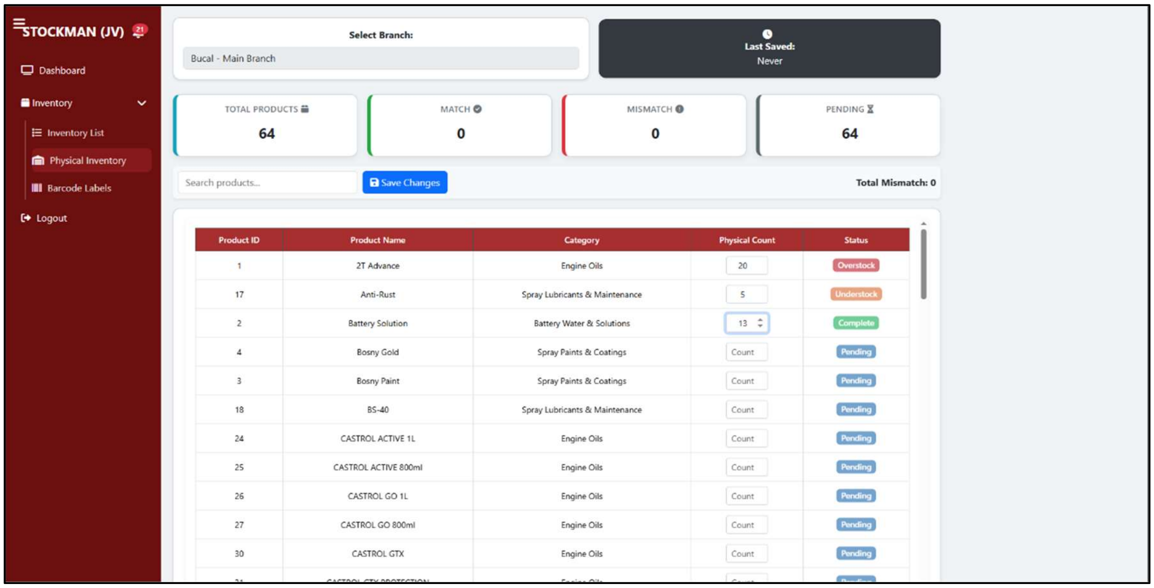

Figure 3-34. Stockman Physical Inventory

Figure 3-33. showcases the physical Inventory interface for stockman, where they can view and update their quantities for their assigned branch. The interface displays key details such as product ID, name, category, system stock, physical count, discrepancies, and status.

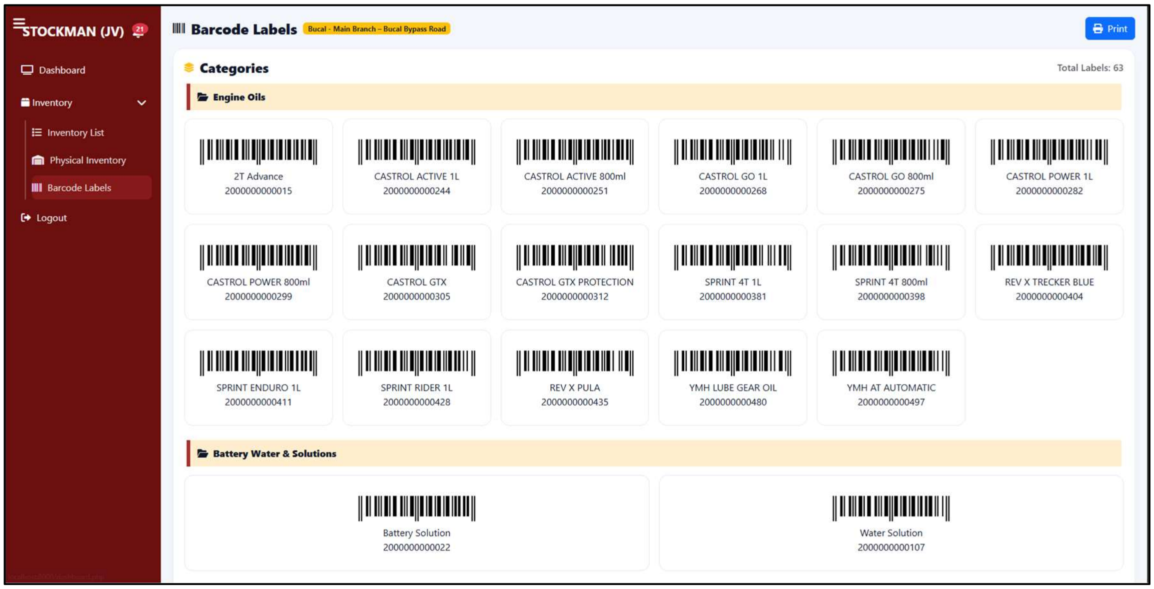

Figure 3-35. Stockman Barcode Labels

Figure 3-34. This figure shows the Barcode Labels interface, which displays product barcodes categorized by type such as solid, liquid, and tire. Each barcode corresponds to a specific product, allowing for easier identification and tracking during inventory management. The stockmen can view, organize, and print barcode labels for their assigned branch to ensure accurate and efficient stock recording.

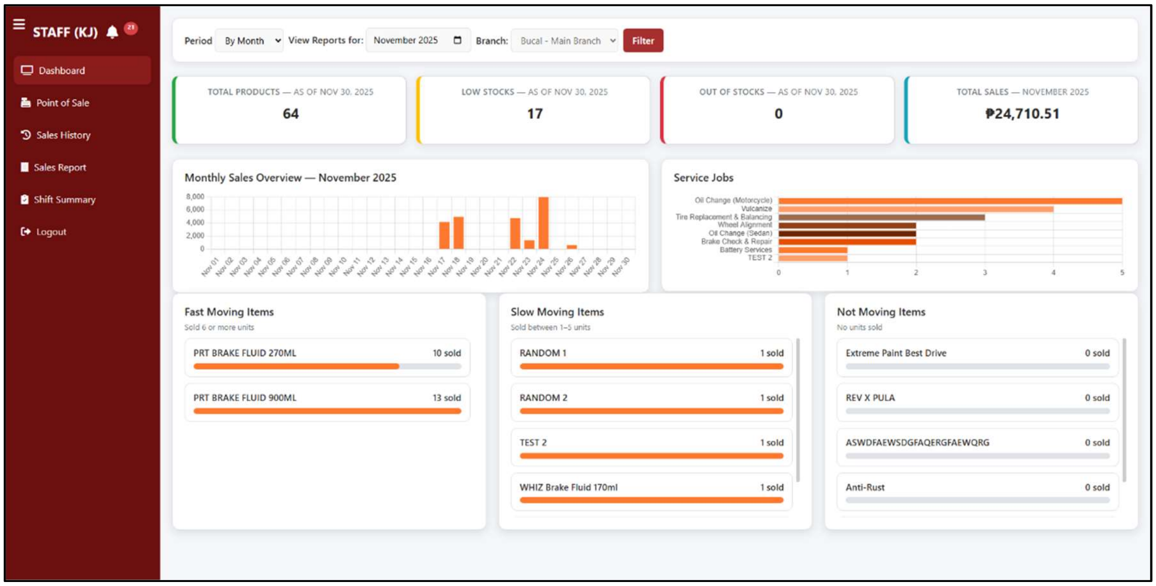

Figure 3-36. Staff Dashboard

Figure 3-35. illustrates the Dashboard interface as viewed by a Staff account in the R.P. Habana Inventory and Sales Management System. This dashboard provides a summarized overview of key performance indicators such as Total Products, Low Stocks, Out of Stocks, and Total Sales for the selected period and branch. It also features analytical sections displaying Monthly Sales Overview, Service Jobs, and product movement classifications including Fast Moving, Slow Moving, and Not Moving Items. The system automatically filters reports based on the branch where the staff is assigned, ensuring data relevance and accuracy for localized sales and inventory performance tracking. This visual summary aids staff members in monitoring sales activity, stock levels, and product performance trends efficiently.

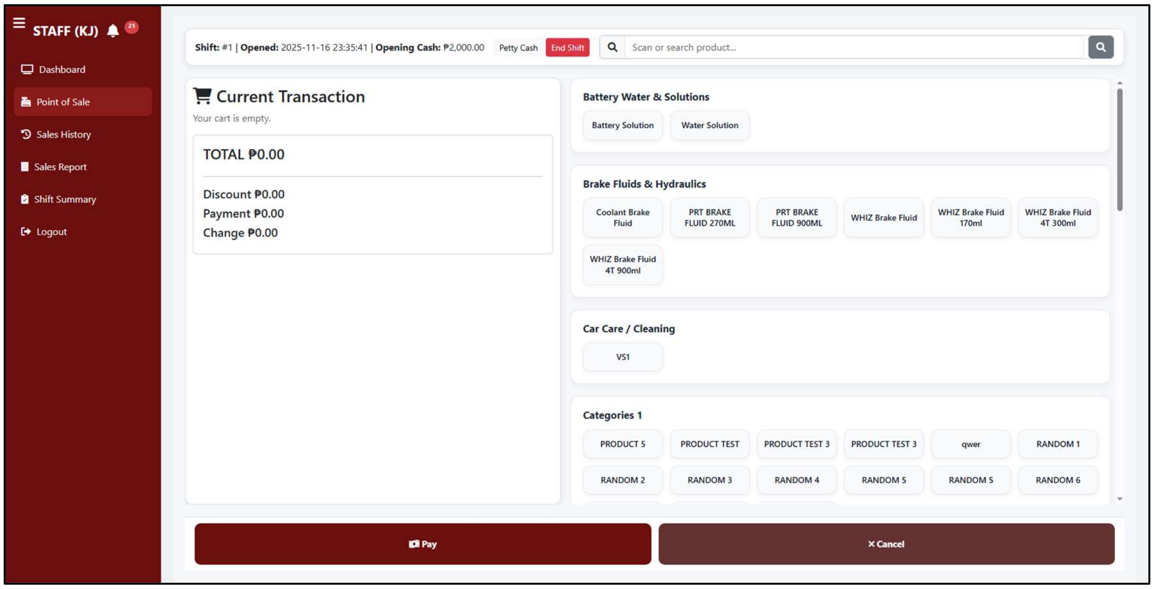

Figure 3-37. Point of Sale

Figure 3-36. displays the Point of Sale (POS) interface as viewed by a Staff account in the R.P. Habana Inventory and Sales Management System. This module facilitates real-time sales transactions by allowing staff to select products and services, adjust quantities, and automatically compute subtotal, VAT, and total amount. Items are categorized into groups such as Liquid, Solid, Tire, and Services for faster selection. The interface includes clear action buttons for Payment and Cancel, enhancing efficiency during checkout. Additionally, transactions processed through this module are automatically linked to the branch where the staff is assigned, ensuring accurate branch-based sales reporting and accountability.

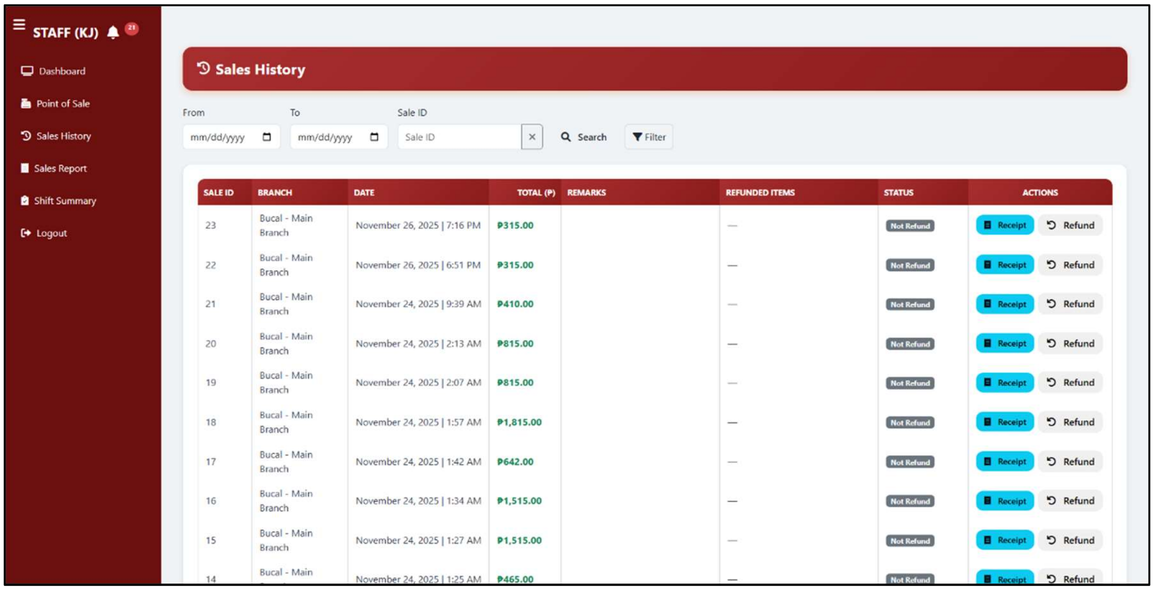

Figure 3-38. Staff Sales History

Figure 3-37. presents the Sales History interface as viewed by a Staff account in the R.P. Habana Inventory and Sales Management System. This module provides a detailed record of all completed sales transactions, including key data such as Sale ID, Branch, Date, Total Amount, Refunded Amount, and Transaction Status. Staff members can filter records by date range or specific Sale ID to quickly locate transactions. The interface also includes functional buttons for viewing Receipts and processing Refunds, ensuring transparency and accountability in daily operations. All sales history entries are automatically filtered to display records from the branch assigned to the staff, maintaining proper data segregation and preventing unauthorized access to other branches' transactions.

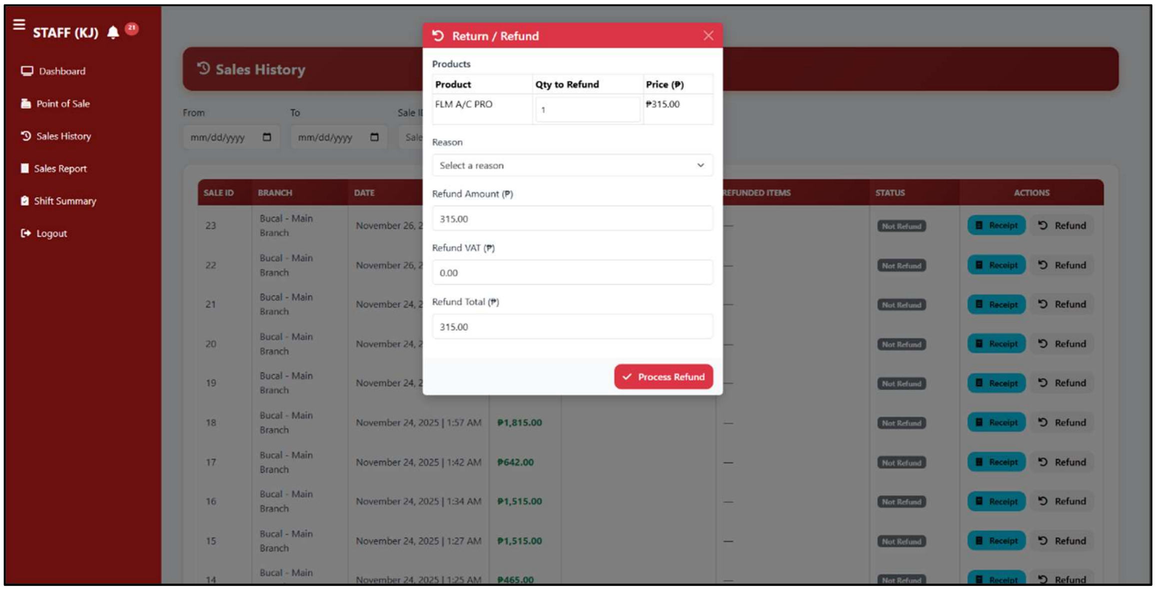

Figure 3-39. Return/Refund Modal

Figure 3-38. shows the Return/Refund modal under the Sales History module as accessed by a Staff account in the R.P. Habana Inventory and Sales Management System. This feature allows authorized staff to process product or service refunds by selecting the corresponding Product or Service, specifying the Quantity to Refund, and entering the appropriate Refund Reason. The system automatically computes the Refund Amount, Refund VAT, and Total Refund to ensure financial accuracy and transparency. Once validated, the staff may click Process Refund to finalize the transaction. All refund operations are logged in the system for monitoring and audit purposes, and refunds can only be performed within the branch to which the staff is assigned, ensuring accountability and controlled refund management.

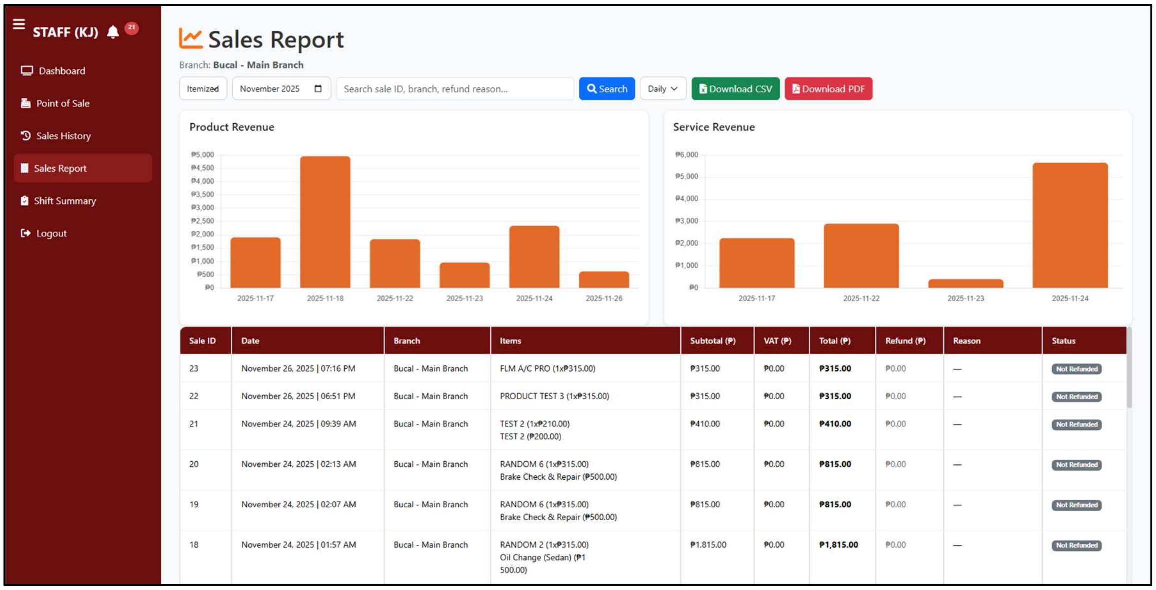

Figure 3-40. Sales Report (Staff)

Figure 3-39. Displays a detailed summary of sales for a selected month and branch, including product information, quantities, totals, and refunds. Designed to help staff easily review sales activity, monitor trends, and support accurate reporting.

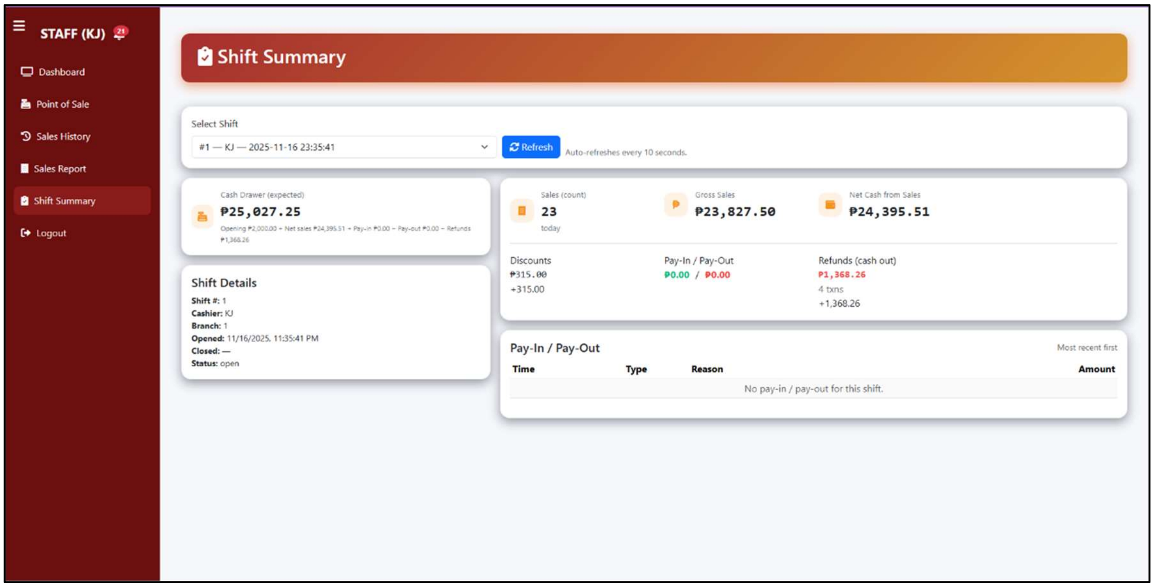

Figure 3-41. Shift Summary

Figure 3-40. display the total cash drawer balance, sales count, VAT, and detailed Pay-In/Pay-Out transactions for transparent shift monitoring."

Phase 3. Develop

In this phase, the researchers determined to use agile methodology to their development approach, developers started to developed the system based on the gathered requirements and feedback on the previous phase. The key modules, features, and database are being structured for its efficiency and securing the data storage. They use programming language and database such as PHP and MySQL. Lastly, the developers also conduct a unit testing to evaluate the system functionality.

Phase 4. Test

Before proceeding to the testing and implementation stages, the researchers conducted several validation activities to ensure the reliability, accuracy, and appropriateness of both the research instruments and the developed system. The validation process ensured that the tools used for data collection, as well as the functionalities of the system, met the standards required for the study. The first activity involved the validation of the research instrument. The survey questionnaire, which was based on ISO 25010:2023 quality characteristics, was subjected to expert validation to confirm its clarity, relevance, and alignment with the performance indicators needed to evaluate the system. The questionnaire was assessed by validators with expertise in system development and software evaluation. They reviewed each item to determine whether it effectively measured the characteristics of interaction capability, performance efficiency, functional suitability, reliability and security. Their recommendations included improving the phrasing of certain statements and adjusting the sequence of items for better coherence. The researchers incorporated these revisions before distributing the final instrument to the respondents. The second phase of validation focused on the system itself. Prior to user-level testing, the initial version of the system was evaluated by our validators, who carefully examined its overall functionality, design structure, navigation flow, responsiveness, security features, and data-handling accuracy. They identified several minor issues such as inconsistencies in layout, button responsiveness, and input validation. Based on their comments and recommendations, the researchers revised the system to improve user experience and ensure that all modules performed according to their intended functions. Lastly, the system underwent client validation. The revised build of the system was presented to the shop owner and selected staff members to determine whether it matched the actual business processes and operational workflow of the auto repair shop. The clients confirmed the relevance of the system's inventory monitoring, POS module, service tracking, and sales management features. They also provided suggestions such as simplifying certain procedures, improving the arrangement of menu options, and adding search functionalities. These improvements were applied before the formal system testing phase.

Following the development and validation stages, a series of system tests were carried out to verify the performance, reliability and overall readiness of the system for actual deployment. These testing procedures ensured that each component functioned correctly and consistently within the intended operational environment. The first testing activity conducted was functional testing. In this phase, each module of the system—such as the point-of-sale module, inventory management, stock adjustment, service tracking, sales history, admin dashboard, and activity logs—was tested individually to confirm that the programmed functions executed as expected. The researchers simulated actual transactions, monitored how the system processed inputs, and checked whether the system generated accurate computations and real-time updates. Any incorrect outputs or system behaviors were documented and corrected immediately. The next testing activity involved usability testing, which assessed how easily users could navigate and operate the system. Selected staff members were asked to perform typical tasks such as adding items to the inventory, processing sales, checking stock availability, and generating reports. Their feedback highlighted areas that needed improvement, such as button sizes, placement of options, labeling clarity, and navigation flow. The researchers refined the interface to ensure that users could perform tasks intuitively, even with minimal training. Compatibility testing was also carried out to determine whether the system functioned properly across different devices and browsers. The system was accessed through desktop and tablet devices using browsers such as Google Chrome and Opera GX to ensure that the interface, layout, and functionalities remained consistent across different platforms. The testing confirmed that the system's layout, text alignment, and button responsiveness remained stable under different display resolutions. Minor interface adjustments were made to optimize the system for smaller screens. To assess the system's performance efficiency, performance testing was conducted. This involved monitoring the system's speed, response time and stability when performing tasks under typical workload conditions. The researchers observed how the system handled multiple consecutive transactions, the loading time for inventory and sales records, and the retrieval speed of data from the database. The system maintained stable performance without experiencing delays, confirming its readiness for real-world usage. Finally, the system underwent user acceptance testing (UAT) to determine whether the final version met the expectations and operational needs of the client. The shop owner and staff carried out actual business processes using the system and evaluated its overall accuracy, reliability and usefulness. They also checked whether the system reflected the correct procedures of their workflow, such as stock deduction, sales computation, service monitoring and notification updates. Their evaluation indicated that the system satisfactorily fulfilled the project objectives and was suitable for adoption in their daily operations.

Phase 5. Release

At release phase, the R.P Habana Inventory and Sales Management System was deployed for actual use. The system became accessible to all branch administrator, and the Head Admin was granted full access to the entire system. The researchers also provided simple training and orientation sessions to ensure that users gained the necessary knowledge to navigate the system properly. This ensured that all users were familiar with the system workflow.

Phase 6. Feedback

For the feedback stage, following the release of the Inventory and Sales Management System for R.P Habana Auto Repair Shop. Interviews, observation, and survey questionnaires were conducted to gather a comprehensive feedbacks from the users and evaluates its effectiveness. This information was used to identify some areas that needed improvement. Additionally, the collected feedbacks was presented in Chapter 4 of the documentation under the Results and Discussion section.

Ethical consideration

Respect for Participants. The Researchers ensured that all participants involved in the study were treated with autonomy and respect. This included recognizing their rights, valuing their time, and maintaining throughout all research interactions. Participants were provided with complete information about the study so they could decide freely whether to participate [26].

The researchers also maintained professionalism during data collection and ensured that business operations were not disrupted. They considered participants' schedules and conducted all interactions ethically and respectfully.

Privacy, Confidentiality, and Data Security. It involves protecting Sensitive data, including sales transactions, inventory levels, and financial reports, and participants personal information are confidential and only accessible to authorized personnel. Researchers comply with the Philippine Data Privacy Act of 2012. It emphasizes that the protection of personal data while allowing it to be used for research purposes [27]. To enhance data security, the system being developed had encryption and authentication mechanisms to prevent unauthorized access.

Non-Maleficence (Do No Harm). This means that there are limits to risks, especially in terms of inconvenience and burdens to the participants [28]. The research avoids things that could disrupt the daily business operations of the shop. To uphold this principle, gathering data collection is conducted without interfering with normal business transactions, and any system implementation is carefully tested to avoid operational issues.

Fairness and Equity. All participants were treated equally to ensure that no group or individual was unfairly disadvantaged. Since the system was intended for use by different levels of users—regardless of role—had equal opportunities to provide feedback and benefit from system improvements. To achieve this, feedback from both the owner and the staff was considered during the system development to create a system that addressed the overall operational needs of the business.

Chapter 4

RESULTS AND DISCUSSION

This chapter of the study presents the analysis and findings of the data gathered in this study. This revealed whether the business owners and employee of R.P Habana are satisfied with the functionality and quality of the system and if it is suitable for their operational needs.

1. To develop a centralized web-based system that accurately manages sales and inventory reports real-time, including product tracking, stock level monitoring, sales transactions, and comprehensive reporting.

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Table 4- 1. Sales and Inventory Management of the R.P. Habana Inventory and Sales Management

Survey Question	E	VS	N	F	P	WM	Interpretation
Real-time tracking of sales transactions	19	3	0	0	0	4.86	Excellent
Ease of monitoring stock levels	19	2	1	0	0	4.82	Excellent
Clarity of real-time sales and inventory reports	21	1	0	0	0	4.95	Excellent
Efficiency of product tracking feature	20	1	1	0	0	4.86	Excellent
Improvement in overall inventory & sales management	20	2	0	0	0	4.91	Excellent
GRAND MEAN			**4.56**				**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

The data in Table 4-1. reveal that respondents rated the Sales and Inventory Management capability of the R.P. Habana Inventory and Sales Management System with an overall weighted mean of 4.88, interpreted as Excellent. This indicates that the system is highly effective in automating sales tracking, stock monitoring, and report generation, enabling users to perform inventory-related tasks with accuracy and minimal effort. Among the indicators, Item 3 obtained the highest mean of 4.95, showing that users strongly agreed the system delivers clear, real-time inventory and sales reports. In contrast, Item 2 scored 4.82, the lowest yet still excellent, suggesting that while stock monitoring is dependable, minor improvements could further enhance its precision.

During testing, staff shared that stock and sales information needed to be updated more consistently, especially during busy hours of operation. They emphasized the importance of having real-time visibility so discrepancies could be avoided.

The researchers automated product tracking, instant sales logging, and report generation in the system. During testing, adjustments were made to the interface and navigation based on staff feedback, ensuring that users could access inventory information quickly and accurately. The system now allows real-time monitoring of stock and sales, reducing manual errors. Supported by Briones [1] and ASEAN Automotive Research Group [11], automated inventory improves consistency and operational accuracy.

2. To create a system that tracks the number of service jobs completed by R.P. Habana Auto Repair Shop, with reporting capabilities on a monthly, quarterly, and yearly basis.

Table 4- 2. Service Job Tracking of the R.P. Habana Inventory and Sales Management

Survey Question	E	VS	N	F	P	WM	Interpretation
Accuracy of tracking completed service jobs	16	6	0	0	0	4.73	Excellent
Ease of generating service job reports	14	8	0	0	0	4.64	Excellent
Ability to identify most profitable services	21	1	0	0	0	4.95	Excellent
Support for decision-making in service operations	22	0	0	0	0	5	Excellent
Improvement of workflow through service job tracking	20	2	0	0	0	4.91	Excellent
GRAND MEAN				**4.85**			**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

The data presented in Table 4-2. reveal that the Service Job Tracking component of the R.P. Habana Inventory and Sales Management System obtained an overall weighted mean of 4.85, interpreted as Excellent. This indicates that respondents strongly agreed on the efficiency of the system in recording, categorizing, and reporting completed service jobs. Among the indicators, Item 4, which states that the system supports better decision-making in managing service operations, garnered the highest weighted mean of 5.00, implying unanimous agreement that the system greatly enhances managerial control and workflow planning. Meanwhile, Item 2, with a mean of 4.64, was the lowest, though still rated "Excellent," showing that while the report-generation feature performs well, respondents perceive room for additional improvement, perhaps through more customizable reporting options.

Staff noted that retrieving past service records was time-consuming and needed a more organized approach. They expressed the need for a single place where all service entries could be viewed and summarized.

The researchers implemented the Service Job Tracking module and refined it based on user input during testing. Fields were reorganized, labels clarified, and input steps simplified to match the staff's workflow. Users could now input and categorize services efficiently, and the system generated organized summaries for monthly, quarterly, and yearly review. This aligns with Rodriguez [13] and ASA [14], emphasizing that structured documentation improves accountability and record accuracy.

3. To implement an automated notification system that provides real-time updates on stock levels, keeping users informed about low-stock and out-of-stock products to ensure timely action and improve inventory management processes.

Table 4- 3. Notification System of the R.P. Habana Inventory and Sales Management System

Survey Question	E	VS	N	F	P	WM	Interpretation
Timeliness of low-stock/out-of-stock notifications	17	4	1	0	0	4.73	Excellent
Effectiveness of automated stock alerts	18	4	0	0	0	4.82	Excellent
Helpfulness of notifications in reducing restocking delays	21	1	0	0	0	4.95	Excellent
Accuracy of real-time stock updates	22	0	0	0	0	5	Excellent

Improvement in inventory management through stock alerts	20	2	0	0	0	4.91	Excellent
GRAND MEAN				**4.85**			**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

The data in Table 4-3. reveal that the Notification System of the R.P. Habana Inventory and Sales Management System earned an overall weighted mean of 4.85, which is verbally interpreted as Excellent. This high evaluation signifies that respondents strongly agreed on the system's ability to deliver timely, accurate, and reliable notifications regarding inventory levels. Among the indicators, Item 4, which highlights the accuracy and reliability of real-time updates, obtained the highest mean of 4.95, indicating that users highly valued the precision and immediacy of notifications generated by the system. Meanwhile, Item 1 (4.73) had the lowest mean, though still excellent, suggesting that users were satisfied but believed further improvements—such as customizable notification settings or mobile alerts—could enhance responsiveness.

Staff indicated that stock levels needed to be monitored more proactively so that replenishment could happen on time. They specifically asked for a system that could highlight critical stock items without requiring constant checking.

The researchers added automated alerts that notify users when stock reaches critical levels. During testing, placement and visibility of alerts were adjusted based on user feedback to ensure they were easily noticed. Staff confirmed that the alerts allowed timely restocking and minimized interruptions. Flores [6] supports that proactive notifications enhance operational responsiveness.

4. To evaluate the feasibility of the proposed system through a cost-benefit analysis.

Cost Benefit Analysis of the Development of R.P. Habana Inventory and Sales Management System

BENEFITS		2025		2026		2027		2028		2029
Reduce Printing and Paper Costs	PHP	-	PHP	3,500.00	PHP	3,500.00	PHP	3,500.00	PHP	3,500.00
Reduce Revenue Losses	PHP	-	PHP	200,000.00	PHP	200,000.00	PHP	200,000.00	PHP	200,000.00
TOTAL BENEFITS	**PHP**	**-**	**PHP**	**203,500.00**	**PHP**	**203,500.00**	**PHP**	**203,500.00**	**PHP**	**203,500.00**
DEVELOPMENT COST										
Developer's Fee	PHP	285,000.00	PHP	37,000.00	PHP	37,000.00	PHP	37,000.00	PHP	37,000.00
User Training Fee	PHP	1,500.00	PHP	-	PHP	-	PHP	-	PHP	37,000.00
Php Web Hosting	PHP	1,800.00	PHP	-	PHP	-	PHP	-	PHP	-
Semaphore SMS API	PHP	700.00	PHP	-	PHP	-	PHP	-	PHP	-
TOTAL DEVELOPMENT COST	**PHP**	**285,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**
OPERATIONAL COST										
Electricty Bill	PHP	11,266.43	PHP	11,266.43	PHP	11,266.43	PHP	11,266.43	PHP	11,266.43
Php Web Hosting	PHP	-	PHP	5,300.00	PHP	-	PHP	-	PHP	-
Semaphore SMS API	PHP	-	PHP	2,800.00	PHP	-	PHP	-	PHP	-
Computer	PHP	-	PHP	84,000.00	PHP	-	PHP	-	PHP	-
TOTAL OPERATIONAL COST	**PHP**	**11,266.43**	**PHP**	**103,366.43**	**PHP**	**11,266.43**	**PHP**	**11,266.43**	**PHP**	**11,266.43**
TOTAL COST	**PHP**	**296,266.43**	**PHP**	**140,366.43**	**PHP**	**48,266.43**	**PHP**	**48,266.43**	**PHP**	**48,266.43**
NET BENEFIT COST	**PHP**	**(296,266.43)**	**PHP**	**63,133.57**	**PHP**	**155,233.57**	**PHP**	**155,233.57**	**PHP**	**155,233.57**
CUMULATIVE NET CASH FLOW	**PHP**	**(296,266.43)**	**PHP**	**(233,132.86)**	**PHP**	**(77,899.29)**	**PHP**	**77,334.29**	**PHP**	**232,567.86**
RETURN OF INVESTMENT (ROI)		**-100.00%**		**44.98%**		**321.62%**		**321.62%**		**321.62%**
PAYBACK PERIOD		**3.5 3 years and 6 months**								

Figure 4-7. Cost Benefit Analysis

As shown in Figure 4-1, the Cost–Benefit Analysis (CBA) presents the financial feasibility of implementing the proposed account recovery feature for the R.P. Habana system. The analysis compares the total development costs with the projected benefits gained from improving system accessibility, reducing account lockout incidents, and minimizing manual intervention from administrators.

The figure highlights issues observed in the current manual recovery process, including delayed password resets, lack of secure verification methods, and the risk of unauthorized requests. These inefficiencies result in time loss, productivity interruption, and potential security vulnerabilities. The developer's fee is justified by Garcia [3], who noted that small businesses often lack technical expertise and require professional development support when implementing digital features. The PHP web hosting cost is supported by HashMicro [19], which states that modern systems require reliable online hosting to keep user authentication and account access features consistently available. Meanwhile, the use of the Semaphore SMS API is supported by Semaphore Philippines [29], which highlights the importance of secure and real-time OTP delivery for user verification during account recovery. Additionally, the inclusion of a user training fee is supported by Farias and Resende [31], who emphasized that proper user training is essential for successful information system implementation, and its corresponding cost for 2025 is based on the DICT's official ICT training fee schedule [32].

Based on the financial summary, the system incurs an initial development cost of ₱296,266.43 in 2025, resulting in a negative net benefit in the first year. In 2026, projected benefits increase due to reduced manual resets, improved security, and faster account restoration, resulting in a positive net benefit of ₱63,133.57. From 2027 onward, the system maintains a consistent annual net benefit of ₱155,233.57. As a result, the cumulative net cash flow—starting at –₱296,266.43 in 2025—turns positive in 2028 at ₱77,334.29. This indicates that the account recovery feature fully recovers its investment within approximately 3.5 years.

The financial results further demonstrate the system's viability through increasing ROI values. The ROI starts at –100% in 2025 due to the initial investment, rises to 44.98% in 2026, and reaches 321.62% from 2027 to 2029. These figures align with the findings of Acosta et al. [18], who emphasized that automated inventory and sales systems significantly reduce manual errors and stock discrepancies, leading to improved financial outcomes. The strong ROI performance in the CBA supports this claim, showing that automation enhances profitability once the initial development cost is recovered.

The CBA demonstrates that implementing a centralized digital inventory and sales management system is financially feasible and operationally advantageous. The system not only recovers its initial cost within the projected payback period but also delivers substantial long-term gains. This confirms that the transition from manual processes to an automated system effectively reduces financial losses, enhances accuracy, and strengthens the long-term sustainability of R.P. Habana's business operations.

5. To evaluate the effectiveness of R.P Habana Inventory and Sales Management System by using ISO-25010 2023 in terms of Interaction Capability, Performance Efficiency, Functional Suitability, Reliability, and Security.

Table 4- 4. Interaction Capability of the R.P. Habana Inventory and Sales Management System

Interaction Capability	E	VS	N	F	P	WM	Interpretation
User-friendly interface	15	4	2	0	0	4.62	Excellent
Easy system navigation	12	10	0	0	0	4.55	Excellent
Easy to learn for first-time users	13	7	2	0	0	4.5	Excellent
Clear system feedback	14	7	1	0	0	4.59	Excellent
Tasks completed with minimal assistance	14	6	2	0	0	4.55	Excellent
GRAND MEAN				**4.56**			**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

As shown in Table 4-5., the Interaction Capability of the system was rated Excellent, with an overall weighted mean of 4.56. This indicates that users found the system intuitive, easy to navigate, and efficient in completing tasks. The highest-rated item (4.62), pertaining to user-friendliness, confirms that respondents had no difficulty interacting with the interface, while the lowest mean (4.50) reflects minor learning curves among new users, possibly due to unfamiliarity with digital workflows.

Users expressed a need for a more organized and efficient interface that reduced the number of steps it took to complete routine tasks. They also highlighted the importance of having frequently used features immediately accessible. The researchers designed a cleaner, more intuitive interface and tested it with users. Adjustments were made to simplify navigation and highlight commonly used actions. Staff reported that tasks could now be completed with fewer steps and less confusion. Laguna Technopark [15] supports that user-centered design improves productivity, matching the staff experience.

Table 4- 5. Performance Efficiency of the R.P. Habana Inventory and Sales Management System

Performance Efficiency	E	VS	N	F	P	WM	Interpretation
Quick system response	17	3	2	0	0	4.68	Excellent
Performs well with many users	17	5	0	0	0	4.77	Excellent
Reliable with large data volumes	17	4	1	0	0	4.73	Excellent
Handles growing data without slowdown	18	4	0	0	0	4.82	Excellent
Runs well on PC, tablet, and mobile	16	4	2	0	0	4.64	Excellent
GRAND MEAN			4.73				Excellent

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

Table 4-6. presents the Performance Efficiency results with an overall weighted mean of 4.73, verbally interpreted as Excellent. The data reveal that the system performs efficiently even under multi-user conditions and large data transactions. Item 4, which focuses on the system's ability to handle growing data without slowing down, received the highest rating (4.82), demonstrating robust optimization. The lowest rating, Item 5 (4.64), still reflects strong satisfaction, confirming device compatibility across platforms.

Staff emphasized the importance of accessing records and information quickly, especially when assisting customers. They identified the need for a system that could retrieve data instantly without lengthy searching.

The researchers structured the system to provide immediate access to inventory, sales, and service data. During testing, users confirmed that tasks could be performed more efficiently than before, reducing time spent on repetitive manual work. TESDA [3] highlights that automated processes improve operational efficiency, consistent with the users' experience.

Table 4- 6. Functional Suitability of the R.P. Habana Inventory and Sales Management System

Functional Suitability	E	VS	N	F	P	WM	Interpretation
Correct access based on user roles	19	3	0	0	0	4.86	Excellent
Accurate inventory and sales records	16	6	0	0	0	4.73	Excellent
Effective monitoring and updating of data	20	0	2	0	0	4.82	Excellent
Accurate POS pricing	22	0	0	0	0	5	Excellent
Real-time inventory updates	20	2	0	0	0	4.91	Excellent
GRAND MEAN			**4.86**				**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

Table 4-7. reveals that the Functional Suitability of the system achieved an overall weighted mean of 4.86, interpreted as Excellent. This reflects that the system's functions are accurate, complete, and appropriate for the users' operational needs. The highest score (5.00) in Item 4 highlights the accuracy of the POS system, while the lowest (4.73) shows minor room for refinement in data consistency.

Users indicated that several important functions they needed daily were not available in their existing workflow. They suggested specific features that would make their tasks easier and more complete.

The researchers included features to cover all key operational requirements. During testing, the researchers refined functions to match actual user needs, ensuring all modules supported daily tasks. Users confirmed that they could perform all necessary actions digitally. Flores [6] emphasizes that functional completeness enhances system reliability and usability.

Table 4- 7. Reliability of the R.P. Habana Inventory and Sales Management System

Reliability	E	VS	N	F	P	WM	Interpretation
Reduces human errors	16	6	0	0	0	4.73	Excellent
Recovers quickly from system issues	14	8	0	0	0	4.64	Excellent
Ensures accurate stock transfers	17	4	1	0	0	4.73	Excellent
Accurately saves and retrieves data	18	4	0	0	0	4.82	Excellent
Prevents data loss in records	19	3	0	0	0	4.86	Excellent
GRAND MEAN			**4.76**				**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

As presented in Table 4-8., the Reliability of the R.P. Habana Inventory and Sales Management System gained an overall weighted mean of 4.76, interpreted as Excellent. This confirms that the system consistently performs as expected, minimizes downtime, and ensures data integrity. The highest-rated item (4.86) indicates strong record retention and protection against data loss, while the lowest (4.64) shows excellent but improvable recovery speed during technical issues.

Staff highlighted the need for information to be consistent and dependable at all times. They stressed the importance of having a system that preserves records accurately and is accessible whenever needed.

The system stores all records digitally with consistent updates and backups. During testing, staff confirmed that records were accurate and retrievable, and the system functioned consistently across modules. TESDA [3] and ASEAN Automotive Journal [14] note that reliable systems are essential for uninterrupted operations.

Table 4- 8. Security of the R.P. Habana Inventory and Sales Management System

Security	E	VS	N	F	P	WM	Interpretation
Restricts access to authorized users (RBAC)	15	7	0	0	0	4.68	Excellent
Encrypts sensitive data	19	3	0	0	0	4.86	Excellent
Enforces strong password rules	18	4	0	0	0	4.82	Excellent
Maintains audit logs for user activity	21	1	0	0	0	4.95	Excellent
Locks account after multiple failed logins	20	2	0	0	0	4.91	Excellent
GRAND MEAN				**4.84**			**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

Table 4-9. shows that the Security feature of the R.P. Habana Inventory and Sales Management System achieved an overall weighted mean of 4.84, interpreted as Excellent. This result confirms that users trust the system's capability to safeguard confidential information through encryption, access control, and monitoring tools. Item 4 earned the highest mean

(4.95), indicating users highly value audit logging for tracking activities and ensuring accountability. The lowest (4.68), while still excellent, implies that access control may benefit from continuous updates to strengthen protection further.

Users raised concerns regarding the protection of sensitive business information and emphasized the need for proper restrictions and tracking of user actions. They wanted a system that ensured accountability and safeguarded data integrity.

The researchers implemented user authentication, role-based access, and secure data storage. Staff testing confirmed that sensitive data was protected while still accessible to authorized users. This aligns with Yadav et al. [15] and Flores [6], who highlight the importance of security in business systems.

Table 4- 9. Overall Results of the Performance of the System.

SYSTEM CRITERIA	WEIGTHED MEAN	INTERPRETATION
Interaction Capability	4.56	Excellent
Performance Efficiency	4.73	Excellent
Functional Suitability	4.86	Excellent
Reliability	4.76	Excellent
Security	4.84	Excellent
Overall Mean	**4.75**	**Excellent**

Legend: 1.00-1.80 Excellent, 1.81-2.60 Very Satisfactory,2.61-3.40 Neutral,3.41-4.20 Fair, 4.21-5.00 Poor

The table presents the overall results of the system's performance based on the respondents' evaluation according to five ISO 25010 system criteria: Interaction Capability, Performance Efficiency, Functional Suitability, Reliability, and Security. The computed overall weighted mean of 4.75, interpreted as Excellent, indicates that the respondents were highly satisfied with the overall quality, functionality, and performance of the system.

Among the identified criteria, Functional Suitability obtained the highest weighted mean of 4.86, suggesting that the system effectively meets user requirements and performs its intended functions accurately and efficiently. On the Other hand, Interaction Capability recorded the lowest weighted mean of 4.56, although still interpreted as Excellent. This implies that while the system is already highly usable and responsive. The consistent Excellent interpretation across all criteria affirms the system's commendable performance, dependability, and efficiency. Overall, the findings demonstrate that the developed system successfully meets both the functional and non-functional requirements, validating its effectiveness, reliability, and suitability for implementation at R.P. Habana Auto Repair Shop.

Users shared that operations would be more efficient if key functions were centralized rather than separated across different tools and documents. They emphasized the value of having one integrated solution that handles all core processes.

The researchers developed an integrated system combining inventory, sales, service tracking, notifications, and reporting. Users confirmed during testing that the system centralized all key operations, improving task efficiency and record accuracy. The ISO 25010 evaluation scores reflect the system's stable, usable, and secure performance. Literature supports that integrated systems improve organizational efficiency, consistent with the improvements observed at the shop.

Chapter 5

SUMMARY, CONCLUSION AND RECOMMENDATIONS

This chapter presents the summary of findings and conclusion made by the researchers based on the results acquired in the study. And provides recommendation for future researchers to enhance the study for better use.

Summary of the Study

The R.P Habana was an automotive service provider with seven branches, offering vehicles repair, maintenance and vehicle components. The shops traditionally relied on the manual-based process of handling inventory records and sales management, the staff usually used tally marks and handwritten notes to record product quantities and sales transactions. This leads to time consuming, and prone to errors. Resulting to inventory discrepancies, inaccurate sales reports among the branches.

To address these challenges, the study developed a centralized web-based inventory and sales management system to streamline the R.P Habana operations. The system enabled a real-time tracking of inventory and sales among multiple branches, automated product and sales logging, and integrate a point-of-sales system that update inventory immediately after each transaction, it also includes a role-based access control for Admin, Stockman, and Staff. ensuring a secure and organized management. This study aimed to provide accurate sales and inventory data, real-time updates for low-stock and out of stock items, track completed service jobs, assessed feasibility through cost benefit analysis, and evaluate effectiveness using ISO-25010 software quality model.

The results of the study show that the newly developed system made a significant difference in how the shop manages its sales and inventory. Before this, the staff relied heavily on manual recording, which often led to inconsistent stock counts and delays in preparing reports—especially when gathering data from different branches. With the introduction of automated product tracking, sales logging, and real-time reporting, updates now happen instantly. Staff no longer need to worry about mismatched records or outdated information, helping everyone work more confidently and efficiently.

The study also found that service jobs were not being documented in a consistent way. Most of the time, staff depended on memory to recall completed services, which made it difficult to review past work or track how busy the shop was during certain months. With the new centralized service log, every completed job is now recorded and organized automatically. This gives the shop a clearer picture of its service trends and makes it easier to understand customer needs and workload patterns.

Another major improvement highlighted in the results is the automated stock alert feature. Previously, low-stock or out-of-stock items were usually noticed only when a transaction was already taking place or during manual counts. The new notification module changes this by constantly checking stock levels and sending timely alerts when items reach low quantity. Respondents rated this feature 4.85 (Excellent), showing strong satisfaction with how reliable and accurate the alerts are. This improvement helps staff avoid stock shortages, respond quickly, and maintain smooth operations throughout the day.

In terms of feasibility, the study found that the system is highly practical and cost-efficient. It was developed using free tools, which eliminated major expenses, and the ongoing costs—such as hosting and API usage—are manageable for the shop. Because the system also reduces the need for paper-based processes and helps prevent costly errors in sales and inventory, it proves to be a financially wise option for long-term use.

Finally, when the system was evaluated using the ISO 25010:2023 Software Quality Model, it received an overall rating of 4.75 (Excellent). Among all the criteria, Functional Suitability gained the highest score of 4.86, reflecting how well the system performs its core tasks like real-time monitoring and automated sales processing. Other aspects, such as interaction capability, performance efficiency, reliability, and security, also received excellent ratings. This shows that the system is stable, responsive, user-friendly, and secure—meeting the everyday needs of R.P. Habana Auto Repair Shop and providing a modern solution to its previous challenges.

Conclusion

Based on the indicated findings, the following conclusions have been drawn:

To improve the operational efficiency of R.P. Habana Auto Repair Shop, the researchers developed a centralized web-based Inventory and Sales Management System to address the limitations found in their manual and Excel-based recording process. Previously, the business relied heavily on manually updated sheets and photo documentation from staff, which often led to delays, inaccuracies, and inconsistent information across all seven (7) branches. These issues made monitoring stock levels difficult and affected the accuracy of sales and inventory data.

The developed system successfully centralized and automated the business's inventory and sales tracking, allowing real-time updates of stock levels, product movements, and sales transactions. Through its integrated Point-of-Sale (POS) module, items sold are automatically deducted from inventory, ensuring accurate and up-to-date stock monitoring at all times. The system also includes a Service Job Tracking feature that records all completed service jobs and generates organized monthly, quarterly, and yearly reports, giving the business a clearer view of service performance and helping improve decision-making.

Furthermore, the implementation of automated low-stock and out-of-stock notifications addressed the recurring issue of delayed replenishment. These real-time alerts help prevent stockouts and ensure that branches can act promptly when supplies need to be restocked. The Physical Inventory feature also supports routine accuracy checks by comparing actual counts against system-recorded quantities, allowing discrepancies to be identified and corrected immediately.

The feasibility analysis demonstrated that the system is cost-effective and sustainable, with operational costs limited to affordable hosting services and API usage. The benefits gained—improved accuracy, reduced manual effort, enhanced monitoring, and more reliable reporting—show that the system provides long-term value to the business.

Finally, the system was evaluated using the ISO 25010:2023 quality standards and met the criteria for Interaction Capability, Performance Efficiency, Functional Suitability, Reliability, and Security. These results confirm that the system performs effectively, responds efficiently, protects data, and supports the overall operational needs of the business.

Overall, the centralized Inventory and Sales Management System successfully achieved all objectives of the study. It improved accuracy, strengthened monitoring across branches, automated key processes, enhanced reporting of sales and service jobs, and provided secure and reliable system performance making it a practical and scalable solution for R.P. Habana Auto Repair Shop moving forward.

Recommendations

Throughout the review of the study, the following recommendations are provided below.

1. **Implement a purchase order.**

   To automate the purchasing workflow. Ensuring stocks are replenished on time and manage fast-moving and slow-moving items.

2. **Add a system-based communication feature.**

   To improve communication between the admin, stockman, and staff for inventory updates, stock transfer and sales-related task.

3. **Implement a cashless payment option.**

   To provide customers a convenient and flexible way to pay.

4. **Integrate a secure payment gateway.**

   To support smoother processing of sales-related transactions and strengthen the accuracy of recorded financial data.

5. **Enabled RFID-Based inventory tracking.**

To enhance inventory monitoring of all branches, streamlining management for faster and more accurate updates.

6. **Implement a Customer Feedback and Review Module.**

To provide consistent insight into service quality and support more meaningful evaluation of overall performance across branches.

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APPENDICES

Appendix A

CLIENT APPROVAL LETTERS

Client Acceptance Letter

Department of Computing and Informatics

February 28, 2025
Riza H. Habana
Owner
R.P Habana Auto Repair Shop, Brgy. Bucal

Dear Mrs. Habana:

Good day!

The undersigned bachelor's degree candidates whose names are listed subsequently are presently enrolled in Capstone Project 1 which requires developing a system and writing a research output that covers Chapter 1 to Chapter 3 as partial fulfillment for the degree of Information Technology.

In line with this, we would like to request your office/business/organization/corporation to serve as our client for our study, "**R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM,**" during the Second Semester of AY 2024-2025. We seek your permission to administer questionnaires and conduct interviews with relevant staff members involved in the study. We assure you that all collected information will remain strictly confidential and used solely for research purposes, in full compliance with the Data Privacy Act of 2012. We appreciate your time and consideration and look forward to your positive response to this academic endeavor.

We humbly ask for you to affix your signature under CONFORME certifying your acceptance of this appointment. Thank you very much.

Respectfully Yours,

Garcia, Jhon Ric, C.

Jerusalem, Roi Angelo, A.

Paje, Jerry, F.

Perez, Kent Steven, D.

Noted By:

Jocelyn Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, MM-ITM
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Director, Department of Computing and Informatics

CONFORME:

Riza H. Habana
Owner, R.P Habana Auto Repair Shop

Old Municipal Site, Brgy. VII, Poblacion, Calamba City, Laguna
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GCED calamba

Continuation of Appendix A…

Client Interview Letter

Department of Computing and Informatics

February 28, 2025

Riza H. Habana
Owner
R.P Habana Auto Repair Shop, Brgy. Bucal

Dear Mrs. Habana:

Good day!

I hope this message finds you well. The undersigned bachelor's degree candidates are currently enrolled in Capstone Project 1 which requires developing a system and writing a research output that covers Chapter 1 to Chapter 3 as partial fulfillment for the degree of Information Technology.

We would like to request an appointment for an interview with you and your staff on **March 3, 2025 at 3:00 PM**, or at a time convenient for you. The purpose of this meeting is to discuss our project, **"R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM,"** and conduct the necessary interviews and questionnaires for the Second Semester of AY 2025. All gathered information will remain strictly confidential and used solely for research purposes, in full compliance with the Data Privacy Act of 2012.

We kindly ask for your confirmation of the said appointment. Thank you very much for your time and consideration.

Respectfully Yours,

Garcia, John Ric, C.

Jerusalem, Roi Angelo, A.

Paje, Jerry, S.

Perez, Kent Steven, D.

Noted By:

Jocelyn Llanderal, MIT
Research Adviser

CONFORME:

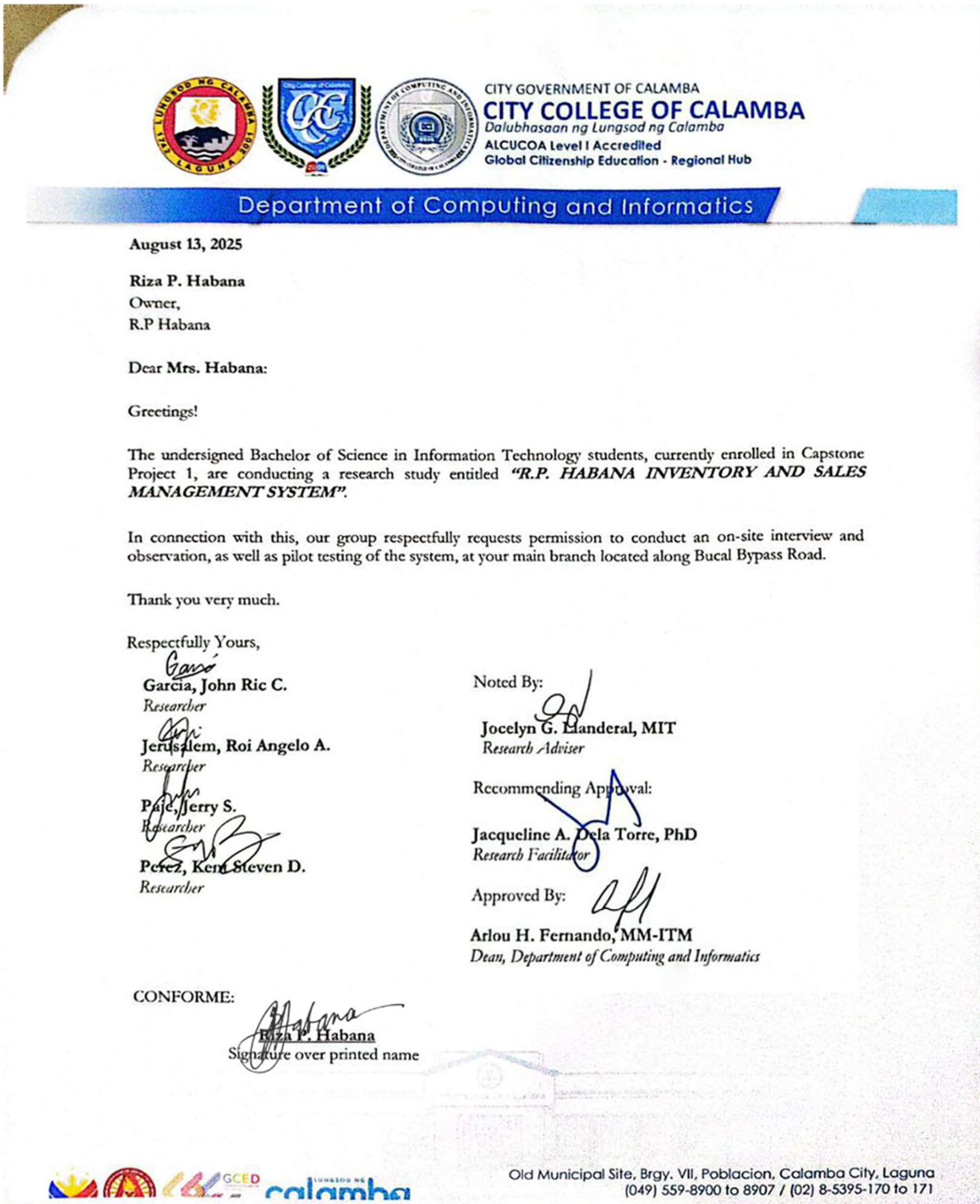
Riza H. Habana
Owner, R.P Habana Auto Repair Shop

GCED calamba

Old Municipal Site, Brgy. VII, Poblacion, Calamba City, L
(049) 559-8900 to 8907 / (02) 8-5395-170
ccc.

Continuation of Appendix A…

Client Interview Letter



CITY GOVERNMENT OF CALAMBA
CITY COLLEGE OF CALAMBA
Dalubhasaan ng Lungsod ng Calamba
ALCUCOA Level I Accredited
Global Citizenship Education - Regional Hub

Department of Computing and Informatics

August 13, 2025

Riza P. Habana
Owner,
R.P Habana

Dear **Mrs. Habana:**

Greetings!

The undersigned Bachelor of Science in Information Technology students, currently enrolled in Capstone Project 1, are conducting a research study entitled ***"R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM"***.

In connection with this, our group respectfully requests permission to conduct an on-site interview and observation, as well as pilot testing of the system, at your main branch located along Bucal Bypass Road.

Thank you very much.

Respectfully Yours,

Garcia, John Ric C.
Researcher

Jerusalem, Roi Angelo A.
Researcher

Paje, Jerry S.
Researcher

Perez, Kent Steven D.
Researcher

Noted By:

Jocelyn G. Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

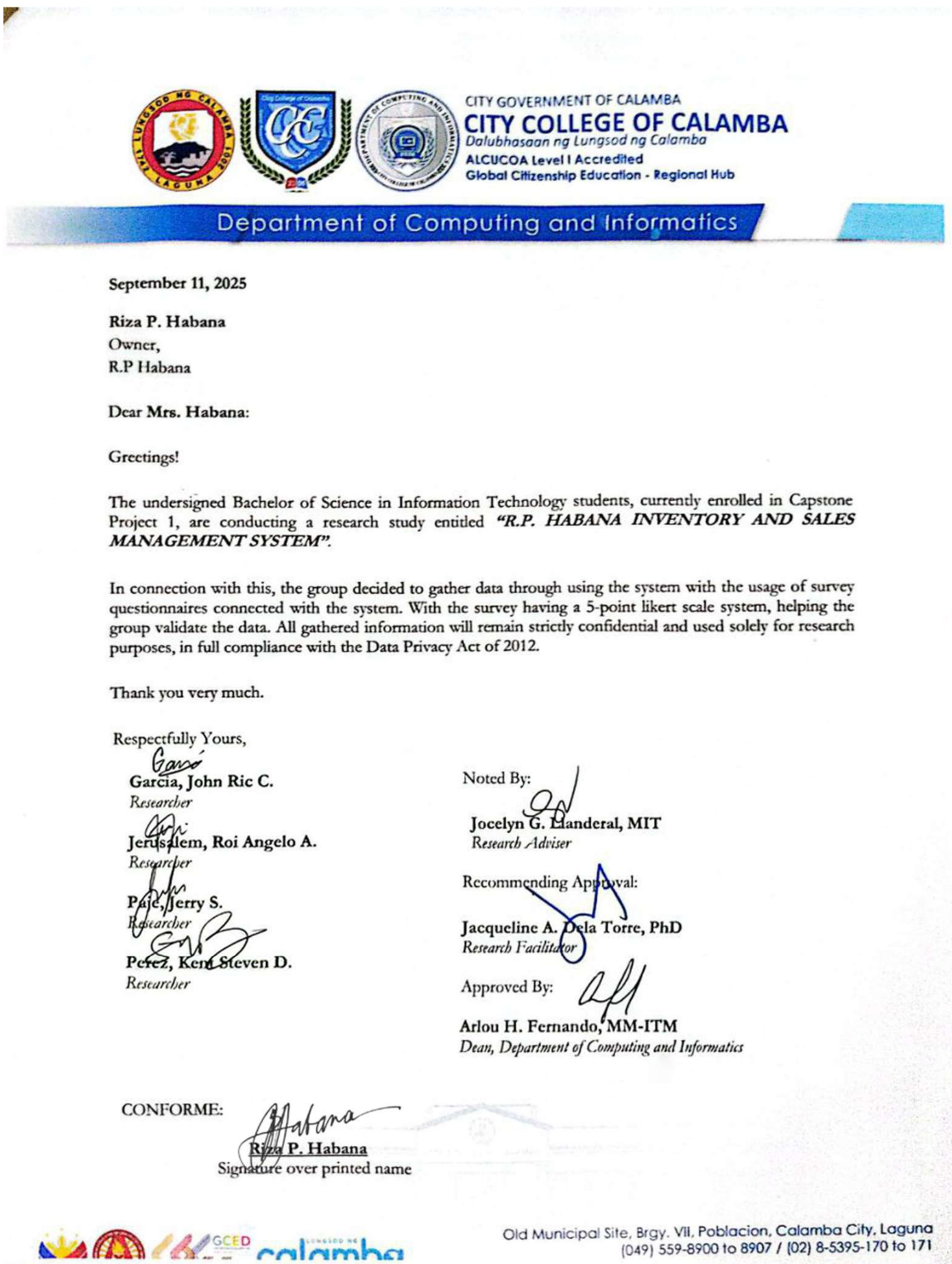
CONFORME:

Riza P. Habana
Signature over printed name

Old Municipal Site, Brgy. VII, Poblacion, Calamba City, Laguna
(049) 559-8900 to 8907 / (02) 8-5395-170 to 171

Continuation of Appendix A…

Consent Letter for Pilot Testing and Software Evaluation



CITY GOVERNMENT OF CALAMBA
CITY COLLEGE OF CALAMBA
Dalubhasaan ng Lungsod ng Calamba
ALCUCOA Level I Accredited
Global Citizenship Education - Regional Hub

Department of Computing and Informatics

September 11, 2025

Riza P. Habana
Owner,
R.P Habana

Dear Mrs. Habana:

Greetings!

The undersigned Bachelor of Science in Information Technology students, currently enrolled in Capstone Project 1, are conducting a research study entitled ***"R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM"***.

In connection with this, the group decided to gather data through using the system with the usage of survey questionnaires connected with the system. With the survey having a 5-point likert scale system, helping the group validate the data. All gathered information will remain strictly confidential and used solely for research purposes, in full compliance with the Data Privacy Act of 2012.

Thank you very much.

Respectfully Yours,

Garcia, John Ric C.
Researcher

Jerusalem, Roi Angelo A.
Researcher

Paje, Jerry S.
Researcher

Perez, Kent Steven D.
Researcher

Noted By:

Jocelyn G. Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

CONFORME:

Riza P. Habana
Signature over printed name

Old Municipal Site, Brgy. VII, Poblacion, Calamba City, Laguna
(049) 559-8900 to 8907 / (02) 8-5395-170 to 171

Appendix B

FORMS (Capstone 1)

OVPRI Research Form 1A: Adviser's Acceptance Form

Research Form 1 - Adviser's Acceptance form

Group No.: **52**

Department:	**Department of Computing and Informatics**
Degree Program:	**Bachelor of Science in Information Technology**
Student Researchers:	**Garcia, John Ric C.** **Jerusalem, Roi Angelo A** **Paje, Jerry S.** **Perez, Kent Steven D.**
Research Working Title:	**R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**

I do hereby accept the invitation of the above-mentioned students to become their **RESEARCH ADVISER** for the **second** semester, Academic Year **2024-2025.** Concomitant with this is my acceptance of the following functions and responsibilities:

1. Attend the orientation meeting scheduled by the Program Director.
2. Assist the student researchers in finalizing the title of the approved topic.
3. Guide and monitor the student researchers in their work in accordance with the timetable or the calendar of the college.
4. Check and improve the submitted proposals, questionnaires, and all aspects of the research paper.
5. Provide and sustain the technical, ethical, direction, and conduct of the student research
6. Monitor the participation of group members during the consultation and in every aspect of research work, and keep a record of consultations conducted
7. Resolve group problem/s arising in the preparation of the research paper
8. Schedule at least one-hour consultation time every week
9. Ensure quality control of the research output of student researchers
10. Prepare the advisees for the final oral defense through the mock defense
11. Endorse for final defense of the final manuscript by affixing signature on the prescribed form
12. Be present during defense although not supposed to participate in the defense.
13. Assist and supervise the student researchers in the final revision of the manuscript as recommended by the panel

Prepared by:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Recommended by:

Arlou H. Fernando, MM-ITM
Program Dean

Conforme: **Jocelyn G. Llanderal, MIT**
Date: ______________________

Approved:

Maryann H. Lanuza, LPT, MSc
Vice President for Research and Innovation

Continuation of Appendix B…

OVPRI Research Form 4A: Language Editor's Acceptance Form

Research Form4 – Language Editor's Acceptance Form

Group No.: **52**

Department	**Department of Computing and Informatics**
Degree Program:	**Bachelor of Science in Information Technology**
Student Researchers:	**Garcia, John Ric C.** **Jerusalem, Roi Angelo A.** **Paje, Jerry S.** **Perez, Kent Steven D.**
Research Working Title:	**R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**
Adviser:	**Jocelyn G. Llanderal, MIT**

I do hereby accept the invitation of the above-mentioned students to become their **LANGUAGE EDITOR** for the **second** semester, academic year **2024-2025.** Concomitant with this is my acceptance of the following functions and responsibilities:

1. Edit thoroughly the manuscript of the student researchers
2. Discuss with the research group/s the corrections, comments, and suggestions made on the manuscript in terms of the prescribed formats, correctness of grammar, and clarity and formality of language used
3. Monitor and assure the incorporation of the corrections and revisions made on the manuscript of student researchers
4. Endorse for soft bind/hardbound the final manuscript by providing certification of manuscript editing.

Prepared by:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Recommended by:

Arlou H. Fernando, MM-ITM
Program Dean

Conforme: **Kevin V. Angay, LPT, PhD**
Date: ____________

Approved:

Maryann H. Lanuza, LPT, MSc
Vice President for Research and Innovation

Continuation of Appendix B…

OVPRI Research Form 5A: Oral Defense Endorsement Form

Research Form 5 – Oral Defense Endorsement Form

Group No.: **52**

Department **Department of Computing and Informatics**
Degree Program: **Bachelor of Science in Information Technology**
Student Researchers: **Garcia, John Ric C.**
Jerusalem, Roi Angelo A.
Paje, Jerry S.
Perez, Kent Steven D.

Research Working Title: **R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**
Adviser: **Jocelyn G. Llanderal, MIT**

This thesis attached hereto entitled **"R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM"** is hereby endorsed for oral defense. Our signatures below attest to the acceptability of the manuscript for oral defense and compliance of all pertinent requirements and obligations of the student researchers.

Jocelyn G. Llanderal, MIT
Adviser

Date

Jacqueline A. Dela Torre, PhD
Research Facilitator

Date

For research facilitators use only

Date of Oral Defense : *May 6, 2025*
Time : *10:00 - 11:00 AM*
Venue : *City College of Calamba*

Continuation of Appendix B…

OVPRI Research Form 6A: Panelist's Acceptance Form

Research Form 6– Panelist's Acceptance Form

Group No.: **52**

Department	**Department of Computing and Informatics**
Degree Program:	**Bachelor of Science in Information Technology**
Student Researchers:	**Garcia, John Ric C.** **Jerusalem, Roi Angelo A.** **Paje, Jerry S.** **Perez, Kent Steven D.**
Research Working Title:	**R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**
Adviser:	**Jocelyn G. Llanderal, MIT**

I do hereby accept the invitation to become a research defense panelist on **May 6, 2025** for the **first** semester of the academic year **2024-2025** at the venue **City College of Calamba.**

Prepared: **Jacqueline A. Dela Torre, PhD**
Research Facilitator
Date Signed

Recommending Approval: **Arlou H. Fernando, MM-ITM**
Academic Dean
Date Signed

Conforme: **Regina G. Almonte, PhD**
Panel Chair
Date:

Approved: **Maryann H. Lanuza, LPT, MSc**
Vice President for Research and Innovations
Date Signed

Noted: **Ronald A. Gonzales, LPT, PhD**
Vice President for Academic Affairs
Date Signed

Continuation of Appendix B…

OVPRI Research Form 6A: Panelist's Acceptance Form

Research Form 6– Panelist's Acceptance Form

Group No.: **52**

Department	**Department of Computing and Informatics**
Degree Program:	**Bachelor of Science in Information Technology**
Student Researchers:	**Garcia, John Ric C.** **Jerusalem, Roi Angelo A.** **Paje, Jerry S.** **Perez, Kent Steven D.**
Research Working Title:	**R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**
Adviser:	**Jocelyn G. Llanderal, MIT**

I do hereby accept the invitation to become a research defense panelist on **May 6, 2025** for the **first** semester of the academic year **2024-2025** at the venue **City College of Calamba.**

Prepared: **Jacqueline A. Dela Torre, PhD**
Research Facilitator
______________________ Date Signed

Recommending Approval: **Arlou H. Fernando, MM-ITM**
Academic Dean
______________________ Date Signed

Conforme: **Jayvee Ryan F. Banal, MSc**
Panel Member

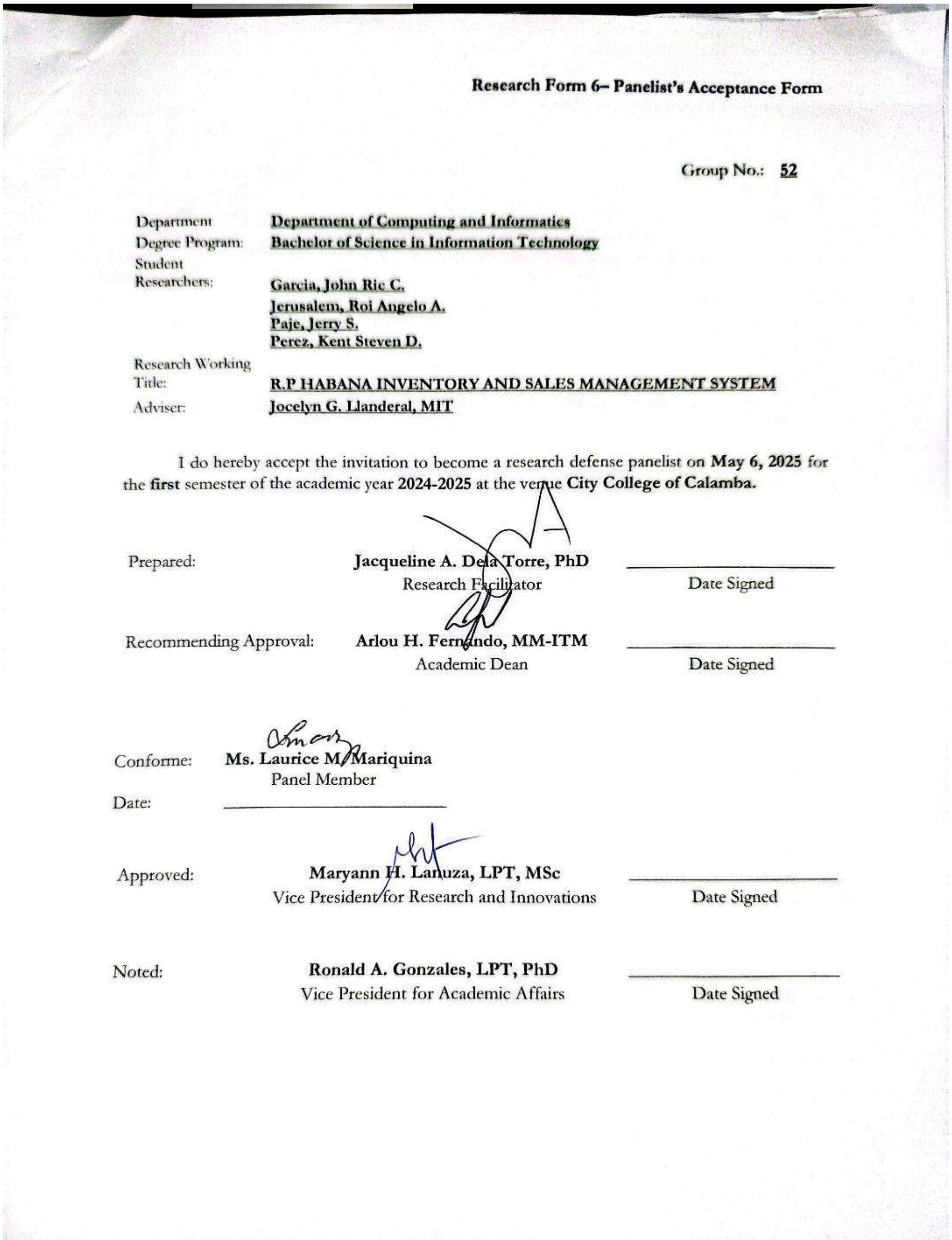
Date: ______________________

Approved: **Maryann H. Lanuza, LPT, MSc**
Vice President for Research and Innovations
______________________ Date Signed

Noted: **Ronald A. Gonzales, LPT, PhD**
Vice President for Academic Affairs
______________________ Date Signed

Continuation of Appendix B…

OVPRI Research Form 6A: Panelist's Acceptance Form



Research Form 6– Panelist's Acceptance Form

Group No.: **52**

Department	**Department of Computing and Informatics**
Degree Program:	**Bachelor of Science in Information Technology**
Student Researchers:	**Garcia, John Ric C.** **Jerusalem, Roi Angelo A.** **Paje, Jerry S.** **Perez, Kent Steven D.**
Research Working Title:	**R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**
Adviser:	**Jocelyn G. Llanderal, MIT**

I do hereby accept the invitation to become a research defense panelist on **May 6, 2025** for the **first** semester of the academic year **2024-2025** at the venue **City College of Calamba.**

Prepared: **Jacqueline A. Dela Torre, PhD** Research Facilitator ________ Date Signed

Recommending Approval: **Arlou H. Fernando, MM-ITM** Academic Dean ________ Date Signed

Conforme: **Ms. Laurice M. Mariquina** Panel Member

Date: ________

Approved: **Maryann H. Lanuza, LPT, MSc** Vice President for Research and Innovations ________ Date Signed

Noted: **Ronald A. Gonzales, LPT, PhD** Vice President for Academic Affairs ________ Date Signed

Continuation of Appendix B…

OVPRI Research Form 10A: Recommendation Compliance Sheet

Research Form 10 – Recommendations Compliance Sheet

Department **Department of Computing and Informatics**
Degree Program: **Bachelor of Science in Information Technology**
Student Researchers: **Garcia, John Ric C.**
Jerusalem, Roi Angelo A.
Paje, Jerry S.
Perez, Kent Steven D.

Adviser: **Jocelyn G. Llanderal, MIT**

Research Title: **R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**

Date of Proposal Defense: **May 6, 2025**
Panel
Chairman: **Regina G. Almonte, PhD**
Member: **Mr. Jayvee Ryan F. Banal**
Member: **Ms. Laurice M. Mariquina**

Comments	Chapter/Section/ Page Number	Actions Taken	Page number reflecting the changes	Panelist who suggested
• Replace the introductory with general overview of the study. • Organize the thought of the project context.	Chapter 1/ Project Context/ Page 1 to 3	Provided the general overview of the study and organized the project context for better clarity and flow.	Page 1 to 3	Regina G. Almonte, PhD
Objectives of the Study • Rephrase/Revise/Replace the objectives.	Chapter 1/ Objectives of the Study/ Page 3 to 4	Rephrase/Revise/Replace the general and specific objectives.	Page 3 to 4	Regina G. Almonte, PhD Mr. Jayvee Ryan F. Banal
Theoretical Framework • It must be the basis to attain the objectives • Independent variable must be aligned to the theories.	Chapter 1/ Research Framework / Page 5	Theoretical framework was remove based on the suggestion of our panel.	----	Regina G. Almonte, PhD
(Continuation at the next pages)				

1

Continuation of Appendix B…

OVPRI Research Form 10A: Recommendation Compliance Sheet

Research Form 10 – Recommendations Compliance Sheet

Comments	Chapter/Section/ Page Number	Actions Taken	Page number reflecting the changes	Panelist who suggested
Conceptual Framework • Insert the inventory reports.	Chapter 1/ Research Framework/ Page 6	Added inventory reports as part of tangible outputs	Page 6	Regina G. Almonte, PhD
Conceptual Framework • Revise the figure caption.	Chapter 1/ Research Framework / Page 7	Updated the figure caption to accurately describe the structure of data in the conceptual framework.	Page 6-7	Mr. Jayvee Ryan F. Banal
Significance of the Study • Arrange from most to least beneficiaries.	Chapter 1/ Significance of the Study/ Page 7 to 8	Change the significance of the study from most to least beneficiaries.	Page 7 to 8	Regina G. Almonte, PhD
Scope and Limitation • Discuss and revise the scope and limitation.	Chapter 1/ Scope and Limitation/ Page 8 to 9	Revised the Scope and Limitation section and included an additional user level, the Stockman, to reflect the updated system roles.	Page 8 to 10	Regina G. Almonte, PhD
Definition of terms • Arrange alphabetically.	Chapter 1/ Definition of Terms/ Page 10 to 11	Arranged the definition of terms by conceptual to operational terms.	Page 11 to 12	Regina G. Almonte, PhD
Literature Review • Establish connections among the paragraphs. • Add themes to attain the objectives. • Revise the synthesis.	Chapter 2 / Literature Review/ Page 12 to 22	Revised the literature review by building clearer connections among the paragraphs, organized the themes that aligned with the research objectives, and refining the synthesis.	Page 13 to 25	Regina G. Almonte, PhD
Research Design • Concentrate on descriptive design	Chapter 3/ Research Design / Page 23	Revise the research design focusing on descriptive design.	Page 26	Regina G. Almonte, PhD
Gantt chart • Focus on software development	Chapter 3/ Gantt Chart/ Page 35	Revised the Gantt chart to focus more specifically on software development.	Page 39	Regina G. Almonte, PhD
(Continuation at the next pages)				

2

Continuation of Appendix B…

OVPRI Research Form 10A: Recommendation Compliance Sheet

Research Form 10 – Recommendations Compliance Sheet

Comments	Chapter/Section/ Page Number	Actions Taken	Page number reflecting the changes	Panelist who suggested
System Design • Revise and fix the diagrams and flowcharts according to the process.	Chapter 3/ System Design/ Page 36 to 55	Revised and fixed the diagrams and flowcharts to accurately reflect the system process.	Page 40 to 48	Regina G. Almonte, PhD Mr. Jayvee Ryan F. Banal Ms. Laurice M. Mariquina

Continuation of Appendix B…

OVPRI Research Form 10A: Recommendation Compliance Sheet

Research Form 10 – Recommendations Compliance Sheet

Prepared by:

John Ric C. Garcia

Roi Angelo A. Jerusalem

Jerry S. Paje

Kent Steven D. Perez

Date: **May 19, 2025**

Checked by:

Jocelyn G. Llanderal, MIT

Date: **May 19, 2025**

Approved by:

Chairman: **Regina G. Almonte, PhD**

Member: **Mr. Jayvee Ryan F. Banal**

Member: **Ms. Laurice M. Mariquina**

Continuation of Appendix B…

FORMS (Capstone 2)

OVPREQA Student Research Form 1: Researcg Consultant's Acceptance Form

Student Research Form 01 – Research Consultant's Acceptance Form

Group No.: 52

Department	Department of Computing and Informatics
Degree Program	Bachelor of Science in Information Technology
Student Researchers	Garcia, John Ric C. Jerusalem, Roi Angelo A. Paje, Jerry S. Perez, Kent Steven.
Research Working Title	R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM
Research Design	Qualitative and Descriptive-Quantitative
Semester, Academic Year	1st Semester, A.Y. 2025-2026

I do hereby accept the invitation of the above-mentioned students to serve as their **RESEARCH CONSULTANT** for their Capstone 1 and 2 courses. Along with this is my full acceptance of the responsibilities and functions corresponding to these roles as stipulated in the students' research manual.

This form shall remain valid for one year and until the research hardbound manuscript has been accepted by the Dean of the Department, or unless another form is accomplished for the change of research consultant.

Prepared by:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Conforme: **Jocelyn G. Llanderal, MIT** Date
Research Adviser

Kevin V. Angay, LPT, PhD Date
Language Editor

Recommendation by:

Arlou H. Fernando, MM-ITM
Department Dean

Continuation of Appendix B…

OVPREQA Student Research Form 4: Panelist Acceptance Form

Student Research Form 04 - Panelist's Acceptance Form

Group No.: 52

Department	Department of Computing and Informatics
Degree Program	Bachelor of Science in Information Technology
Student Researchers	Garcia, John Ric C. Jerusalem, Roi Angelo A. Paje, Jerry S. Perez, Kent Steven.
Research Working Title	R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM
Research Design	Qualitative and Descriptive-Quantitative
Research Adviser	Jocelyn G. Llanderal, MIT
Semester, Academic Year	1st Semester, A.Y. 2025-2026

I do hereby accept the invitation to become a research defense panelist on **November 18, 2025** for the **FINAL RESEARCH DEFENSE** at the venue City College of Calamba.

Prepared by:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Conforme: **Regina G. Almonte, PhD** Date:
Panel Chair

Jayvee Ryan F. Banal, MSc Date:
Panel Member

Ms. Laurice M. Mariquina Date:
Panel Member

Recommendation by:

Arlou H. Fernando, MM-ITM
Department Dean

Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting

Student Research Form 06 – Thesis Writing Monitoring Form

Group No.: 52

Department: Department of Computing and Informatics
Degree Program: Bachelot of Science in Information Technology
Student Researchers: Garcia, John Ric C. Mobile Number: 09567692614
Jerusalem, Roi Angelo A. Mobile Number: 09939903687
Paje, Jerry S. Mobile Number: 09854910625
Perez, Kent Steven D. Mobile Number: 09282876871
Adviser: Jocelyn G. Llanderal Mobile Number: ____________
Research Title: R.P HABANA INVENTORY AND SALES MANAGEMENT SYSEM
Semester, Academic Year: 1st Semester, A.Y. 2025-2026
Research Facilitator: Jacqueline A. Dela Torre, PhD
Language Editor: Kevin V. Angay, LPT, PhD
Data Analyst (if applicable): ____________

Date of Consultation	Place / Platform of Consultation	Topic/ Issue(s) / Concerns/Recommendations	Phase of Thesis Writing (Concept Paper/Proposal/Final)	Signature of Research Consultants
January 24, 2025	• JMC Building • Face to Face	• Meeting with the Research Facilitator Orientation for Capstone 1 Subject	Concept Paper	Jacqueline A. Dela Torre, PhD
January 30, 2025	• Rizal Building • Face to Face	Initial meeting with adviser • Orientation and acceptance as official adviser.	Concept Paper	Jocelyn G. Llanderal, MIT
January 31, 2025	• JMC Building • Face to Face	• Discussion of concept paper	Concept Paper	Jacqueline A. Dela Torre, PhD
February 5, 2025	• Rizal Building • Face to Face	Checking on Concept Paper • If the admin can view the inventory • The title along with its functions, has a limited scope or coverage. • Be specific on the expected outputs	Concept Paper	Jocelyn G. Llanderal, MIT
February 7, 2025	• JMC Building • Face to Face	• Discussion of updated deadlines for the concept paper and title defense	Concept Paper	Jacqueline A. Dela Torre, PhD
February 10, 2025	• Online Meeting • Google Mee	• Include a sampling method for the concept paper	Concept Paper	Jocelyn G. Llanderal, MIT
February 10, 2025	• Google Meet • Online	• Discussion on the submission of Chapter 2 and Objectives, that must be signed by the adviser	Proposal	Jacqueline A. Dela Torre, PhD

Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting

Student Research Form 06 – Thesis Writing Monitoring Form

February 13, 2025	• Rizal Building • Face to Face	Consultation • Meeting with adviser about client interview plans	Proposal	Jocelyn G. Llanderal, MIT
February 19, 2025	• Rizal Building • Face to Face	Discussion • Manual process of business • Monthly sales estimate • updated Chapter 2	Proposal	Jocelyn G. Llanderal, MIT
February 20, 2025	• Online	• Sharing of possible modules/function that can be found on the system.	Proposal	Jocelyn G. Llanderal, MIT
February 21, 2025	• Rizal Building • Face to Face	Objectives & RRL • Add sales transaction, data analytics, and reports in objectives • Elaboration of incomplete information on RRL • Advised to make a benchmarking for RRL	Proposal	Jocelyn G. Llanderal, MIT
February 27, 2025	• Rizal Building • Face to Face	Chapter 1 & Chapter 2 • Add more tangible and intangible outputs in conceptual framework • Definition of terms must be in alphabetical order • Revision of related literature, provides the methodology, modules, and function used in the recent study.	Proposal	Jocelyn G. Llanderal, MIT
February 28, 2025	• JMC Building • Face to Face	• Discussion regarding the proposal defense and submission date of chapter 1 and 2.	Proposal	Jacqueline A. Dela Torre, PhD
March 6, 2025	• Online Chat • Messenger	Chapter 1 and 3 • Provides the exact address of the shop in the project context. • Explain methodology used in the research. • Explain what researchers do during each phase of software development process. • Provide a system flowchart.	Proposal	Jocelyn G. Llanderal, MIT
March 10, 2025	• Rizal Building • Face to Face	• Checking of progress for system and documentation.	Proposal	Jocelyn G. Llanderal, MIT

Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting

Student Research Form 06 – Thesis Writing Monitoring Form

March 11, 2025	• Rizal Building • Face to Face	Chapter 1-3 • Elaboration on incomplete information in the project context. • Justify the inefficiencies the business face. • Put the user design on chapter 3. • Provide an ethical consideration for the study.	Proposal	Jacqueline A. Dela Torre, PhD
March 14, 2025	• Google Meet • Online	• Discussion on the updated submission schedule for chapter 1- 3 and 30% of the working system.	Proposal	Jacqueline A. Dela Torre, PhD
March 21, 2025	• JMC Building • Face to Face	• Checking and monitoring of chapter 1-3, Panelist acceptance form, and monitoring form.	Proposal	Jasqueline A. Dela Torre, PhD
April 23, 2025	• Rizal Building • Face to Face	• Meeting with adviser for checking Chapter 1-3.	Proposal	Jocelyn G. Llanderal, MIT
July 30, 2025	• Rizal Building • Face to Face	• Consultation of updated document and 80% of the system.	Finals	Jocelyn G. Llanderal, MIT
August 3, 2025	• Google Meet • Online	• Online Consultation - System review.	Finals	Jacqueline A. Dela Torre, PhD
August 8, 2025	• Google Meet • Online	• Announcement of Class Schedule.	Finals	Jacqueline A. Dela Torre, PhD
August 13, 2025	• Rizal Building • Face to Face	• Consultation of System and Papers.	Finals	Jocelyn G. Llanderal, MIT
September 4, 2025	• Google Meet • Online	• Capstone online meeting for submission date.	Finals	Jacqueline A. Dela Torre, PhD
November 17, 2025	• Rizal Building • Face to Face	• Mock Defense.	Finals	Jocelyn G. Llanderal, MIT
November 24, 2025	• Rizal Building • Face to Face	• Consultation for compliance.	Finals	Jocelyn G. Llanderal, MIT
November 25, 2025	• Rizal Building • Face to Face	• Checking of revised documentation and system.	Finals	Jocelyn G. Llanderal, MIT
November 25, 2025	• Google Meet • Online	• Online meeting for submission of manuscript and system turn over	Finals	Jacqueline A. Dela Torre, PhD

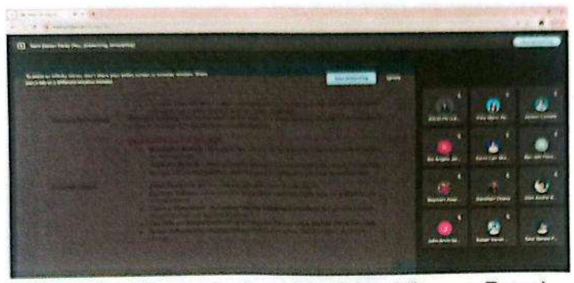
Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting

Student Research Form 06 – Thesis Writing Monitoring Form

**Student Researchers and their respective research adviser are expected to have at least 10 consultation meetings per semester.*
**All research consultations to any research consultants (data analysts, validator, language editors, panelists) shall be reflected in the monitoring form.*

Documentations with proper captions:

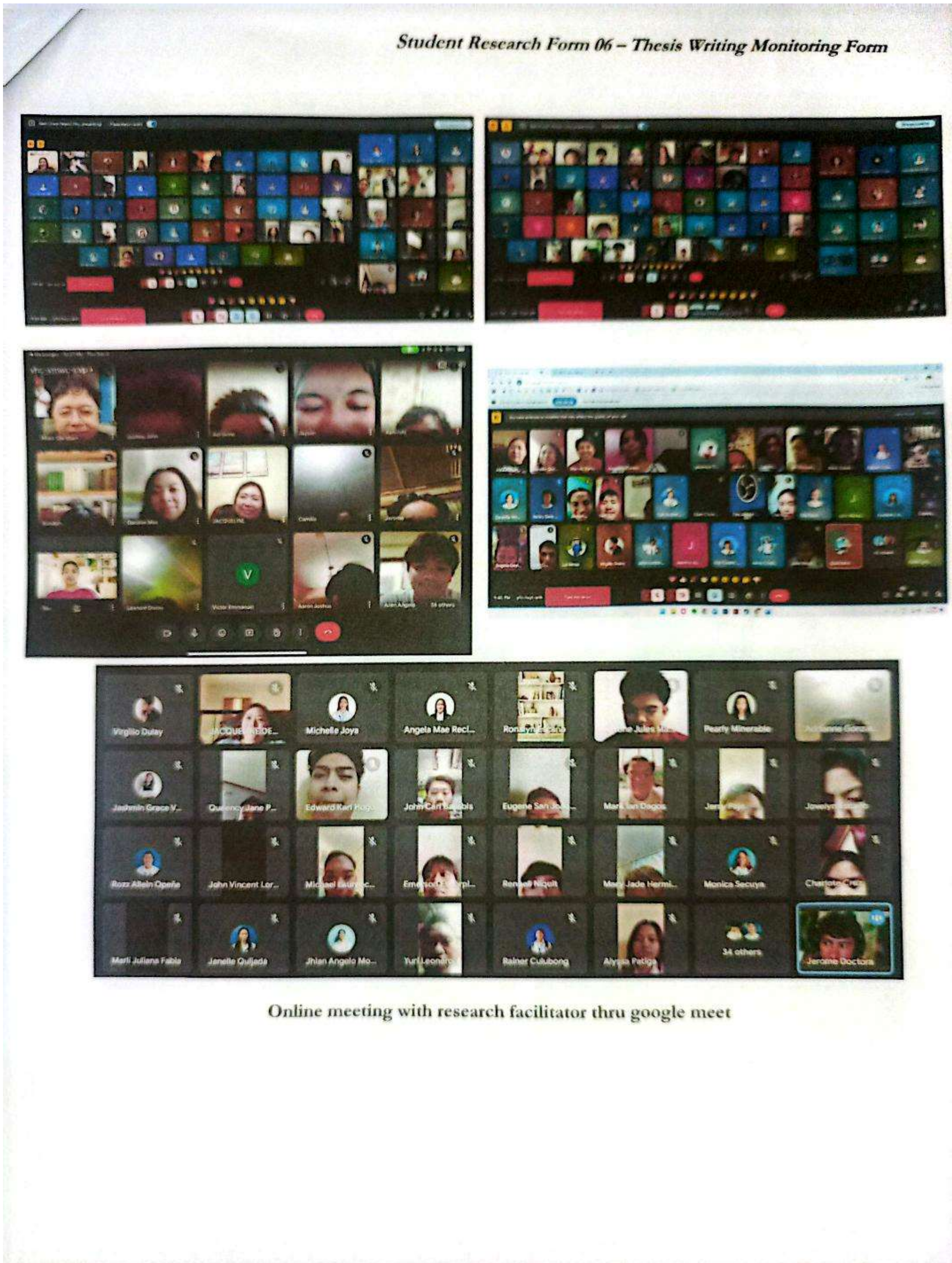


Capstone title consultation with adviser (Concept Paper)

Online Consultation with the adviser thru messenger (Chapter 1-3 Proposal)

Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting



Student Research Form 06 – Thesis Writing Monitoring Form

Online meeting with research facilitator thru google meet

Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting

Student Research Form 06 – Thesis Writing Monitoring Form

Consultation with capstone adviser (face to face)

Continuation of Appendix B…

OVPREQA Student Research Form 6: Thesis Writing Moniroting

Student Research Form 06 – Thesis Writing Monitoring Form

Noted:

Preliminary Term	**Jacqueline A. Dela Torre, PhD**	Date:
Midterm	**Jacqueline A. Dela Torre, PhD**	Date:
Final Term	**Jacqueline A. Dela Torre, PhD**	Date:

Continuation of Appendix B…

OVPREQA Student Research Form 7: Recommendation Compliance Sheet

Student Research Form 07 – Recommendations Compliance Sheet

Group No.: **52**

Department: **Department of Computing and Informatics**
Degree Program: **Bachelor of Science in Information Technology**
Student Researchers: **Garcia, John Ric C.**
Jerusalem, Roi Angelo A.
Paje, Jerry S.
Perez, Kent Steven.
Adviser: **Jocelyn G. Llanderal**
Research Title: **R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM**
Semester, Academic Year: **1st Semester, A.Y. 2025-2026**
Date of Final Defense: **November 18, 2025**
Panelist
Chairman: **Regina G. Almonte, PhD**
Member: **Jayvee Ryan F. Banal, MSc**
Member: **Ms. Laurice M. Mariquina**

Comments	Chapter/Section/ Page Number	Actions Taken	Page number reflecting the changes	Panelist who suggested
• Revise the synthesis	Chapter 2/Synthesis/ Page 24 to 25	• Revised the synthesis in Chapter 2 to shorten the discussion and improve clarity.	Page 24 to 25	Jayvee Ryan F. Banal, Msc
• revise and revise data analysis plan's statistical column.	• Chapter 3/Data Analysis Plan/34	• Updated the Data Analysis Plan, specifically the statistical treatment column, to better match the objectives and methods used.	Page 33	Regina G. Almonte, PhD
• revise gantt chart: specify the objectives • revise context: review flows • revise DFD levels • revise flowcharts focus on major process	Chapter 3/Gantt Chart/Context Diagram/DFD/Flowcharts/ Page 40/ 41/43/46/47/48/ 49	• Improved the Gantt Chart by clearly stating each objective and aligning the timeline with project tasks. • Refined the Project Context by reviewing the process flows for accuracy and coherence. • Revised the DFDs and ensured all diagrams follow the	Page 39/40/42/45/ 46/47/48	Regina G. Almonte, PhD Jayvee Ryan F. Banal, Msc

Continuation of Appendix B…

OVPREQA Student Research Form 7: Recommendation Compliance Sheet

Student Research Form 07 – Recommendations Compliance Sheet

		correct logical flow of the system. • Updated the Flowcharts to highlight only the major processes and remove unnecessary steps.		
• discuss testing procedure and must have validation process before that phase.	Chapter 3/Testing Phase/page 80	• Expanded the Testing Procedure by adding a validation process before system testing.	Page 79 to 81	Regina G. Almonte, PhD
• revise and discuss how the objectives were attained. Basis – CBA: ROI & Payback Period.	Chapter 4/ Results and Discussion/ page 83 to 97	• Discussed how each objective was attained, using CBA results such as ROI and Payback Period as the basis.	Page 84 to 99	Regina G. Almonte, PhD Jayvee Ryan F. Banal, Msc
• revise conclusion and recommendation and must align with the objectives	Chapter5/Conclusion/recommendation / Page 98 to 103	• Revised the Conclusion and Recommendations to ensure they directly align with the study's objectives and findings.	Page 103 to 105	Regina G. Almonte, PhD Jayvee Ryan F. Banal, Msc
GUI • Improvements and use color palletes	System	• Improves GUI by changing the color theme. • Uses Color palette for better branding	System	Jayvee Ryan F. Banal, Msc
Sales Report • Add sales report in Staff Account	System	• Give permission to the Staff accounts to view sales report of their own branch.	System	Regina G. Almonte, PhD Jayvee Ryan F. Banal, Msc Ms. Laurice M. Mariquina
Inventory Report • Add Inventory Report in Admin Account and Stockman Account	System	• Added inventory report that generates and show Inventory Movements as well	System	Regina G. Almonte, PhD

Continuation of Appendix B…

OVPREQA Student Research Form 7: Recommendation Compliance Sheet

Student Research Form 07 – Recommendations Compliance Sheet

		as Summary Reports. • Inventory movements show how the products move through out a specific period. • Summary Report shows the Beginning and Ending Stocks of all the products in the specific period of time.		Jayvee Ryan F. Banal, Msc
Add Products • Expiration Date Input must only visible if a specific Category do expires, else hidden	System	• Make expiration input fields only show when a specific category of products does require an expiration date, else, it is hidden.	System	Regina G. Almonte, PhD
POS • Services in POS have Quantity, remove it • Discount must have limitation as well as validation of it (e.g. 100-500 pesos only)	System	• Removes the quantity for POS and limits to only one per transactions. • Improved input validation by limiting the discount field from 0-500 only.	System	Regina G. Almonte, PhD
Stock Transfer • Force an automatic transfer request when staff click an item and it is out-of-stock already	System	• Explain to the panelist stock transfer procedure	System	Jayvee Ryan F. Banal, Msc Regina G. Almonte, PhD
Physical Inventory • In stockman, System stocks and how many the discrepancies are hidden. It only shows there's discrepancies • In Admin, System stocks and how many the discrepancies are shown.	System	• Removed the permission for stockman to see System stocks and number of discrepancies columns.	System	Regina G. Almonte, PhD
Services • Connect this with inventory and sales	System	• Added another action in manage services to assign specific materials	System	Regina G. Almonte, PhD

Continuation of Appendix B…

OVPREQA Student Research Form 7: Recommendation Compliance Sheet

Student Research Form 07 – Recommendations Compliance Sheet

		needed in a particular service. • Provide separate revenue for products and services.		
Error Message • This must be clear, what the error is about	System	• Improve the show toast messages.	System	Jayvee Ryan F. Banal, Msc
Data Validation • Improve Data Validation (POS)	System	• Improve Data validation per input fields.	System	Jayvee Ryan F. Banal, Msc

Continuation of Appendix B…

OVPREQA Student Research Form 7: Recommendation Compliance Sheet

Student Research Form 07 – Recommendations Compliance Sheet

Prepared by:

Garcia, John Ric C.

Jerusalem, Roi Angelo A.

Paje, Jerry S.

Perez, Kent Steven D.

Date:____________

Checked by:

Jocelyn G. Llanderal
Research Adviser
Date:____________

Approved by:

Chairman: **Regina G. Almonte, PhD**

Member: **Jayvee Ryan F. Banal, MSc**

Member: **Ms. Laurice M. Mariquina**

Appendix C

LETTERS AND CERTIFICATIONS

Letter for System Validation

CITY GOVERNMENT OF CALAMBA
CITY COLLEGE OF CALAMBA
Dalubhasaan ng Lungsod ng Calamba
ALCUCOA Level I Accredited
Global Citizenship Education - Regional Hub

Department of Computing and Informatics

August 14, 2025

Arlou H. Fernando, MM-ITM
Program Dean,
Department of Computing and Informatics,
City College of Calamba

Dear **Prof. Fernando**:

We, the researchers of City College of Calamba, currently enrolled in Capstone 2, are conducting a research study entitled "R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM". As part of our development process, we are required to have three IT experts to review and validate our proposed system to ensure it meets the necessary technical standards and functionalities.

We respectfully believe that your knowledge and expertise is suited to serves as one of our IT experts. Your professional insights, and feedback would greatly help us to refine our system, to ensure that is it both efficient and reliable. We would be honored if you could accept our request to be one of our IT experts for this validation.

Thank you very much.

Respectfully Yours,

Garcia, John Ric C.
Researcher

Jerusalem, Roi Angelo A.
Researcher

Paje, Jerry S.
Researcher

Perez, Kent Steven D.
Researcher

Noted By:

Jocelyn G. Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

CONFORME:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

GCED calamba

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Letter for System Validation

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CITY COLLEGE OF CALAMBA
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Global Citizenship Education - Regional Hub

Department of Computing and Informatics

August 14, 2025

Jasper M. Garcia, MIT
Faculty Member,
Department of Computing and Informatics,
City College of Calamba

Dear **Prof. Garcia**:

We, the researchers of City College of Calamba, currently enrolled in Capstone 2, are conducting a research study entitled "R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM". As part of our development process, we are required to have three IT experts to review and validate our proposed system to ensure it meets the necessary technical standards and functionalities.

We respectfully believe that your knowledge and expertise is suited to serves as one of our IT experts. Your professional insights, and feedback would greatly help us to refine our system, to ensure that is it both efficient and reliable. We would be honored if you could accept our request to be one of our IT experts for this validation.

Thank you very much.

Respectfully Yours,

Garcia, John Ric C.
Researcher

Jerusalem, Roi Angelo A.
Researcher

Paje, Jerry S.
Researcher

Perez, Kent Steven D.
Researcher

Noted By:

Jocelyn G. Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

CONFORME:

Jasper M. Garcia, MIT
Faculty Member, Department of Computing and Informatics

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Letter for System Validation

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Department of Computing and Informatics

October 15, 2025

Jessabelle Perez
Sales Accounting Jr. Manager,
Moment Group Inc.

Dear **Ms. Perez**:

We, the researchers of City College of Calamba, currently enrolled in Capstone 2, are conducting a research study entitled "R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM". As part of our development process, we are required to have three validators to review and validate our proposed system to ensure it meets the necessary technical standards and functionalities.

We respectfully believe that your knowledge and expertise is suited to serves as one of our validators. Your professional insights, and feedback would greatly help us to refine our system, to ensure that is it both efficient and reliable. We would be honored if you could accept our request to be one of our experts for this validation.

Thank you very much.

Respectfully Yours,

Garcia, John Ric C.
Researcher

Respectfully Yours,

Garcia, John Ric C.
Researcher

Jerusalem, Roi Angelo A.
Researcher

Paje, Jerry S.
Researcher

Perez, Kent Steven D.
Researcher

Noted By:

Jocelyn G. Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

CONFORME:

Jessabelle Perez
Sales Accounting Jr. Manager, Moment Group Inc.

Continuation of Appendix C…

Letter for Research Instrument Validation

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Department of Computing and Informatics

May 26, 2025

Riza P. Habana
Owner,
R.P Habana

Dear **Mrs. Habana:**

Greetings!

The undersigned Bachelor of Science in Information Technology students, currently enrolled in Capstone Project 1, are conducting a research study entitled ***"R.P. HABANA INVENTORY AND SALES MANAGEMENT SYSTEM"***.

In connection with this, we respectfully seek your expertise to validate the attached research instrument intended for the evaluation of our proposed system. We kindly request that you review the instrument and affix your signature to certify its validation.

Thank you very much.

Respectfully Yours,

Garcia, John Ric C.
Researcher

Jerusalem, Roi Angelo A.
Researcher

Paje, Jerry S.
Researcher

Perez, Kent Steven D.
Researcher

Noted By:

Jocelyn G. Llanderal, MIT
Research Adviser

Recommending Approval:

Jacqueline A. Dela Torre, PhD
Research Facilitator

Approved By:

Arlou H. Fernando, MM-ITM
Dean, Department of Computing and Informatics

CONFORME:

Habana
Riza P. Habana
Signature over printed name

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Continuation of Appendix C…

Letter for Research Instrument Validation

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Direction: Please read and rate your experiences by putting a (✓) mark on the rows column that corresponds to the assessment. Use the scale below:

Scale: **(5) Excellent** **(4) Very Satisfactory** **(3) Satisfactory** **(2) Fair** **(1) Poor**

Survey Questionnaire:					
Section 1: Sales and Inventory Management (Objective 1)	**5**	**4**	**3**	**2**	**1**
The system allows me to accurately track sales transactions in real-time.					
The system can easily monitor stock levels through the system.					
The system provides clear, real-time inventory and sales reports.					
The product tracking feature of the system is efficient and reliable.					
The centralized system improves our overall inventory and sales management.					
Section 2: Service Job Tracking (Objective 2)	**5**	**4**	**3**	**2**	**1**
The system effectively tracks completed service jobs by category.					
The system can easily generate service job reports on a monthly, quarterly, and yearly basis.					
The reports help me identify the most profitable services per branch.					
The system supports better decision-making in managing service operations.					
Tracking service jobs through the system has improved our workflow and efficiency.					

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Continuation of Appendix C…

Letter for Research Instrument Validation

Section 3: Notification System (Objective 3)	5	4	3	2	1
The system ensures timely notifications about low-stock items and out-of-stocks items.					
The automated alerts help me take timely action on low-stock and out-of-stocks items.					
The notification system keeps me updated and reduces delays in restocking.					
The real-time updates provided by the system are accurate and reliable.					
The stock alerts significantly enhance our ability to manage inventory effectively					
Section 4: Cost-Benefit Evaluation (Objective 4)	**5**	**4**	**3**	**2**	**1**
The benefits of the system outweigh its costs.					
Using the system has reduced the time spent on manual inventory tasks.					
The system has led to cost savings in our inventory and sales operations.					
The investment in this system was worth the improvements gained.					
The return on investment of the system is satisfactory.					

Continuation of Appendix C…

Letter for Research Instrument Validation

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Software Quality Model from ISO 25010: 2023					
A. Interaction Capability	**5**	**4**	**3**	**2**	**1**
The system is user-friendly and easy to navigate.					
The interface design supports quick and efficient interaction.					
The system is easy to learn, even for first-time users.					
The system provides clear feedback when actions are performed.					
The system allows user to complete task with minimal assistance or training.					
B. Performance Efficiency	**5**	**4**	**3**	**2**	**1**
The system is responsive and reacts to end-user input quickly					
The system works well when many users are on it at the same time					
The system maintains stable performance even when handling large volumes of data (e.g., product inventory data, sales transaction)					
The system's database is scalable, without requiring major costs and downtime.					
The system is optimized for various devices (PC, tablet, and mobile) without experiencing delays or performance issues.					
C. Functional Suitability					
The system allows the users to access their designated page based on their user roles (admin, staff, stockman).					
The system provides accurate inventory and sales transaction records					
The system allows users to monitor and update inventory and sales.					
The POS system ensures accurate pricing and real-time inventory updates with every transaction.					
The system's buttons perform the function their meant to do.					

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Continuation of Appendix C…

Letter for Research Instrument Validation

D. Reliability	5	4	3	2	1
The system reduces human errors compared to manual inventory and sales processing.					
The system quickly recovers from unexpected crashes or issues.					
The system enables accurate stock transfers to R.P. Habana branches, ensuring reliable communication.					
The system accurately saves and retrieves inventory and sales data.					
The system reliably maintains inventory and sales records, preventing data loss.					
E. Security	**5**	**4**	**3**	**2**	**1**
The system allows only authorize users based on Role-Based Access Control (RBAC).					
The system encrypts the data and information. (password, branch number, product id, etc.).					
The system enforces a strong password policy requiring uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and special characters (!@#$%^&*).					
The system maintains audit logs to track user activities and security events.					
The system locks the user account after failed log-in attempts.					

Continuation of Appendix C…

Certificate of Software Validation

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Department of Computing and Informatics

CERTIFICATE OF SOFTWARE VALIDATION

This is to certify that I have validated the interview protocol of the study entitled

"R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM"

Prepared by

Garcia, John Ric C.

Jerusalem, Roi Angelo A.

Paje, Jerry jr. S.

Perez, Kent Steven D.

and I have found it complete and satisfactory in terms of face and content validity.

Arlou H. Fernando, MM-ITM

Name of the Validator

Academic Dean, City College of Calamba

Affiliation

____

Date

Continuation of Appendix C…

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JASPER M. GARCIA, MIT
Name of the Validator

IT Professor, City College of Calamba
Affiliation

____
Date

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Jessabelle S. Perez

Name of the Validator

Jr. Manager, Moment Group Inc.

Affiliation

____

Date

Continuation of Appendix C…

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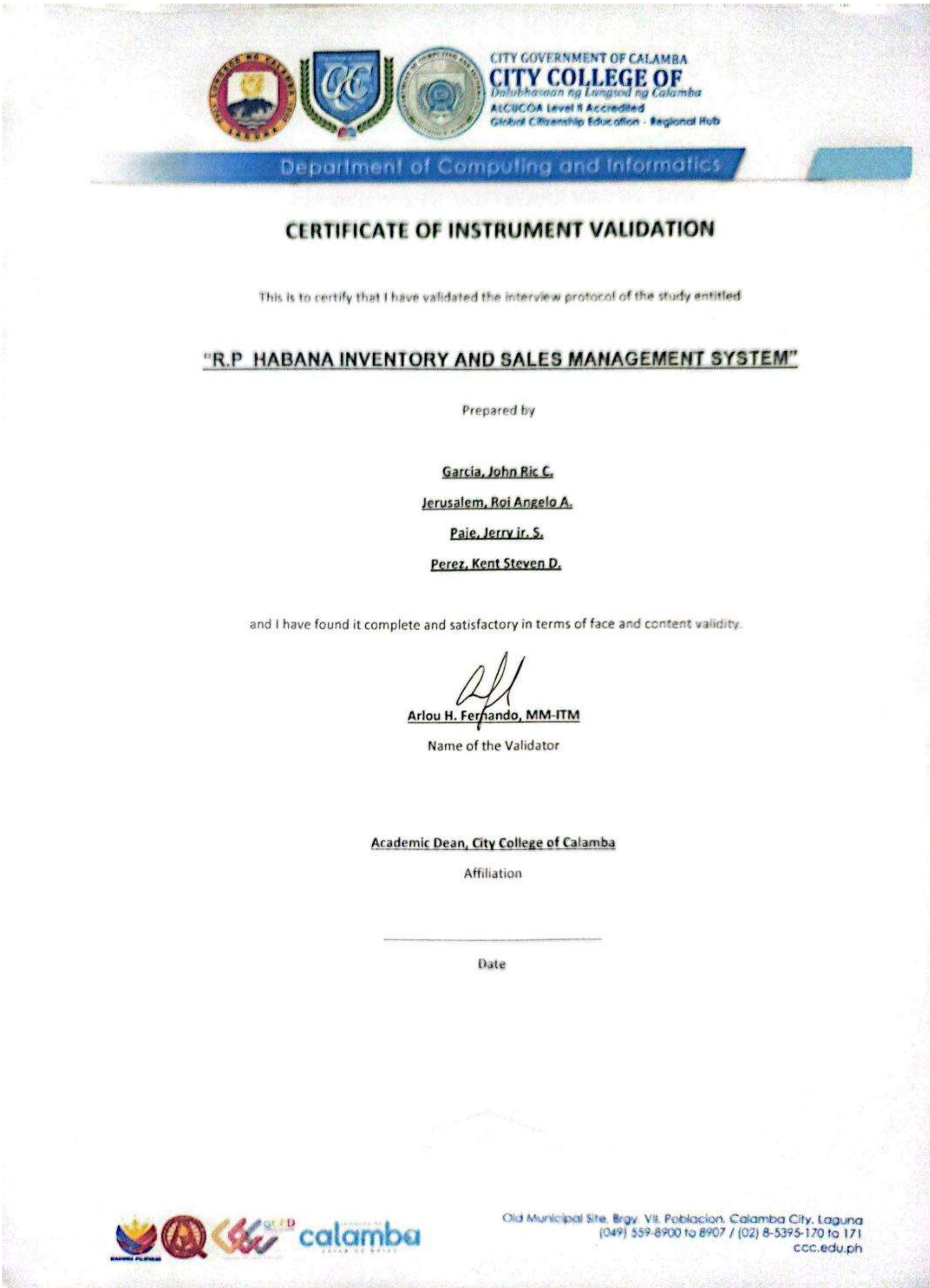

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Arlou H. Fernando, MM-ITM
Name of the Validator

Academic Dean, City College of Calamba
Affiliation

Date

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Perez, Kent Steven D.

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JASPER M. GARCIA, MIT

Name of the Validator

IT Professor, City College of Calamba

Affiliation

____

Date

Continuation of Appendix C…

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Jessabelle S. Perez

Name of the Validator

Jr. Manager, Moment Group Inc.

Affiliation

____

Date

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Appendix D

INTERVIEW QUESTIONS AND TRANSCRIPTS

Interview Question for R.P HABANA

INTERVIEW QUESTION ON R.P HABANA SHOP

Current Inventory and Sales Process

1) How does your inventory and sales management work?
2) Do you have any tools or system software use for tracking?
3) What are the challenges you faced when recording and monitoring the sales and inventory manually?
4) How do you track the inventory stocks of all branches?

Data Management

5) How do you sort, record and organize the inventory and sales data?
6) are there any difficulties you faced when updating the inventory stocks?
7) How often do you monitor the inventory and sales reports?

Accuracy

8) Can you provide some examples of inventory and sales-related errors?
9) How do you verify the accuracy of your reports?
10) How does human errors affect the inventory records?
11) How do you plan to reduce the manual errors in inventory and sales management?

Comparing Manual Process to System

12) What are the advantages you see in using a system instead of manual process?
13) What features/modules would you like to see in the web-based system?
14) Are the staff comfortable in using a system for inventory and sales management system?
15) Do you think a centralized web-based system would help for your smooth business operations?

Continuation of Appendix D…

IT Experts' Comment and Suggestions

IT EXPERTS' COMMENTS AND SUGGESTIONS

Group 52

R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM

IT EXPERT	COMMENTS/SUGGESTIONS
Ms. Jessabelle S. Perez	**Main Suggestions** 1. Add a **Start Float** and **End Float** function in the Point-of-Sale (POS) system to ensure accurate monitoring of the cash drawer at the beginning and end of each shift. This helps prevent discrepancies and improves transparency in cash handling. **Point-of-Sales and Sales History** 2. Provide a dedicated Shift Summary module for staff, allowing them to review daily shift activities such as: opening balance, total sales, refunds processed, pay-ins/pay-outs, and closing balance. 3. Ensure that the receipt layout includes all necessary transaction details such as: business name and address, receipt number, date/time, cashier name, list of items with quantity and price, payment method, and total amount due.

Continuation of Appendix D…

IT Experts' Comment and Suggestions

IT EXPERTS' COMMENTS AND SUGGESTIONS

Group 52

R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM

IT EXPERT	COMMENTS/SUGGESTIONS
Dean. Arlou H. Fernando, MM-ITM	**Main Suggestions** 1. Provide a dedicated landing page for customers, admin and staff where they can view business information and easily navigate to the sign-in portal, dedicated only for admin and staffs. 2. Simplify the password reset process by implementing OTP-based verification for enhanced convenience and security. **Dashboard & Interface** 3. Utilize submenus for modules related to Inventory and Backup & Restore to maintain a cleaner and more organized sidebar layout. 4. Include a date range or timeline filter for dashboard summary cards (e.g., Total Products, Low Stock Items, Out-of-Stock Items, Total Sales) to allow users to view data more accurately. **Inventory & Stock Management** 5. Implement an automatic product barcode generator and include a submenu within the Inventory module that allows users to print barcode labels for available products. 6. Introduce a unique Transfer Ticket ID and Stock-In Request ID to improve tracking and traceability of product and inventory movements. 7. Add expiration alerts for products approaching their expiry date to prevent spoilage and loss. **User and Security** 8. Improve the user account creation process by using a guided modal form to make the process clearer and more user-friendly. 9. Enhance user logs to record login activities, account modifications, and key inventory-related actions to strengthen accountability and security.

Continuation of Appendix D…

IT Experts' Comment and Suggestions

	Point-of-Sales and Sales History 10. Add a required remarks or reason input field when processing refunds for transparency and documentation. 11. Simplify the POS interface to maintain a clean and functional workflow; avoid designs that resemble online e-commerce storefronts. **System Backup & Receipts** 12. Enhance the Activity Logs UI by adding filter options and simplifying the table structure to ensure the logs remain informative and easy to understand for both staff and administrators.

Continuation of Appendix D…

IT Experts' Comment and Suggestions

IT EXPERTS' COMMENTS AND SUGGESTIONS

Group 52

R.P HABANA INVENTORY AND SALES MANAGEMENT SYSTEM

IT EXPERT	COMMENTS/SUGGESTIONS
Jasper M. Garcia, MIT	**Main Suggestions** 1. Improve the system's opening page to make it more professional and visually appealing. 2. Simplify the interface layout to avoid overwhelming users, ensuring clearer navigation of features. **Dashboard & Interface** 3. Redesign the dashboard to display only essential information (e.g., sales summary, alerts, pending tasks). 4. Add clear legends or labels for color indicators (e.g., red = critical stock, green = sufficient stock). 5. Increase visibility of the search button by enlarging it and placing it consistently across modules. **Inventory & Stock Management** 6. Make stock alerts more noticeable by using a pop-up alert or modal for low/out-of-stock items. 7. Streamline product creation by requiring only essential details. Add support for product QR/barcode features. 8. Centralize product records so branches share the same product list while updating only their respective stock quantities. **User and Security** 9. Improve the user account creation process by using a guided modal form instead of a plain input form.

Continuation of Appendix D…

IT Experts' Comment and Suggestions

	10. Enhance user logs to include login records, account changes, and critical inventory actions for accountability. **System Backup & Receipts** 11. Add a unique receipt tracking code for all transactions for easier audit trails. 12. Implement automatic daily system backup for product stocks and sales records to ensure data security and continuity.

Continuation of Appendix D…

Interview Transcript with Client

Researcher: Our first question ma'am is, ilang branches po ang business nyo?

Client: We have seven branches na ngayon.

Researcher: Ano po yung pinakamalaki nyong branch or main branch nyo po?

Client: Dito sa buca, sa may bypass.

Researcher: yung customer nyo po per day? Kahit sa main branch nyo po? Alam nyo po ba kung ilan?

Client: Depende.

Researcher: Pero tingin nyo po ba? Approximate number of customers po?

Client: siguro mga 5 more or less yon.

Researcher: sa isang branch lang po iyon? What if po sa total po or lahat po ng branch nyo?

Client: Dipende talaga sya sa branch.

Researcher: Pero yung number of customers po? Manageable naman po ba?

Client: Kaya naman.

Researcher: Next po ay yung number of employees po. Ilan po ba ang employees nyo sa laha po ng branches?

Client: Ano sya, mga 20.

Researcher: 20 po? 20 below po or Above?

Client: 20 Above sya.

Researcher: Next po ay ask may system po ba kayo na ginagamit?

Client: wala e

Researcher: Paano nyo po na ta-track yung expenses nyo or inventory ng business nyo?

Client: Manu-mano e, manually namin syang ginagawa

Researcher: Ahh manu-mano po? Ano po ba yon? Through papers po ba?

Client: Oo e, sa papel ko sya ginagawa eh

Researcher: Pati po ba yung sa sales ma'am?

Continuation of Appendix D…

Interview Transcript with Client

Client: Oo e, kasi hindi naman madami. Dipende sya sa totoo lang. May mga shop na iilan lang yung benta. Lalo na pag mayroong motor parts yung shop ayun, pero pag wala, hindi gaano.

Researcher: Uhmm… Lahat po ba ng shop nyo ma'am ay nag ooffer ng iisang services?

Client: Oo, pero may shop ako na dinagdag dito sa may bucal, mayroon syang service for alignment pero hindi sya yung pang malakasan talaga na alignment, yung ibang branches ko wala sya non.

Researcher: Ano po or mayroon po ba kayong common problems na na-eencounter?

Client: Oo e, dipende sa gawa ng employee ko, minsan kasi na ba-back job sya. Lalo nap ag hindi sanay gumawa.

Researcher: Eh sa inventory po ba?

Client: Oo e, minsan nga e napapalpak inventory namin since manually namin syang ginagawa. Kasi sila pinag-iinventory ko tapos tinitignan ko sya per week. Minsan may palpak sya.

Researcher: Ano po ba yon? May kulang po ba or sobra sa inventory?

Client: Oo, may kulang. Kapag may sobra walang problema syempre. Magkakaroon ng problema pag may kulang sa inventory

Client: Actually, may binili akong machine, ano nga tawag don? POS, Kaso hindi ko sya nagagamit.

Researcher: Ahh point of sale system po? Bakit po hindi nyo nagagamit?

Client: Ahh wala, asa akin pa kasi sya, dahil nga hindi ako nakakapag-stay sa isang shop since working pa rin ako bilang teacher.

Researcher: Ano pa po kaya yung common problems na ma-share nyo aside po sa inventory? Sa employee po? Pag mamanage po ng employee?

Client: Ahh mayroon minsan may dishonesty sila e, and to tell you rin, Malaki advantage ng mayroong parking kapag magtatayo ka ng business na mayroon kami.

Researcher: Lahat po ba mayroon parking?

Client: Ahh yung iba wala, kasi diba yung kalsada pinapa-expand sya minsan kung saang hindi ka pwedeng magpark doon.

Continuation of Appendix D…

Interview Transcript with Client

Researcher: Eh pano kaya yung sa payroll nila?

Client: Paanong payroll?

Researcher: If may 20 above po kayo na employee, paano nyo po na aayos payroll nila?

Client: ahh yung sa sahod nila?

Researcher: Opo

Client: Ahh ano sya per day. Sa computer, ine-encode ko yung pinaka sahod nila.

Researcher: Excel po ba gamit nyo roon?

Client: Yes, Excel ang gamit ko. Mayroon akong ginawang formula.

Client: Kasi ganito yan, diba nag vulcanize sila sainyo halimbawa 100 pesos yung singil nila, 40 pesos doon ay sakanila na, 60 pesos ay sa'min.

Researcher: Ahh okay po. Tingin nyo po ba kailangan pa po sa system ng isang mas maayos na payroll system?

Client: Pwede. Pero mas gusto ko talaga sa items, doon kasi ako pinaka-hirap, mag itemized or mag-inventory ng products.

Researcher: ito po ma'am last na tanong ko na po para po ma-make sure ko, ilan po yung consistent customer nyo per day? 5 po talaga?

Client: Above or Below 5 sya. Minsan nga na 0 yung isang items.

Researcher: Manageable po ba? Like sa employee po? Hindi naman po nagkakaroon ng problem sa mismong process ng business operation nyo?

Client: Mayroon naman kaming employee na 2-3 per branch.

Researcher: Kaya na po ba yon? Like yung sa overall job na kailangang gawin po sa business? Hindi naman po ba nagkakaroon ng problema?

Client: Ahh hindi naman.

Researcher: Ahh bali ang pinaka-issue nyo po ma'am ay yung sa inventory po?

Client: Oo doon sa inventory, kasi kaya ko naman yung tao e, ayos naman yung tao ko e.

Researcher: So I assume ma'am, iba-iba po yung pinapasa nilang inventory sainyo since marami po kayong branch?

Continuation of Appendix D…

Interview Transcript with Client

Client: Iba-iba rin kasi laman ng shop.

Researcher: Ahh okay po, iba-iba po pala.

Client: Oo, kasi may mga shop na may motor parts, may shop naman na wala. May battery sa isang shop, sa iba naman wala, dipende sa lugar.

Researcher: Paano po pala nila binibigay sainyo yung inventory po?

Client: Bali may ganito ako oh, papel or compiled sheets of inventory?

Researcher: Ahh okay po, aware po ako sa ganyan since nag work din po ako and nag handle ng inventory.

Researcher: Ano po ba ginagawa nila? Nag sesend na lang po ba sila sainyo ng picture po ng inventory?

Client: Oo, pag na punta ka naman sakanila, tinitignan ko rin. Minsan din may hawak ako, xinexerox ko. Halimbawa, sa isang araw nila, ito yung kinita, ito yung benta ganon. Piniprint ko yan, mayroon rin akong hawak sa inventory tapos ni leless ko. Kasi dipende minsan, isang buwan mag inventoy kami buong shop, may kasama akong tao, tapos para ma-audit namin, may record sila, may record ako. Tapos mag minus sila sa bawat benta nila.

Researcher: Tingin nyo ma'am may way para mapabilis yung process ng inventory nyo?

Client: POS, kaso nga lang mahal.

Researcher: Hindi po ba kayo na ooverwhelm yung branch kapag maraming customer?

Client: Hindi naman, alam naman namin kasi kung saang shop ang malakas sa hindi, halimbawa sa batino. Maganda ang ktiaan don, kaya minsan ang tao doon ay apat, tapos may secretary ako doon.

Researcher: May mga returning customers po ba kayo? And humihingi po ba sila ng discount or mayroon rin po ba kayong way para ma-store yung information nila? Para kunwari po, returning customer sila? Nag offer po ba kayo ng discount sa ganon?

Client: Ahh hindi na eh.

Researcher: Pero may returning customer po kayo?

Client: Oo mayroon naman. Kaso nga lang mahirap magbigay ng card for discount e, kasi tao lang namin humahawak non, and pabago-bago sya minsan kada branch. O kaya naman

Continuation of Appendix D…

Interview Transcript with Client

pag alam nilang customer na nila yon, sila na mismo nag leless. Tatawag na lang yun sa boss nila, para iinform.

Researcher: Kasi ma'am isa po sa naiisip ko na isama sa system ay maglalagay po kami ng customer profiling po para po centered na po doon, and ma-coconfirm nyo po if naging customer nyo na po si customer and ilang beses na po sya nagging customer, para po hindi lang po sa memories ni employee yung customer, kasi pwede naman po nila sabihing naging customer na nila pero ang totoo po ay kakilala lang po talaga.

Client: Oo ganon nga.

Researcher: May mga times po ba na may nag papa-appointment sainyo?

Client: Oo, may nagpapagawa din, nagpapaschedule rin, lalo na sa alignment.

Researcher: Nagkakaroon po ba ng problema sa scheduling?

Client: Ahh hindi naman, iilan pa lang naman at bago pa lang yung alignment, tapos madalas through messenger or contact number sila nag papa-appointment.

Researcher: May facebook page rin po ba kayo?

Client: Oo, pero bago pa lang to and minsan lang din to ginagamit.

Researcher: Bali base po sa assessment namin ma'am, ang system po na pwede naming gawin para sa business nyo ay inventory system po. Since sa nakikita po namin sa assessment namin sainyo, mahirap po sainyo na i-manage yung inventory ng seven branches ng business nyo and working pa rin po kayo bilang teacher.

Continuation of Appendix D…

Interview Transcript with Client

Date of interview: May 06, 2025

Researcher: About po sa products niyo po tulad po ng oil, gulong po may supplier po ba kayo ng

mga yon?

Client: Marami.

Researcher: marami, siguro ilan po?

Client: sabihin natin lima or higit pa.

Researcher: Iba-iba po yun?

Client: oo, iba-iba.

Researcher: Same po yun? Like lahat po ng suppliers nyo ay same sa lahat ng branches po na meron

kayo?

Client: Oo.

Researcher: Tas tuwing kailan naman po kayo um-order sa supplier ninyo?

Client: kapag wala ng i-stock or a month or week depende, depende sa sitwasyon ng kalakasan o

katumalan, depende sa benta. Pag walang i-stock mag order

Researcher: Nabanggit rin po last interview na may sitwasyon po na yung stock po dito sa branch na

ito mag rerequest po kayo sa ibang branch?

Client: yes po

Researcher: pano po nagiging process nito?

Client: May riders naman tayo na nag dedeliver.

Researcher: For example po. Hindi ba po mababawasan po ng inventory yung stock don sa isang

branch. Binabayaran po ba iyon or hindi na po?

Continuation of Appendix D…

Interview Transcript with Client

Client: hindi, ika-cancel nalang yon don. Kumbaga sa inventory ibabawas lang don kung san napunta

yung items.

Researcher: Tauhan niyo rin po ba yung rider sa pag dedeliver ng products sa bawat branch?

Client: Oo

Researcher: Since, Nag bebenta po kayo ng products? Nasa sainyo na po kung sasagutin niyo po ng

specific or percentage? And since confidential naman po ito nasa sainyo na rin po kung sasagutin niyo

po. Rerespetuhin naman po naming desisyon niyo po. Gusto lang po naming malaman yung example

po ng mark-up price Ninyo po?

Client: sa brand new gulong? 253.50 hanggang limang daan.

Researcher: Ano po yung mga services na in-offer nyo? Hindi po ba iba-iba yan sa bawat branch?

Client: same lang!, isa lang naman ang shop ko na ano, kagaya ng vulcanizing services. Bucal lang

naman ang may wheel alignment, wheel balancing sa iba wala vulcanizing lang.

Researcher: Sa lahat po ng branch nag chi-change oil po kayo?

Client: Oo, sa mga ano lang motor. Kasi maliliit lang naman yung ibang branch namin, sa bucal lang

naman yung Malaki, kumbaga rito yung may mga alignment, change oil ng mga four wheels. Bali sa

ibang branch maliit lang mga vulcanizing lang yun tas palitan ng gulong.

Researcher: May times po ba na nali-late kayo ng order kasi hindi nyo po na bantayan yung stocks

nyo na mababa na po?

Continuation of Appendix D…

Interview Transcript with Client

Client: Syempre meron, meron Talaga. Kapag wala naman stock syempre kakausapin natin customer

kung makakapag antay kasi walang i-stock. Depende kasi yan kung ang items natin ay mabenta lang

naman kung fast moving. Hindi naman tayo nag i-stock kasi medyo hindi ganon ka fast moving yung

items.

Researcher: May mga fast moving items po kayo?

Client: Oo, depende sa mga size kagaya ng sa mga fortuner, pero yung mga pang BMW na gulong

wala tayo niyan, kasi hindi naman masyadong hanapin nyan. May mga order pero hindi tayo mag istock kasi masyado mataas ang puhunan tas mati-tengga lang naman. Pag kailangan or mabilisan na hanapin ayun lang ini-stock namin.

Date of Interview: August 30, 2025 12:28PM

Researcher: Good afternoon po! May mga follow-up questions po kami, may mga kailangan lang kaming malaman para mas maging maayos po yung system.

Client: Hello, sige ano yun? Para habang andito tayo, itanong mo na yung mga kailangan ninyo.

Researcher: So ayun na nga po. Since auto supply ang business ninyo, paano niyo po hinahandle ang mga items na matagal na sa shelf o malapit na ma-expire?

Client: Sa totoo lang, kahit marami kaming binebenta dito, bihira naman mag-expire ang products namin. Karamihan sa binibenta namin ay gulong, langis, at iba pang mechanical parts.

Researcher: Bali yung mga products po na yun ay matagal mag-expire?

Client: Oo, tulad ng langis, matagal ang shelf life. Yung mechanical parts naman wala talaga expiration. Pero sa gulong, may DOT code yan. May mga customer na marunong magbasa nun, nakikita nila kung gaano na katagal yung tire.

Researcher: So kapag po yung gulong ay medyo luma na based sa DOT code, ano po ang nagiging approach ninyo sa pagbebenta?

Continuation of Appendix D…

Interview Transcript with Client

Client: Maseselan kasi ang karamihan sa mga buyer. Gusto nila bago o yung pasok pa sa DOT code. Pero syempre may mga buyer pa rin ng mga lumang gulong.

Researcher: So binebenta nyo pa rin po sila?

Client: Oo, binebenta pa rin pero palugi na lang. Kailangan ma-dispose para hindi masayang. Syempre mas mababa ang kita. Pero bihira naman mangyari kasi mabilis ang galaw ng stocks, lalo na yung mga common sizes.

Researcher: Ibig sabihin po, hindi naman madalas mangyari ang spoilage or full expiration ng items?

Client: Oo, madalang talaga. Usually nangyayari lang yan sa mga sizes na hindi gaanong in demand—halimbawa sa malalaking sasakyan, like truck tires.

Researcher: Base po sa sagot ninyo, we're planning to add discount feature sa POS system na gagawin para sa inyo. Para kung may buyer pa rin ng mas lumang gulong, you can apply discount and ma-track pa rin ng system ang sales.

Client: Pwede naman yun. Pero doon lang kami nagbibigay ng discount ha? Usually sa services namin, depende na yun sa mekaniko ko kung bibigyan nila ng discount yung customer, lalo na kapag kilala na nila.

Researcher: Noted po. By the way, regarding pricing po — kapag halimbawa bagong dating na stocks or normal pricing, magkano po ang dinadagdag ninyo sa original cost? Percentage po ba or fixed amount?

Client: Madalas fixed amount. Mga around ₱100 to ₱300 ang dinadagdag namin, depende sa product, brand, at laki ng item.

Researcher: Ah, noted po. Kasama na po ba yung VAT dun sa markup na yun?

Client: Hindi pa. Yung VAT hiwalay pa namin ina-apply. Iba pa yun sa final computation ng presyo kapag i-eencode na sa resibo.

Researcher: Ma'am Thank you so much po ulit sa time! Kung may nakalimutan man po ako, pasensya na po, mag set na lang po ulit ako ng other meeting.

Continuation of Appendix D…

Interview Transcript with Client

Date of Interview: September 6, 2025 2:38PM

Interviewee: Mrs. Riza P. Habana and its Staffs

Researcher: Good afternoon po Ma'am Riza. Gusto lang po sana namin mag conduct ng konting observation dito sa shop nyo at para rin po masubukan nyo at nila kuya at ate na mga staff nyo po yung aming system na dinedevelop.

Client: Good afternoon mga to! Bali makiki-alalayan na lang ang karamihan sakanila, ako naman kasi nasubukan na ang system, pero kalian pa yon, gamay ko ang pag gamit ng laptop, pero ang karamihan sa empleyado ko ay hindi.

Researcher: Wala naman pong problema don maam, simulant na po na'tin.

Staff 1: Itanong ko lang, para saan ba yung ginagawa nyo?

Researcher: Para po ito sainyong lahat, hindi lang po kay Ma'am Riza, pati na rin po sainyong mga staff nya. Ito po kasi ay para mapadali ang inyong trabaho, in terms of monitoring po ng inventory and sales.

Client: Maganda to, kasi ano sya… Mapapadali yung pag hihirapn natin sa inventory *laughed*

Researcher: Lahat po kayo ay mag nanavigat ngayon, tuturuan naman po namin kay. We're simply making the inventory and sales centralized and automatic po.

Researcher: Bali yung dating manual nyo po na binibilang ang inventory and sales each branch, automatic na po sya and real-time using po nitong system na ipapagamit ko po sainyo.

Staff 2: *ohhhh…* Ang ganda nito bossing ah, mas madali nga sya, nakikita ko kung ilang stocks na lang yung isang item.

Researcher: Naku! Salamat po kuya, pero hindi pa po tapos yan, makikita nyo naman po dito yung POS nyo, or yung Point-of-Sale system nyo, yung parang sa mga cashier po.

Staff 1: Ano ba ito? Kapag ka nag tuloy sa benta dito sa system nyo? Automatic mababawasan yung stocks? Tama ba?

Researcher: Yes po, ganon na nga po. Ayun na rin po yung ibig sabihin namin na automatic. Hindi nyo na po kailangan bilangin pa or i-compute sa sales nyo para malaman kung ilan po ang nabawas sa stocks nyo.

Staff 2: Galing naman, nagagawa nyo yun. *Laughed*

Continuation of Appendix D…

Interview Transcript with Client

Staff 1: Ang nagustuhan ko rito, titignan na lang dito kung ilan yung stocks, instant na.

Researcher: Masaya naman po kami at nagustuhan nyo kahit marami pa po kaming kailangan idagdag.

Client: To! Baka naman pwede nyo rin kami gawan ng magandang logo?

Researcher: Wala pong problema dyan Ma'am. Madali naman na po yun at maganda rin po ang business nyo.

Client: Maraming Salamat hah. May mga kailangan pa ba kayo? Picture? Data?

Researcher: Ayun! Bali ma'am, since marami po kayong data, ask ko lang po kung baka po pwede makahingi ng updated lists po?

Researcher: Para po kasi sya pag mag lagay na po kami ng data nyo po, ilalagay na po namin sya sa system po kapag tapos na. Para kapag po nakuha nyo na ito, ready to use na po.

Client: Wala namang problema to, kaso kasi marami yon, wala kasi akong time para ayusin yung 2024 na ibinigay ko sainyo since teacher nga ako diba?

Researcher: Ayun na lang po ba gamitin namin?

Client: Oo to, alam ko na rin naman na pwedeng mag edit ng product dyan sa system nyo. Once na tapos na yan, kahit nga kami na ang mag edit non. Keri na ng mga trabahador ko yon.

Researcher: Maraming Salamat po ma'am!

Date of Interview: September 13, 2025 4:30PM

Interviewee: Mrs. Riza P. Habana

Researcher: Ma'am, magandang araw po. Since we're developing the system that will enhance your business operation, lalo na po in terms of monitoring and updating inventory stock. We would like to ask na ang poor inventory management po ba ninyo ay nag reresult sa lost sales?

Client: Oo, minsan talaga. Yung understock, yun talaga ang mostly na dahilan ng lost sales. May mga customer na pupunta, hinahanap ang part, tapos wala pala sa stock.

Researcher: Ilang beses po bang nangyayari ito sa isang buwan? madalas po ba o occasional lang?

Continuation of Appendix D…

Interview Transcript with Client

Client: Depende sa branch. Sa mga maliliit na shop, maybe once or twice a week. Sa main branch baka less, pero may mga special parts na talaga bihira, pag wala na, hindi agad nababalik.

Researcher: Kapag nangyayari yung ganon po, ano po usually ang ginagawa ni customer? We assume po na naghahanap sila ng ibang shop?

Client: Yon, madalas talaga na nag-hahanap sa ibang shop. Kaya alam na agad ng mga tauhan ko na sayang ang kita ng shop.

Researcher: Sa tingin nyo po, gaano kalaki ang epekto nito sa kita? May rough estimate po ba kayo ng lost sales?

Client: Mahirap i-exact, pero regardless kung anong klaseng item, siguro around 200-250 pesos per month. Eh mayroon akong 7 branches, that's around 1700. Malaki na rin yan every year.

Researcher: Nai-encounter niyo rin po ba overstock, at ano po ang epekto nun?

Client: Oo, overstock din madalas. Yung mga hindi mabilis maikot, nagtatagal lang sa shelf. Nagkakaroon ng damaged items, lalo na kapag battery or ibang sensitive parts. Tapos naka-tie up yung pera namin dun.

Researcher: May spoilage or obsolescence din po ba dahil sa overstock? Nagiging obsolete ba yung mga produkto?

Client: Meron. May mga bagong model na lumalabas, yung old models na sobra sa stock, hirap maibenta. Minsan ibinibenta namin ng discount pero maliit lang ang margin. Mawala yung profit.

Researcher: Next naman po ay, napansin po namin kung gaano karaming papel po ang ginagamit nyo sa manual inventory. Alam nyo po ba kung magkano ang nagagastos nyo for papers and printing ng inventory sheet nyo?

Client: *hmm…* kaya ko naman magbigay siguro ng estimate lang ba, pero usually kasi mukha lang sya magastos pero y ung mismong inventory sheet kasi namin, nagagamit na sya ng around 2-3 months.

Researcher: Ohhh… matagal nyo rin po pala nagagamit, pero mga na sa magkano po ba? Sabihin na nating yearly po?

Client: Siguro na sa 2000 sya.

Researcher: Ahhh 2000 yearly po.

Continuation of Appendix D…

Interview Transcript with Client

Client: Ay hindi, kasi naalala ko, nag mahal din kasi ang ink, so I think mga na sa 3000-3500 pala sya yearly.

Researcher: Ayan po ang estimated po na inyong nagagastos sa Inventory sheet po, despite na may 7 branches po kayo?
Client: Oo naman, kasi yung pinaka inventory sheet, ako lang ang may hawak non. Dinadala ko lang yung bawat shop pag mag iinventory na kami, nakakapagod nga mag paikot-ikot dito sa mga branches namin e.

Researcher: Ayun lang po Ma'am, maraming Salamat po sa time!

Appendix E

COMPUTATIONS

Weighted Mean Computation for Survey Responses

Process:

STEP 1: After counting the number of responses for each scale value, multiply each by its corresponding scale value.

STEP 2: then after getting the sum of the specific statement divided it to the total number of participants to get the weighted mean.

RATING	WEIGTHED MEAN	DESCRIPTIVE VALUE
5	4.21-5.00	EXCELLENT
4	3.41-4.20	VERY SATISFACTORY
3	2.61-3.40	NEUTRAL
2	1.81-2.60	FAIR
1	1.00-1.80	POOR

Functional Suitability	E	VS	N	F	P	WM	INTERPRETATION
FS 1	19	3	=ROUND(((B57*5)+(C57*4)+(D57*3)+(E57*2)+(F57*1))/SUM(B57:F57),2)				
FS 2	16	6	ROUND(**number**, num_digits)			4.73	Excellent
FS 3	20	0	2	0	0	4.82	Excellent
FS 4	22	0	0	0	0	5	Excellent
FS 5	20	2	0	0	0	4.91	Excellent
OVERALL						4.86	Excellent

STEP 3: then after getting the weighted mean of each specific statement add it and divided it to the total number of items

Functional Suitability	E	VS	N	F	P	WM	INTERPRETATION
FS 1	19	3	0	0	0	4.86	Excellent
FS 2	16	6	0	0	0	4.73	Excellent
FS 3	20	0	2	0	0	4.82	Excellent
FS 4	22	0	0	0	0	5	Excellent
FS 5	20	2	0	0	0	4.91	Excellent
OVERALL					=ROUND(AVERAGE(G57:G61),2)		llent
					ROUND(**number**, num_digits)		

Continuation of Appendix E…

Cost Benefit Analysis Computation

PROCESS:

Added all the benefit values to compute the Total Benefits. The amounts for reduced printing costs, and reduced revenue losses were summed using the formula =SUM(B3:B4) to get PHP 374,300.00 for each year.

Cost Benefit Analysis of the Development of R.P. Habana Inventory and Sales Management System

BENEFITS	2025		2026		2027		2028		2029	
Reduce Printing and Paper Costs	PHP	3,500.00	PHP	3,500.00	PHP	3,500.00	PHP	3,500.00	PHP	3,500.00
Reduce Revenue Losses	PHP	370,800.00	PHP	370,800.00	PHP	370,800.00	PHP	370,800.00	PHP	370,800.00
TOTAL BENEFITS	=SUM(B3:B4)		**PHP**	**374,300.00**	**PHP**	**374,300.00**	**PHP**	**374,300.00**	**PHP**	**374,300.00**

Added all the development cost items to compute the **Total Development Cost.**

The **Developer's Fee** serves as the primary cost component. Using the formula =SUM(B7:B7), the total development cost amounts to **₱285,000.00**, representing the full financial requirement for the system's development.

DEVELOPMENT COST										
Developers Fee	PHP	285,000.00	PHP	37,000.00	PHP	37,000.00	PHP	37,000.00	PHP	37,000.00
TOTAL DEVELOPMENT COST	**=SUM(B7:B7)**		**PHP**	**37,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**

Added all operational cost items to compute the Total Operational Cost. The electricity bill, internet bill, web hosting, SMS API, and computer cost were summed using the formula =SUM(B10:B14), resulting in a **Total Operational Cost of PHP 182,965.00** on the first year.

OPERATIONAL COST										
Electricty Bill	PHP	78,865.00	PHP	78,865.00	PHP	78,865.00	PHP	78,865.00	PHP	78,865.00
Internet Bill	PHP	84,000.00	PHP	84,000.00	PHP	84,000.00	PHP	84,000.00	PHP	84,000.00
Php Web Hosting	PHP	5,300.00	PHP	-	PHP	-	PHP	-	PHP	-
Semaphore SMS API	PHP	2,800.00	PHP	-	PHP	-	PHP	-	PHP	-
Computer	PHP	12,000.00	PHP	-	PHP	-	PHP	-	PHP	-
TOTAL OPERATIONAL COST	=SUM(B10:B14)		**PHP**	**162,865.00**	**PHP**	**162,865.00**	**PHP**	**162,865.00**	**PHP**	**162,865.00**

Computed the Total Cost by adding the Development Cost and the Operational Cost. The formula =SUM(B15, B8) was used to combine the Total Development Cost and the Total Operational Cost. Since the Development Cost is **₱285,000.00**, the Total Cost is **₱467,965.00**

Continuation of Appendix E…

Cost Benefit Analysis Computation

TOTAL DEVELOPMENT COST	**PHP**	**285,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**	**PHP**	**37,000.00**
OPERATIONAL COST										
Electricty Bill	PHP	78,865.00	PHP	78,865.00	PHP	78,865.00	PHP	78,865.00	PHP	78,865.00
Internet Bill	PHP	84,000.00	PHP	84,000.00	PHP	84,000.00	PHP	84,000.00	PHP	84,000.00
Php Web Hosting	PHP	5,300.00	PHP	-	PHP	-	PHP	-	PHP	-
Semaphore SMS API	PHP	2,800.00	PHP	-	PHP	-	PHP	-	PHP	-
Computer	PHP	12,000.00	PHP	-	PHP	-	PHP	-	PHP	-
TOTAL OPERATIONAL COST	**PHP**	**182,965.00**	**PHP**	**162,865.00**	**PHP**	**162,865.00**	**PHP**	**162,865.00**	**PHP**	**162,865.00**
TOTAL COST	**=SUM(B15,B8)**		**PHP**	**199,865.00**	**PHP**	**199,865.00**	**PHP**	**199,865.00**	**PHP**	**199,865.00**
NET BENEFIT COST	**PHP**	**(93,665.00)**	**PHP**	**174,435.00**	**PHP**	**174,435.00**	**PHP**	**174,435.00**	**PHP**	**174,435.00**
CUMULATIVE NET CASH FLOW	**PHP**	**(93,665.00)**	**PHP**	**80,770.00**	**PHP**	**255,205.00**	**PHP**	**429,640.00**	**PHP**	**604,075.00**
RETURN OF INVESTMENT (ROI)		**-20.02%**		**87.28%**		**87.28%**		**87.28%**		**87.28%**

Calculated the Net Benefit Cost by subtracting the Total Cost from the Total Benefits. The formula =B5 – B16 was used to get the Net Benefit Cost. Since the Total Benefits (PHP 38,000.00) are higher than the Total Cost (PHP 374,300.00), the system results in a **net benefit of PHP 93,665.00** on the first year.

Appendix F

Client Sample Forms

R.P Habana Inventory sheet

No.																									
									1	2	3	4	5	6	7	8	9	10	1	12	13				
1	OILS							0														0		0	
2	2T ADVANCE	14						14														0		14	
3	BATTERY SOLUTION	2						2														0		2	
4	Bosny Paint	4						4														0		4	
5	Bosny Gold							0														0		0	
6	Extreme Paint Best Drive Paint	14						14														0		14	
7	Thunder	4						4														0		4	
8	TOP SPORTS	7						7														0		7	
9	VS1	2						2														0		2	
10	WD40							0														0		0	
11	WATER SOLUTION							0														0		0	
12	ZEALANT TIRE big	38						38														0		38	
13	ZEALANT TIRE small	29						29														0		29	
14	Magic Gatas 500ml							0														0		0	
15	Magic Gatas 200ml							0														0		0	
16	Tire black							0														0		0	
17	Metal Polish							0														0		0	
18	Anti Rust							0														0		0	
19	BS-40							0														0		0	
20	Moto Gear Oil							0														0		0	
21	WHIZ brake fluid							0														0		0	
22	0	2						2														0		2	
23	Kawasaki							0														0		0	
24	Butane Gas							0														0		0	
25	0							0														0		0	
26	Brake Fluid / WHIZ / Nat'l 170ml							0														0		0	
27	Brake Fluid/Nat'l 4T 300ml	3						3														0		3	
28	Brake Fluid/Nat'l 4T 900ml	3						3														0		3	
29	0							0														0		0	
30	Bosny Paint							0														0		0	

Appendix G

Documentation

Client Interview

Continuation of appendix G…

Client Interview

May 02, 2025

Continuation of appendix G…

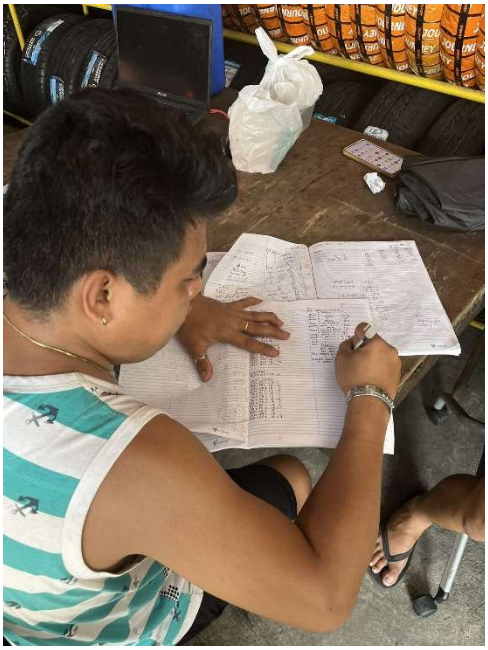
Pre-Oral Defense

May 06, 2025

Continuation of appendix G…

Client Interview

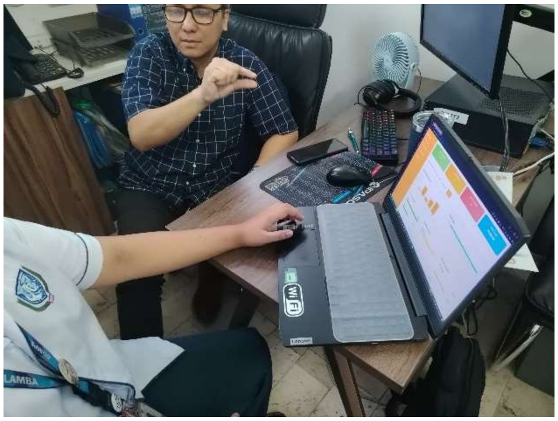
May 18, 2025

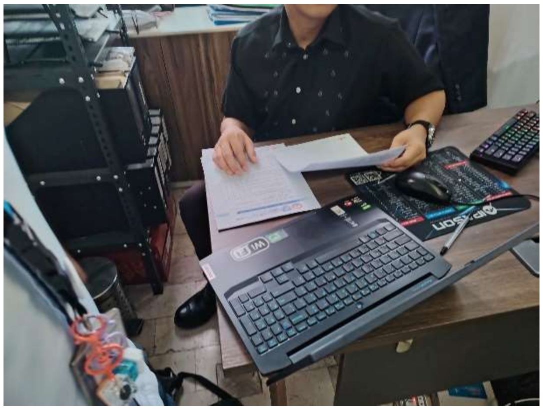


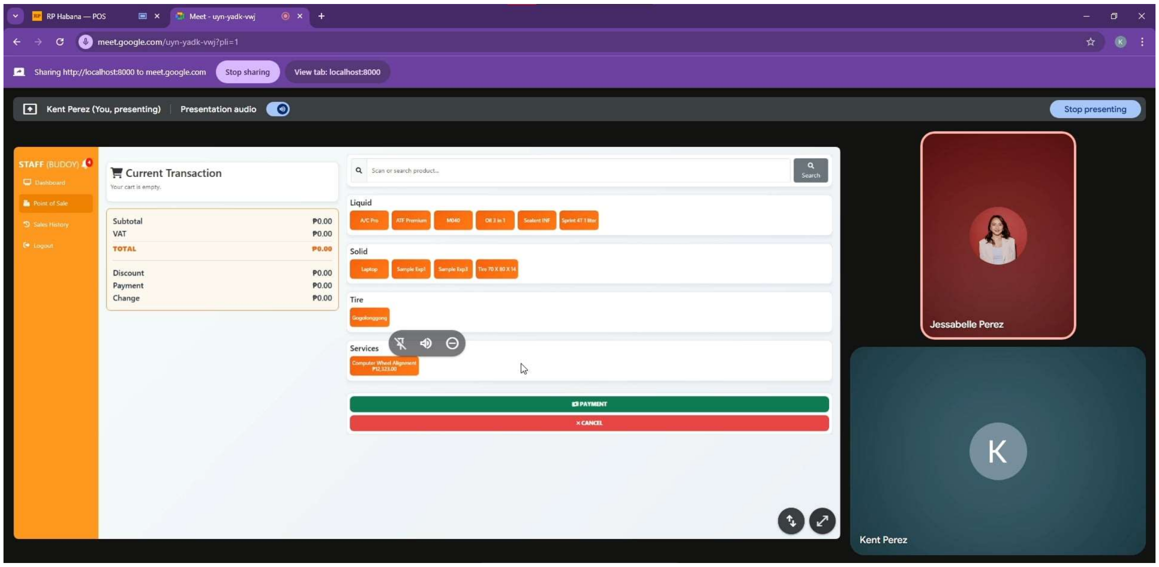
Continuation of appendix G…

System Validation with IT Experts

September 11 - October 15, 2025



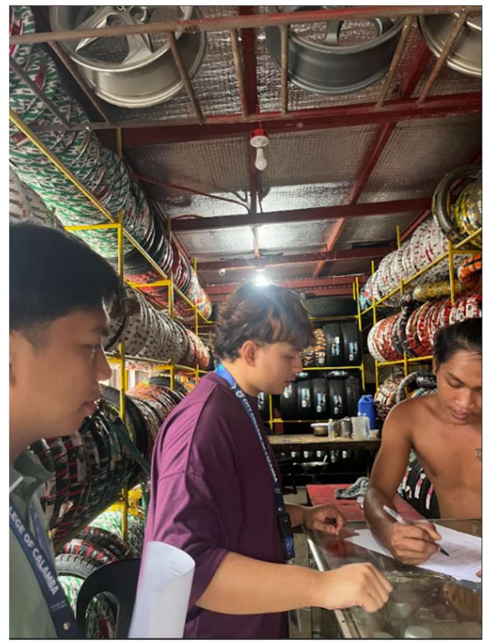




Continuation of appendix G…

Client Testing, Software Survey and Evaluation

October 17-18, 2025



Continuation of appendix G…

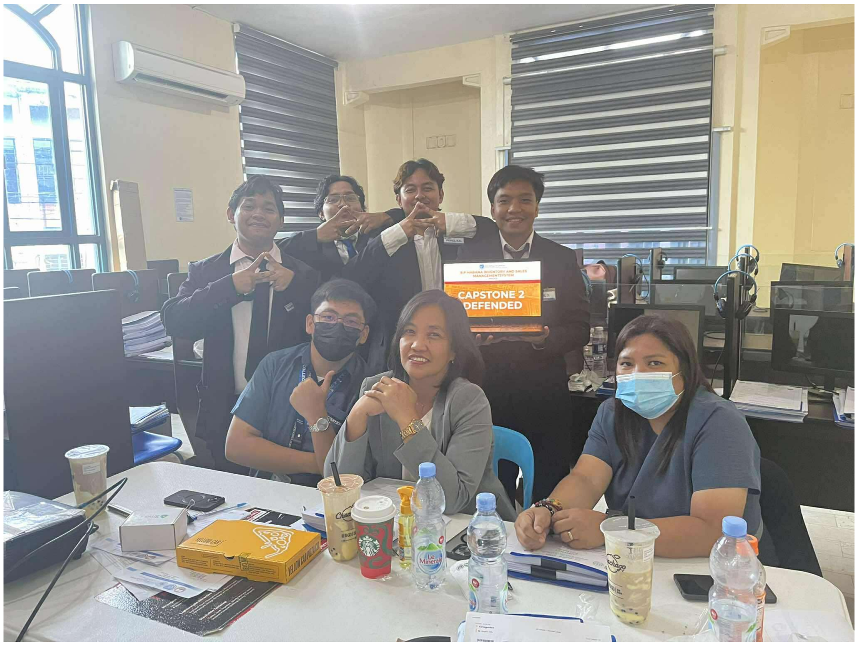
Client Testing, Software Survey and Evaluation

October 17-18, 2025

Continuation of appendix G…

Final Defense

November 18, 2025



Appendix H

USER MANUAL